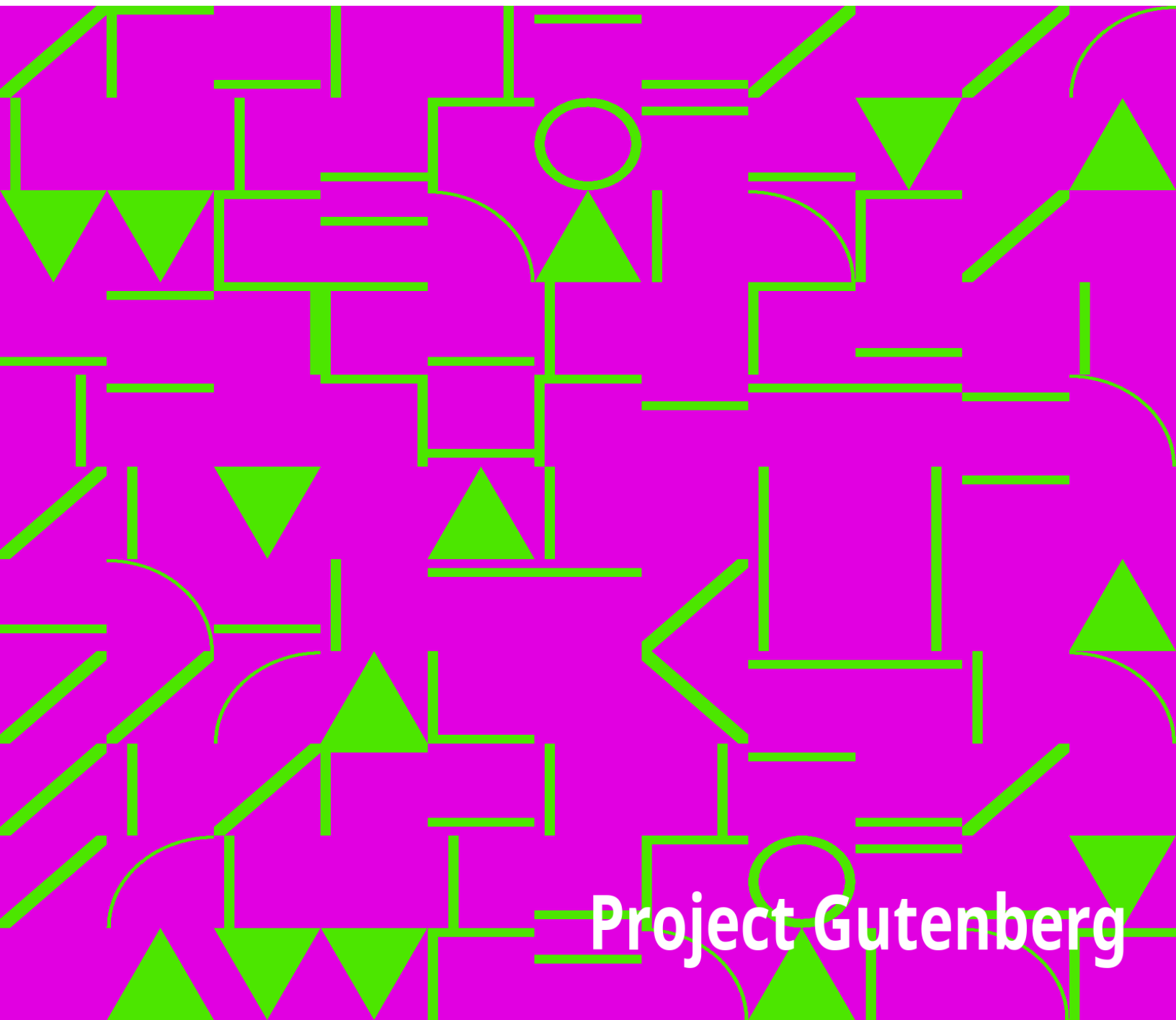


Northern Nut Growers Association Report of the Proceedings at the Fifteenth Annual Meeting

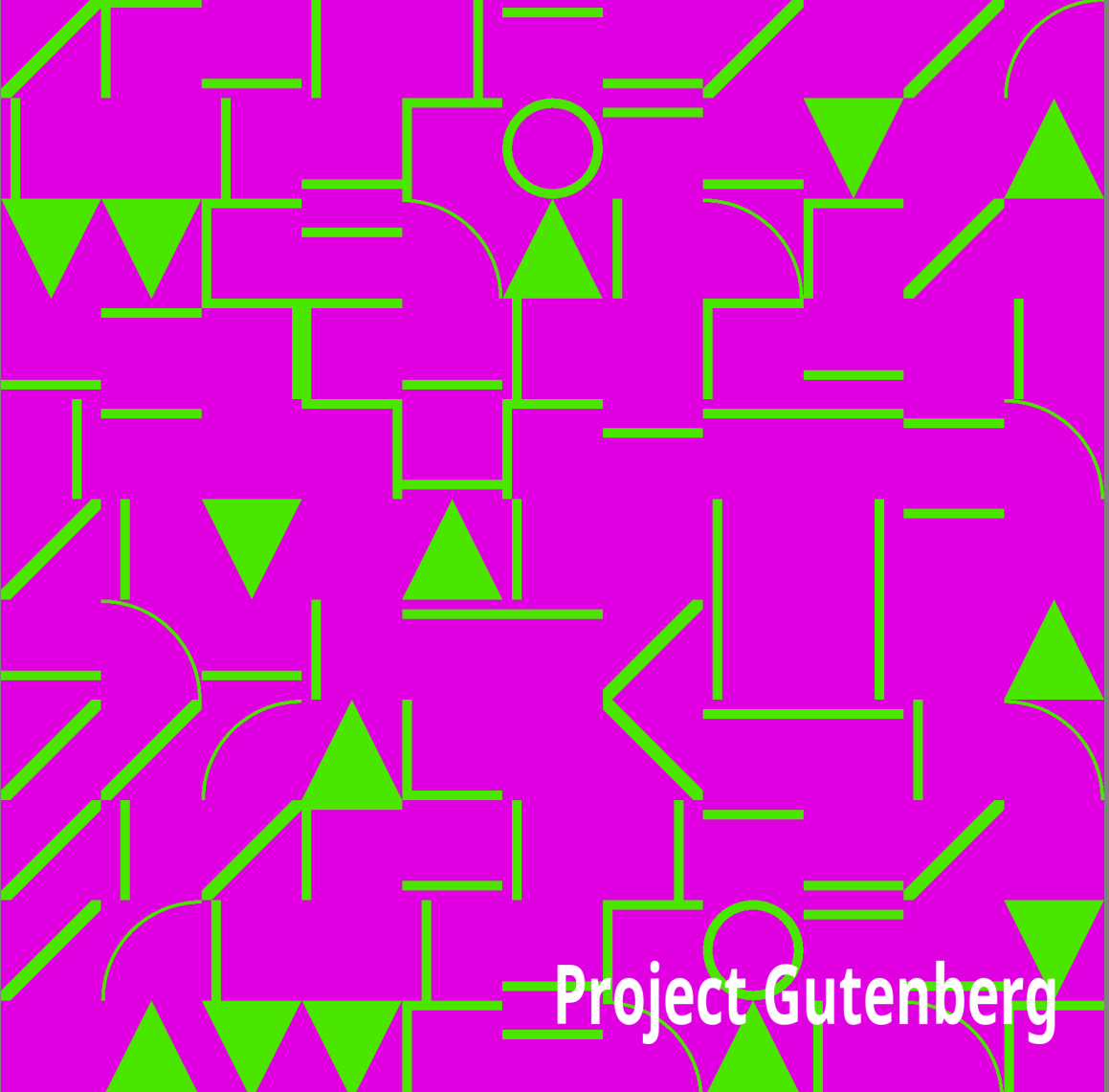
Northern Nut Growers Association

An abstract geometric pattern consisting of various shapes like squares, triangles, circles, and lines, all outlined in black and filled with a light blue color, set against a white background.

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Northern Nut Growers Association Report of the Proceedings at the Fifteenth Annual Meeting

Northern Nut Growers Association



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NUT GROWERS ASSOCIATION REPORT OF THE PROCEEDINGS AT
THE FIFTEENTH ANNUAL MEETING ***

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NORTHERN NUT GROWERS ASSOCIATION

(INCORPORATED)

REPORT

OF THE PROCEEDINGS AT THE

FIFTEENTH ANNUAL MEETING

NEW YORK CITY SEPTEMBER 3, 4 and 5, 1924

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(Compiled November 12, 1924)

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Hartzel, B. F., Shepherdstown

Mish, A. F., Inwood

WISCONSIN

Holden, Dr. Louis Edward, Beloit

* Life Member.

CONSTITUTION

ARTICLE I

Name. This society shall be known as the NORTHERN NUT GROWERS ASSOCIATION.

ARTICLE II

Object. Its object shall be the promotion of interest in nut-bearing plants, their products and their culture.

ARTICLE III

Membership. Membership in the society shall be open to all persons who desire to further nut culture, without reference to place of residence or nationality, subject to the rules and regulations of the committee on membership.

ARTICLE IV

Officers. There shall be a president, a vice-president, a secretary and a treasurer, who shall be elected by ballot at the annual meeting; and an executive committee of six persons, of which the president, the two

last retiring presidents, the vice-president, the secretary and the treasurer shall be members. There shall be a state vice-president from each state, dependency, or country represented in the membership of the association, who shall be appointed by the president.

ARTICLE V

Election of Officers. A committee of five members shall be elected at the annual meeting for the purpose of nominating officers for the following year.

ARTICLE VI

Meetings. The place and time of the annual meeting shall be selected by the membership in session or, in the event of no selection being made at this time, the executive committee shall choose the place and time for the holding of the annual convention. Such other meetings as may seem desirable may be called by the president and executive committee.

ARTICLE VII

Quorum. Ten members of the association shall constitute a quorum, but must include two of the four elected officers.

ARTICLE VIII

Amendments. This constitution may be amended by a two-thirds vote of the members present at any annual meeting, notice of such amendment having been read at the previous annual meeting, or a

copy of the proposed amendment having been mailed by any member to each member thirty days before the date of the annual meeting.

BY-LAWS

Article I

Committees. The association shall appoint standing committees as follows: On membership, on finance, on programme, on press and publication, on nomenclature, on promising seedlings, on hybrids, and an auditing committee. The committee on membership may make recommendations to the association as to the discipline or expulsion of any member.

ARTICLE II

Fees. Annual members shall pay three dollars annually, or four dollars and a half including a year's subscription to the American Nut Journal. Contributing members shall pay ten dollars annually, this membership including a year's subscription to the American Nut Journal. Life members shall make one payment of fifty dollars, and shall be exempt from further dues. Honorary members shall be exempt from dues.

ARTICLE III

Membership. All annual memberships shall begin either with the first day of the calendar quarter following the date of joining the association, or with the first day of the calendar quarter preceding that date as may be arranged between the new member and the Treasurer.

ARTICLE IV

Amendments. By-laws may be amended by a two-thirds vote of members present at any annual meeting.

ARTICLE V

Members shall be sent a notification of annual dues at the time they are due, and if not paid within two months, they shall be sent a *second notice*, telling them that they are not in good standing on account of non-payment of dues, and are not entitled to receive the annual report.

At the end of thirty days from the sending of the second notice, a *third notice* shall be sent notifying such members that unless dues are paid within ten days from receipt of this notice, their names will be dropped from the rolls for non-payment of dues.

REPORT OF THE PROCEEDINGS

at the

FIFTEENTH ANNUAL CONVENTION

of the

NORTHERN NUT GROWERS' ASSOCIATION, INC.

September 3, 4 and 5, 1924

Held in the

MUSEUM OF THE BOTANICAL GARDENS

BRONX PARK, NEW YORK CITY

EXCURSIONS

Baldwin, Long Island, Sept. 4 Stamford, Connecticut, Sept. 5

Chairman—PRESIDENT HARRY R. WEBER

FIRST DAY—MORNING SESSION

THE PRESIDENT: The meeting will please be in order, and we will have the secretary read his report.

THE SECRETARY: Secretary's Report for 1924.—Fourteen years ago, on November 17, 1910, two women and ten men, seers and prophets, met for organization in this building at the invitation of Dr. N. L. Britton, at that time and now, Director of the New York Botanic Gardens. We meet here again today by reason of his unfailing kindness.

Of the twelve persons present at that first meeting, three are here again, Dr. Britton, Dr. Morris and myself, and two are known to be dead, Prof. Craig of Cornell University, and Mr. Henry Hales, of Ridgewood, New Jersey.

The association has held an annual convention each year of its existence except during the war, in 1918, when no formal meeting was held. An annual report has been published every year, except that the report of the proceedings of the first meeting was incorporated in the report of the second meeting, and the ninth report, that for 1918, has not yet been issued.

The present secretary has held the office every year except in 1918 and 1919, during military service, when Mr. Bixby took his place.

From an educational and scientific standpoint I think the association may be said to have fulfilled creditably its original declaration of purpose, "the promotion of interest in nut-bearing plants, their products and their culture." Many choice nuts have been brought to notice and perpetuated. The establishment of nurseries where grafted nut trees of choice varieties may be obtained has been encouraged. The art of grafting and propagating nut trees has been brought to a high degree of success by members of the association. Experimental orchards, both of transplanted nursery trees and

of topworked native trees, have been established in widely separated parts of the country.

Acting on the suggestion and request of members of the association, Mr. Olcott established the American Nut Journal, one of the most important of our accomplishments. Finally, and perhaps best of all, a number of horticultural institutions have taken up seriously the study of nut culture and the planting of experimental orchards. Testimony to this will be found in letters to be read by the secretary and in the presence on our program today of representatives of several horticultural and other institutions of learning. I believe that the association can take credit to itself for having, by its publications and other means of influence, in large degree brought about this interest and action.

As for any commercial success in nut-growing, brought about by our activities, when we compare nut-growing in our field with pecan-growing in the South, and with walnut, almond, and perhaps filbert-growing, on the Pacific Coast, our results are meagre indeed. Of course commercial production, the building of a new industry of food supply for the people, is our ultimate goal. Why are our results in this direction, after fourteen years of effort, so small? Is it because we have devoted ourselves too exclusively to the scientific and educational aspects of our problems and neglected, either from over-cautiousness or from inertia, to encourage commercial plantings? There are some of our members who think that we have. They say that we should have been bolder in assuring people of success to be attained in nut tree planting.

As for me I do not think that we have been too cautious. We who are so accused, can point to the disastrous results of following the advice of commercially interested persons, results which have had much to do with

retarding and discouraging nut planting and counteracting the labors of our association.

But now, however, I believe that we have reached a state of knowledge where we can confidently recommend the commercial planting of nut orchards. We recommend the Indiana pecan in many states; the improved black walnuts over a much wider area, and the chestnut in many localities where it is not a native tree. The top-working of native hickories and black walnuts also can be confidently recommended. In every case, however, the adaptability of the kind of nut to the locality should be passed upon by an expert. In every case, also, even in that of top-working native hickories and walnuts, intelligent and generous care is essential for any degree of commercial success.

It is probable also, that the planting of the European filbert can be recommended under conditions of intelligent care.

Now what of the association's future? The field is boundless but the working cash is wanting. Faith is unlimited but works are conditioned by want of appeal to commercial powers. It is almost a vicious circle, no commercial appeal no money, no money no development to appeal to commerce. But we do make progress and it is accelerated progress. In time we must necessarily arrive at our goal. Our lines of advance are sketched out and our progress along these lines depends on the energy of the workers and the means with which they have to work.

I shall ask the association to establish a rule as to when members are in good standing and when they should be dropped from the rolls for non-payment of dues.

I shall also ask for a clear understanding, in the form of an amendment to the by-laws, on the question of annual dues and their combination with the

American Nut Journal.

It is desirable that we have a ruling as to a fiscal year.

The delay in the issuance of the annual report was due to my unwillingness to contract debts for the payment of which funds were not in sight.

The treasurer's report will show that we have a surplus in the treasury to date of about \$50. The report of the treasurer is too long to be read at this time, so I will simply repeat that it shows on hand a cash surplus of \$50. I will turn the detailed report over to the auditing committee for their action.

TREASURER'S REPORT

STATEMENT OF ACCOUNTS OF THE NORTHERN NUT GROWERS' ASSOCIATION FROM OCT. 3, 1923, TO AUG. 31, 1924, BOTH INCLUSIVE

NOTE—Owing to delay in mails, the report given below is a later one than that used by the secretary. The one here included should have reached the secretary previous to convention, and it is the final, correct statement.

RECEIPTS

Membership—Plan No. 1 \$ 2.00

Membership—Plan No. 2 19.25

Membership—Plan No. 6 111.00

Membership—Plan No. 7 149.50

Membership—Plan No. 9 8.25

Membership—Plan No. 10 7.75

Total receipts from membership \$297.75

Transfer of Funds from Former Treasurer 104.13

Contributions 235.00

Sales of Literature 10.01

Interest .10

Total \$646.99

EXPENDITURES

Cash on hand \$.80

Middletown National Bank, Middletown, Conn. (Deposit) 170.64

Litchfield Savings Society, Litchfield, Conn. (Deposit) 4.23

Charged to Loss. 2 Subs, to Amn. Nut Journal on former

Treasurer's account 3.00

Expenses:

Postage, Express and Insurance \$ 9.79

Government Envelopes and Stamps 15.63

Adhesive Stamps 8.54

Postal Cards 1.25

Postal Cards and Printing 3.25

Registry Fee and Money Order Fee .18

Telegrams 1.18

Reporting Proceedings of Rochester Convention 50.00

Transcript of Proceedings of Rochester Convention 85.00

Reporting, etc., Proceedings of Washington Convention 60.00

Blank Account Book for the Association 5.00

Seal for the Association 7.00

1000 Letterheads 8.50

1500 Letters 8.50

500 Letters, double sheet 8.00

1500 Circulars 6.50

500 Reports, (92 pp., including cover) 184.00

500 Manila Envelopes 2.00

Printing 1.50

Addressing and Mailing 2.50

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Respectfully submitted,

H. J. HILLIARD, Treas.,

Northern Nut Growers Ass'n, Inc.

* * * * *

THE PRESIDENT: We will now be addressed by Dr. Britton, Director of the Botanical Gardens in which we are assembled.

DR. BRITTON: Mr. President and Members of the Northern Nut Growers' Association: By curious coincidence, in looking over the records of the New York Botanical Society's reports, I find the printed account of the organization meeting of your association. It is printed in the Journal of the New York Botanical Gardens, No. 132, for December, 1910. The article is written by George B. Nash. I believe I will read this report and if, perchance, the document is not in your files, I will turn this copy over to your president for preservation.

ORGANIZATION MEETING, NORTHERN NUT GROWERS ASSOCIATION

A meeting was held in the museum building on November 17, (1910) for the purpose of organizing an association devoted to the interests of nut-growing. The meeting was called to order shortly after 2 p. m. by Dr. N. L. Britton, who welcomed those present and wished them success in their undertaking. During his remarks he referred to a recent visit to Cuba where he succeeded in collecting nuts of the Cuban walnut, *Juglans insularis Griseb.* Specimens of these were exhibited and some of them presented to Dr. R. T. Morris for his collection of edible nuts of the world, deposited at Cornell University.

Dr. W. C. Deming was made chairman of the meeting and a temporary secretary was elected. The chairman read a number of letters from various parts of the country expressing an active interest in the formation of an organization such as was proposed. A committee of three was appointed by the chair to draft a constitution. This committee, consisting of Mr. John

Craig, Dr. R. T. Morris and Mr. T. P. Littlepage, submitted a report recommending that the name of the organization be the Northern Nut Growers' Association, that residents of all parts of the country be eligible to membership, and that the officers be a president, a vice-president and a secretary-treasurer. An executive committee of five was also provided for, two of said committee to be the president and secretary-treasurer. The annual dues were placed at \$2.00, and life membership at \$20.00. The recommendations of the committee were adopted.

An interesting exhibition of nuts, and specimens illustrating methods of grafting, formed a feature of the meeting. Chestnuts, walnuts, and hickory nuts, including the pecan, were illustrated in much variety. Mr. T. P. Littlepage had a series of nuts of the pecan which he had collected from a number of selected trees in Kentucky and vicinity. One of these, almost globular in form, was of particular excellence, being of clean cleavage and delicious flavor.

Dr. R. T. Morris was elected president; Mr. T. P. Littlepage, vice-president; and Dr. W. C. Deming, secretary-treasurer.

George V. Nash.

DR. BRITTON: May I say to you that our good wishes for your association, expressed at that time, are simply repeated now, and we hope that you will make yourselves at home and as comfortable as possible. We have made arrangement for the convention to leave here about one o'clock, for luncheon at Sormani's as guests of the Botanical Society. The autos will be at the door promptly, so I trust that you will adjust the session so as to be free to leave then.

THE PRESIDENT: We wish to extend our thanks to Dr. Britton for his kind remarks and for his hospitality.

We will now have the secretary read reports from our state vice-presidents.

THE SECRETARY: These are very interesting. The first one is from Mrs. Ellwanger, our state vice-president for New York.

(Reading in part) "My walnut trees are doing well and have many more nuts than ever before. The filberts planted two years ago, also have some, and the chestnuts, those the blight have left me, are covered with burs. There are beech nuts, too.—I intend to keep on planting chestnut trees, in spite of the blight."

Mr. C. S. Ridgway, Lumberton, New Jersey, writes as follows:

"There are very few nut trees in our vicinity. In fact, very few except what I have—some large old pecans at Mt. Holley, but the fruit is so small they are not gathered."

The next letter is from Mr. Howard Spence, of Ainsdale, Southport, England. Mr. Spence writes:

"During the last year I have got one of our horticultural research stations interested in the subject of walnut culture and just recently the headquarters of the Ministry of Agriculture and Fisheries also. The latter are using a small pamphlet on nut culture generally, to which I have contributed some facts. But a point of more definite interest at the moment is that the Minister has agreed to instruct all their inspectors over the country to make a collection of all walnuts of merit and forward them to me for classification and identification of varieties which may be worth perpetuating. As almost all the large number of trees in this country are seedlings I am hopeful that some interesting material may be located."

Here is a letter from Mr. Richard H. Turk, Vice-President for the state of Washington:

"Your request for a report from this Pacific Coast state came as a surprise. The Western Walnut Growers' Association is very strongly organized as regards Oregon and Washington, and it is difficult to persuade our nut growers here to join an association with its base of operations so far removed as the Northern Nut Growers' Association. I believe that I have been responsible for an additional membership of at least one or two which I think can be considerably augmented this fall.

Filbert growing has firmly caught hold of the enthusiasm of the people here. The acreage has reached 2,000 acres as compared to a bare 150 acres of six years ago. I estimate a planting of 1,500 additional acres to this quick bearing nut, this season. I have trees enough in my nursery to plant 600 acres but regard the majority of the plants as being too small. Planters plant even the smallest one-year layers out a distance varying from ten to twenty-five feet. I regard this as a waste of time, money and energy. Trees with two year old roots are none too big. The variety most planted is the Barcelona, closely followed by Du Chilly, and is supported by pollinizers for these two varieties at the rate of one pollinizer to every nine of the commercial sort. Intent eyes are watching every new seedling in search of new and superior varieties. Some have been found and will be propagated. Nut growers are but warming to the idea. I am putting out eight thousand four-year old seedling filbert trees in orchard form to be tested for qualities desired in a better filbert.

Tree filberts instead of bushes is a new idea that is fast gaining headway against the old method of removing the suckers by hand each season. *Corylus colurna*, the Turkish species, and *Corylus chinensis*, the Chinese tree hazel, are most favored as stocks. It has been found that these trees are

easily grafted to filberts, that they are extremely hardy and grow twice as fast as the filbert, and that the vigor of the stock enlarges the size of the nut, regardless of variety. Foremost in the recommendation of grafted tree filberts, I have correspondents in many foreign countries and have arranged for the delivery of several thousand pounds of these nuts to grow seedlings of.

The tree hazel is of the future as yet, and one must recognize the demand for layered stock until replaced by what appears to be better. To add at least thirty acres to my present filbert plantings this year is my desire. I am planting at least 400 trees to the acre as interplants in a grafted walnut orchard. No use in wasting time before the trees begin to bear profitable crops. Three and four years at most for man-sized returns when using a ten foot planting.

One planting of Du Chilly filberts last year produced an average of close to 40 pounds per tree on nine-year-old trees and an average of 10 pounds on four-year-old trees. The spread of the latter trees was scarce four feet, and I counted 22 nuts on a branch eight inches in length. Mr. A. W. Ward reports an average crop of 200 nuts to each two-year-old filbert tree in his four-acre planting this season. These are also Du Chillys that are fast building up a sentiment favoring them before the lower-priced Barcelona variety. The Barcelona is a more vigorous tree and shells out of the husk 75% whereas the Du Chilly is but 40% self husking, but that will not offset the differential of five to ten cents per pound in favor of the great, oblong nuts.

The *walnut* acreage of Washington and Oregon is approximately 12,000 acres and is now taking a new hold with all the additional planting being made up of *grafted* trees. The VROOMAN FRANQUETTE variety grafted on the California black walnut stock is the tree used in these plantings. Formerly, seedlings of the so-called second generation type were quite

popular, but when it became evident that seedlings would not transmit the superior qualities of the parent, that method of propagation was thrown into the discard. Eight thousand acres of the acreage now out, are seedling trees that must be topworked before Oregon will be truly famous for the quality of the nuts it produces. These seedling trees are paying at present under our present high prices after many years of barrenness.

My own 900 seedling trees I top-worked last year to the Vrooman Franquette variety, placing as many as thirty grafts in some trees and obtained an average of 70 per cent successful grafts. These grafts have made wonderful growth this season, and are quite capable of bearing large quantities of nuts next season. My crew of walnut grafters are becoming well known over a radius of 100 miles, and the work they are doing is a road to profit for many an owner of unproductive nut trees.

This fall I intend publishing some of the leading articles of the nut-growing authorities of this section, in conjunction with a catalogue well illustrated and containing my experience as a nut grower. Anyone contemplating planting walnuts or filberts may well send in their reservation of copy. Generally speaking, nut tree nurserymen and nut tree planters have not had time nor desire to add to the literature on this subject. I believe that when the nurserymen get behind the move to plant nut trees there will be some very interesting developments. There is one good thing in sight, and that is that it will not be the old-fashioned seedling that they will push this time. I think that you people of the East have got to make another determined effort to drive home the impossibility of seedlings ever being satisfactory. Outside the association a nut tree is a nut tree regardless of seedling and grafted trees, and one is expected to bear just as many fine large nuts as the other and just as soon. After losing twenty to thirty thousand dollars in delayed returns from a seedling walnut orchard, is it any wonder that I oppose the planting of more seedlings by the unwary?

In concluding this report I wish to state that I have talked nuts before a score of different meetings during the last year, and in the press of Oregon and Washington have done much to encourage the prospective grower."

THE SECRETARY: It seems to me that this report is one that will be very useful to nut growers in the East and very suggestive to beginners in nut growing. I would like to ask Mr. Reed if he has any comments to make on the report.

MR. REED: As I know conditions in the Pacific Northwest Mr. Turk has given an accurate report. The one criticism that I might make would be, perhaps, that there seems to be a probability of over-enthusiasm. This often occurs in any part of the country with respect to new things. It has been most conspicuous with the pecan in the South, and the almond industry in the West. As the pioneers in the nut industry in Oregon and Washington are acquiring greater experience they are increasingly more cautious with regard to such matters as varieties, planting sites, planting distances, interpollination, and others of kindred nature.

The industry in the Northwest is still comparatively small. It is centered mainly in the Willamette Valley of Oregon and to some extent in a narrow strip running north towards Seattle. The best informed are planting only in fertile, moist, properly drained soils so situated that air drainage is good. The local soils are much more variable than would be suggested by casual observation. Also, greater attention is being paid to air drainage in that part of the country than in the East. Several years ago there was a sudden drop in temperature from 32 degrees above to 24 degrees below zero, at McMinnville, Oregon. This proved fatal to trees and plants of many kinds, particularly those on flat bottoms or on hillsides from which, for any reason, the cold air was prevented from blowing to lower levels.

In addition to the species of nuts discussed by Mr. Turk, something might be said regarding the possibilities of chestnut culture in the Pacific Northwest. Numerous trees, planted singly or even in small groups found there, grow so well as to indicate plainly that the genus is capable of adapting itself to existing environment. However, both planters and consumers are generally prejudiced against the chestnut. This is easily explained for the reason that either sufficient numbers of varieties have not been planted together to ensure interpollination, or Japanese chestnuts have been planted. Early planters were evidently not aware that most varieties are largely self-sterile, and they did not know that the average Japanese chestnuts are fit for consumption only when cooked. Had these two facts been taken into consideration by them, it is not improbable that there would now have been an entirely different situation regarding the chestnut in that part of the country.

THE SECRETARY: I have a few more reports. Is it the sentiment of the meeting that I go on reading them?

MR. REED: I would like to hear the reports.

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THE SECRETARY: *Knight Percy, from Salem, Oregon*, writes:

"Both filbert and walnut planting have continued in Oregon during the past year. There has been a steady increase in the acreage of these two nut crops during the past five years but, fortunately, no planting boom.

The older walnut orchards are almost all seedling groves and many of these seedling groves are producing a very attractive revenue. Practically all of the new plantings are of grafted trees, it having been amply demonstrated that, while seedlings are often revenue producers, the grafted orchards bring

in more revenue and at no greater cost of operation. Seedling orchards are offered for sale, but very few grafted plantings are on the market. The Franquette continues to be the principal tree planted; probably 95% of the new plantings being of this variety.

A co-operative walnut marketing association has been formed, and this year for the first time carlot shipments of Oregon nuts will be sent East.

The filbert, a younger member of the Oregon horticultural family than the walnut, is being planted as heavily as the walnut, if not more heavily. Probably 60,000 trees were planted in the Willamette Valley of Oregon last year. Production of filberts has not yet become heavy enough to supply home markets. It will probably be some time before Oregon filberts reach eastern markets.

No other nuts are grown commercially in the state, although the chestnut does well here."

Mr. T. C. Tucker, State Vice-President from California, writes:

"The principal consideration in relation to the California nut situation is a recognition of the tremendous increase in planting within the last ten years. Many of these newly planted orchards have already come into bearing. The marketable almond tonnage of California has increased until it is now over three times that of ten years ago. The walnut tonnage has doubled during the same period.

New plantings are going forward very slowly at the present time due to the conditions prevailing in the fruit industry in general.

Economic conditions, coupled with the keenest kind of foreign competition have interfered materially with the sale of almonds in this

country, with the result that almond growers have been losing money every year for the past four years. At the same time the tremendously increased domestic tonnage has resulted in keeping the prices to the consumer very low in relation to pre-war prices and costs. The consumer has been getting the benefit of maintaining the domestic almond producers in the business. The fact that domestic tonnage cannot be kept down, as soon as a profit is in sight, warrants the American public in maintaining a sizable industry in this country by means of a protective tariff, even though it may appear on the surface as though it might mean increased prices. The experiences of the last four years have demonstrated beyond a doubt that increases in import duties have not resulted in increased prices to the consumer. They have, in fact, increased the competition to a point where prices have dropped rather than risen.

The same situation applies to walnuts, except possibly as regards losses to growers during recent years. The fact that walnuts ordinarily take longer to come into bearing than almonds has prevented any rapid increase in production such as has taken place with almonds. They are, however, facing many of the same conditions of keen competition from countries where costs of production are very, very low.

Conditions this year point to both almond and walnut crops of approximately the same size as last year. That means the walnut crop will be around 25,000 tons and the almond crop around 10,000 tons. The condition of the walnut crop seems to be about normal. Where irrigation is not available they are suffering from lack of water. Almonds this year are showing in many districts the disastrous effects of the unusually dry season. This will show up most strongly, however, in reduced tonnage for next year, and stick-tights for this year. These latter, however, are not saleable, so the consumer need not worry but that the almonds received in the markets will

be good, edible almonds. What the final outcome of the drought will be it is a little too early to tell.

Pecans and filberts are produced in such small quantities in California that they do not affect the market in any way except possibly locally. There is nothing to indicate any abnormal condition affecting either of these in the few places where they are grown. No large plantings of either of these nuts are being made, since there seems to be considerable question as to how successful they will be from a commercial standpoint.

Chestnuts are not being planted as fast as they might be, especially in those sections of the state to which they are well adapted. With the rapid disappearance of the chestnut forests of the eastern states, through the ravages of the chestnut bark disease, there is no reason why chestnuts could not be grown in California, especially in many of the foot-hill districts. This, of course, presupposes that the chestnut bark disease can be kept out of the state, and we believe it can be. The general price situation, however, is such as to discourage any extensive plantings at this time. The interest that is being taken in possible future plantings, however, is such that it appears reasonable to believe that the next few years will see materially larger plantings made, provided there is any improvement in agricultural economy conditions."

Mr. James Sharp, Vice-President from Kansas, writes:

"The only nut native here is black walnut, and the crop is heavy. There are some Stabler and Thomas planted here, and some grafted on native black are bearing. We have something like fifty grafted pecans planted of all varieties, but none bearing yet. The pecan is a native south and east of here in Kansas, and the crop is good, I understand. We also have a few grafted sweet chestnuts growing in Kansas which are bearing well, and more are being planted. I have one English walnut growing near my house, which

had male blooms last spring, but no nuts. We do not think they will be a success in Kansas but we hope to grow some nuts on our tree next year, the first in Kansas."

Mr. U. H. Walker, Nacla, Colorado, who says he is probably the only one in that state attempting to grow nut trees, instead of fruit, writes of his attempts. His place is at an altitude of 5,800 feet, where he can at times look down into the clouds, and on clear days can look up into perpetual snow. Mr. Walker has black walnut trees that have produced crops each year for the last ten years, three pecan trees and two persimmons. He has been experimenting with nut trees obtained from the government for the last ten or twelve years, and is willing to plant and care for any trees which the members of the association would like to have tried out in the center of the Rocky Mountain district.

Prof. V. R. Gardner, Michigan Agricultural College, in a letter to C. A. Reed, says: "We are getting a very nice collection of hardy nuts started on our Graham Station grounds near Grand Rapids. These are for the most part young trees being planted in orchard form. We are also doing some top-grafting and as soon as we shall be able to accumulate more data upon which to base recommendations, I am inclined to think that we will put on a number of nut grafting demonstrations in the state. I am sure there will be a demand for it.

If your meetings could be held later in the year, perhaps some time during the winter, I think it would be easier for some of the station men to attend them."

MR. REED: Might I add that Prof. Gardner was at one time Assistant in Horticulture at Corvallis, in the heart of the walnut district of Oregon. From there he went to Missouri as State Horticulturist. During the three years at that place he top-worked a considerable number of walnut trees with scions

of supposedly hardy varieties of Persian walnuts, especially the Franquette, and such varieties of Eastern black as he could obtain. The Persian practically was killed out during the first winter. The black walnut tops are now coming into bearing, and considerable attention is being attracted to them throughout the Mid-West. Prof. Colby may know something further regarding the work in Missouri.

THE SECRETARY: I hope you notice how many more reports we are getting from the men connected with the horticultural departments of the state institutions. Here is a letter from H. H. Bartlett, Director of the Botanical Gardens at Ann Arbor, University of Michigan:

"Our Botanical Garden in its present location is relatively new, having been established only in 1914. The development of permanent plantings has been mostly in the last two or three years, so you see we have as yet done nothing with nut trees other than to assemble what varieties we could get hold of. I must confess that the poor little things look much as if the wrath of heaven had overtaken them. We had 8 degrees of frost on the night of May 22d, when all the trees were in young leaf. All the nut trees were badly killed back, some below the graft, so I've had to pull some out. Since they had only a miserable start last year, they look pretty sad now. However, I'll replace where necessary, and hope for better luck next time.

If there should be an opportunity in the course of the discussion to state that we are prepared to receive and take care of nut trees that originators wish to try out in this region, I shall appreciate it. We are receiving occasional nut-bearing plants from the Office of Seed and Plant Introduction of the Department of Agriculture, and are very glad to act as a testing station for new introductions or productions.

In order not to give a false impression as to the extent of our work, I feel impelled to say that we haven't yet a nut tree in bearing, and only one over

three feet high."

Mr. Conrad Vollertsen writes that he will not be able to be here as he had planned. He states that all of his 31 varieties of filbert trees, except one, have fairly good nut crops. His place, as you know, is in Rochester, N. Y.

Mr. F. A. Bartlett, of Stamford, Conn., writes:

"You may be interested to know that some of my nut trees are giving some results this year. A number of varieties of filberts are fruiting, three varieties of black walnuts, almonds, Chinese chestnuts, heartnuts, besides the native hickory and butternuts."

MR. REED: According to Mr. Bartlett the Lancaster heartnut, which was introduced by Mr. Jones, is starting out in highly encouraging manner at his place near Stamford. It has grown well and is now a handsome, symmetrical tree. Indications are that it will bear well.

THE SECRETARY: Mr. Bartlett takes good care of his trees. We shall hope to pay a visit to his place.

I have a letter from Mr. Hicks, Westbury, Long Island. He will be with us today, and he proposes in his letter that we make an excursion to his place on Long Island.

Mr. J. W. Killen, Felton, Delaware, in a letter to Mr. Reed, writes as follows:

"This year we are maturing some nuts on the cordiformis and sieboldiana types of the Japanese walnut (young trees 3 to 5 feet high) that had no staminate blossoms. These we are producing by crossing with the pollen from one of our best Persians. We are looking for something interesting from there nuts when planted and the trees come into bearing. But all this

takes time and patience. We had more chestnuts last fall than ever before, and the prices averaged higher, about 20 cents per pound, wholesale. Our best chestnuts are looking good now. Will soon be opening; usually begin about the 5th to the 10th of September, to open up.

"We have not succeeded very well in propagating Mollissima (Chinese chestnut) but we find the quality of the nuts very good. All of our American sweet and all of the European type, including Paragon, Numbo, Dager, Ridgely, etc., have been gone for years, and left our Japs just about as healthy looking as they were 20 years ago, yet they were all set in the same block."

THE SECRETARY: It is encouraging to know that Mr. Killen has a strain of chestnuts that will grow there without being destroyed by blight.

MR. REED: Blight is not serious with his trees.

THE SECRETARY: It is with mine. But Mollissima has resistance.

MR. REED: The real pest in Mr. Killen's chestnut planting is the weevil. The nuts have to be marketed promptly in order to avoid destruction by this insect.

THE SECRETARY: I have a letter from Mr. Littlepage, who regrets that he will not be able to be with us.

Another letter is from Mr. Riehl, who regrets that because of his age he will not be able to take the long trip from Godfrey, Ill., to New York City. He writes to us of the place of the chestnut in northern nut culture, as follows:

"Blight and weevil are the greatest enemies of this nut. Blight in all probability will destroy practically all native chestnut where it is native, and

in all such districts the planting of chestnut orchards for profit will be useless until varieties are found or produced that are immune to that disease. In time this, no doubt, will be done. If I were fifty years younger and lived in a blight section, it would appeal to me to do something in that line.

Where the chestnut does not grow naturally it can be grown without fear of the disease. I have the largest chestnut orchard in the West, of all ages from seedlings to sixty years, with no blight.

Even were there no blight it would not be advisable to plant chestnut orchards where it is native because of the weevil. The weevil appears to be worse on the large improved varieties than on the smaller native. Of course any one planting a chestnut orchard now would plant the newer, larger varieties, as they will always outsell the smaller. No one who has not talked with handlers of chestnuts can have any idea of the handicap the weevil is to sales and prices. Where the chestnut is not native the nuts produced will be free of weevils.

The place to plant chestnut orchards is where the chestnut is not native, on soils that are not wet. Such situations exist in the central west and westward to the Pacific coast. I have had reports of chestnut trees growing and bearing in all this territory, and have had favorable reports of trees that I sent there of my improved varieties.

There is a good market at good prices for good, homegrown chestnuts. My own crops so far have sold readily at 25 to 40 cents per pound wholesale, and the demand is always for more after the crop is all sold.

Of all the nuts that I have experimented with I have found the chestnut to come into profitable bearing sooner and more profitably than any other."

DR. MORRIS: Some of the state vice-presidents have spoken of native chestnuts of good kinds. One obstacle, however, in the distribution of good chestnuts, has been the state laws which prevent us from sending chestnuts from one state to the other. I would like to ask Mr. Reed if it would be possible to make some arrangement at Washington whereby scions might be sent under government inspection to the West and to other parts of the country where blight does not exist. On my property at Stamford I had several thousand choice chestnut trees. The blight appeared and I cut out 5,000 trees that were from fifty years to more than a hundred years old. Among them there was one sweet American chestnut superior to the others. It had a very large, high-quality nut, and very beautiful appearance, having two distinct shades of chestnut color. The tree was the first to go down with the blight but I have kept it going ever since by grafting on other chestnut stock. I would like mighty well to have that chestnut grow in other parts of the country. It would be an addition to our nut supply.

Furthermore I have among a large number of hybrids, two of very high quality between the American sweet chestnut and the chinkapin. I gave these to Mr. Jones. He found, however, that he had no market for them because of the fear of blight. I would like to present scions of this to anybody outside the chestnut area where chestnuts are being grown, provided I can do this under government methods. We should find a way to do this.

THE SECRETARY: And not by boot-legging.

MR. REED: As Prof. Collins is more likely to be informed in regard to quarantine laws than I am he is the proper one to answer that question. I may say, however, that the federal department is unlikely to interfere in any way with the carrying out of state quarantine laws. Prof. Collins is now in the room. Dr. Morris, will you kindly re-state the question to him?

DR. MORRIS: In brief, I have some very superior chestnuts. They will be valuable for horticultural purposes in other parts, or in non-blight regions, of the country. I have kept them going by care and attention. I would be very glad to send those out of Connecticut, provided that the way may be found, by sending them through Washington to other states. It would be necessary, however, to have the scions treated in such a way as to make sure that the endothia spores had been destroyed.

THE PRESIDENT: I suggest that Prof. Collins give the matter some thought, and when he gives his paper he will be able to inform us about that. We will now ask Mr. Reed for a report as to promising seedlings.

MR. REED: There are quite a number of new things which might be mentioned. One is a group of Chinese walnuts now in their second or third year in the nursery of Mr. Jones, at Lancaster. In this lot there are many beautiful young trees grown from nuts obtained for Mr. Jones by Mr. P. W. Wang, of Shanghai. They are from North China, the territory which I visited more than two years ago and from which I also obtained considerable seed. Of the latter we have now several hundred seedlings ready for distribution. Personally I would like them to be distributed among members of this association. Mr. Jones has 300 or 400 of the Wang trees which he proposes to sell as seedlings. Others will be used as stocks for grafting varieties of *regia*.

Dr. Morris has already referred to the Chinese chestnuts. Mr. Dorsett, of the U. S. Department of Agriculture, has recently arrived in China for a two-years' trip. He will doubtless send many chestnuts.

Another particularly interesting group of nut trees is a lot of hazel-filbert hybrids produced by Mr. Jones. These are between the Rush and the Barcelona, or other European varieties. He now has plants three to five years of age in bearing. They average as high as a man's head. Practically

all are in bearing with attractive clusters of nuts, and some are fruiting heavily. The Rush variety, as most members know, is a native hazel of unusually prolific habits of bearing. The nuts are of fair size and quality.

Recently I have seen some interesting pecan trees in the East. Two of these are on the Eastern Shore of Maryland, one in the outskirts of Easton and the other at Princess Anne; the former is a trifle the larger, measuring 15 ft 5 inches in girth at breast height, the latter measuring 4 feet and 2 inches at the same distance and estimated to be 110 feet high. It was grown from a nut said to have been planted in 1800. The nuts from these trees are small but well filled and much appreciated by their respective communities.

THE PRESIDENT: We have the secretary down for a paper.

THE SECRETARY: This paper opens a symposium on topworking hickory trees.

TOP WORKING HICKORIES IN THE NORTH

By W. C. Deming, Connecticut

I do not recall a single modern improvement of importance in the art of grafting nut trees in the North that is not due to either Mr. Jones or Dr. Morris, except that to Mr. Riehl belongs, I believe, the credit of the idea of waxing the entire graft, which is now the accepted procedure. Therefore I speak before these two gentlemen with diffidence. I do so in the hope that perhaps I may recall something which they have forgotten to make known, or that what I say may elicit from them available emendatory remarks. My experience of fourteen years on my own place, and of five years grafting for others, is the basis of my observations.

Compatibility of Species and Varieties

This question will be particularly discussed by Mr. Bixby who has been conducting careful experiments that should soon settle the question for the commoner hickories. A few scattering observations of my own may be useful.

It is generally believed that any species of the genus hickory will catch on any other, though not necessarily that the union will be blessed. It is self evident that any hickory will thrive on any variety of the same species,

shagbark on shagbark, pecan on pecan, though even here close observation will probably disclose differences of compatibility. Probably any hybrid hickory will thrive on either of its parents. In some cases this may turn out to be a test of hybridity. For instance, the Barnes is one of the few shagbarks known to thrive on mockernut. It shows other evidences of mockernut blood.

I have found no hickory, so far, that does not appear to thrive on the shagbark, except the pecan. Even here there are differences. I have one Major pecan on shagbark that is over twenty-five feet high that has a very healthy appearance and that has shown staminate bloom for two or three years. I have also an Indian pecan that looks fairly prosperous. The Iowa pecans, the Marquart, Greenbay, Campbell, Witte, and others, catch readily and grow vigorously, at least for the first years. There are many data, however, on the adaptability of the pecan to the shagbark and the consensus of opinion is that ultimate results are poor. This is probably because the shagbark starts early and makes its season's growth in about six weeks, while the pecan naturally has a much longer growing season. However, these observations have been made, mostly, in the South and it may be different in the North. The question is not yet finally decided.

The Stanley shellbark, *H. laciniosa*, is completely at home on the shagbark, apparently, but has not yet borne with me.

The Hatch bitternut grew luxuriantly on shagbark for a year but blew off.

The Zorn hybrid made a growth of one foot on shagbark but then was winter killed, apparently.

I have a back pasture full of vigorous pignuts, *H. glabra*, which for eleven years I have been grafting with faith which now seems childlike, that soon I would have fourteen acres of bearing hickory trees. Yet as a result of

all these years of grafting the only hickories that I have found to thrive are the Brooks, which appears to be vigorous, the Terpenney, which is vigorous and bearing nuts in its fourth year, and possibly the Barnes. Not a single pecan survived more than a year, though many started. The Beaver hybrid makes a long spindling growth and then, in the first or second year, the leaves turn yellow and mosaic and the growth dies. The Kirtland, Kentucky, Hales, Taylor and several others, have all with me, proved failures on the pignut. Mr. Bixby's experiments appear to be showing somewhat different results.

The question of the compatibility of species and varieties is really a very important one because in some localities either the pignut or the mockernut is the prevailing species, and we wish to know with what species and varieties they may be successfully grafted. For instance, if the Barnes, which is an excellent shagbark, will do well on both the pignut and the mockernut, where so many other varieties fail, and the Brooks is at home on the pignut, these are highly important facts to be known by the man with fifteen acres of hilly woodland full of young pignuts and mockernuts.

Size of Stocks

I prefer stocks of moderate size, up to three inches in diameter. One gets greater results for the labor with these than with larger trees. Of course a tree of any size may be topworked but the labor is disproportionately greater, especially in the after care.

Cutting Back Stocks for Topworking

I doubt if it is important to cut back stocks during the dormant season, except that then there is more time. With larger trees this counts for a good

deal, but in the smaller ones I like to cut them off just where I want to graft at the time of doing so. However, they may be cut off when dormant at the point of selection for grafting and later grafted without further cutting back. This reduces, or does away with the risk of bleeding. Except in very small stocks it is better to leave a number of the lower branches to prevent bleeding. When bleeding does occur it may be checked by making one or more cuts with the knife or saw into the sapwood of the trunk below the graft. Better results come when the cutting back is of the top branches and not the lower ones because of the stronger flow of sap toward the top of the tree. In my opinion a side branch should always be left at the point where the stock is cut off to maintain a circulation of sap. Otherwise the stub will often die back and the graft fail. Also, the cambium close to a side branch will be observed to be thicker and I infer that the circulation of sap is more active. I prefer to cut off the top half, or two-thirds of the tree and graft into the top and the side branches near the top.

Hickories in full foliage may usually be cut back without evident harm. Occasionally a tree will be apparently shocked to death. Sometimes when a tree in foliage is cut back severely the remaining leaves will turn black and partly, or completely, die, but the tree will throw out vigorous new growth later.

Trees up to three inches in diameter may have the whole top cut off, at the risk of occasionally shocking a tree to death. Such complete cutting back must be done in the dormant season or there will be severe and prolonged bleeding. This method has the advantage of forcing a tremendous growth in the grafts which will need careful support. This is much more easily done however, than when the grafts are in the top of the tree. Cutting back in the dormant season and painting with paraffine has not worked well for me as the paraffine has not adhered well for any length of time to the freshly cut surfaces. Probably this could be easily remedied if it were a real

advantage. In the case of small stocks and branches where there is no bleeding and the paraffine adheres well green callus will often be seen spreading out beneath the paraffine over the cut surface.

Stocks should be vigorous. Dwarfed, stunted, submerged, hide bound trees make poor stocks. This is important, I believe.

Scions

The condition of the scion is the most important element for success in top-working hickory trees. The technique of grafting has been so simplified as to make it fairly easy, and native stocks are usually vigorous. But unless the scions have full vitality success will be limited. They should be plump and not pithy. A limited success is possible with scions of feeble growth, or those subjected to devitalising influences in keeping or handling, but the largest success will be had with well grown scions, cut from vigorous trees or grafts, whose buds are completely dormant, and have a fresh, green appearance on cutting. When the cambium layer shows a yellowish or brownish tint the scions are useless. Slender wood may make good scions but is more difficult to keep in good condition. Heavy wood from vigorous, young, grafted trees, or from cut back trees, makes the best scions and is the easiest to keep. Wood more than 1 year old and as large as one can handle makes good scions. Dr. Morris, with the use of the plane, has succeeded with astonishingly large scions and even branches. Sometimes buds are absent from these large scions or are very inconspicuous. They may be searched for with a lens.

Preferably scions should be cut when entirely dormant. Buds that show signs of breaking should be removed. Scions cut after growth starts may be used with success if there are dormant buds. This "immediate grafting," as Dr. Morris calls it has not been fully studied. It may be of great value. It is

quite successful with the apple and the pear. It appears to depend chiefly on the presence of dormant buds of vitality.

The later in the season the dormant scions are cut the shorter the time they have to be kept, though probably this is not of importance if the method of keeping is right.

Keeping Scions

The larger the scion the easier it is to keep it. Dr. Morris cuts whole branches and keeps them in the sawdust of his icehouse. I have cut them two inches in diameter and kept them lying uncovered on the barn cellar floor into the second summer looking fresh and green. The smaller the scion the more susceptible it is to moisture environment. Scions must be kept where it is neither too moist nor too dry. Usually the mistake is made of keeping them too moist. The buds may start if the scions are too moist even when the temperature is quite low. This happened for me when I stored scions for a week or two in the very cold bottom of an icebox. The most successful grafters keep scions with a sort of intelligent neglect. Dr. Morris buries them in the sawdust of his icehouse and it seems to make no difference if ice is there or not. I once tried keeping them in an icehouse over the ice and they became soaking wet. I have noticed that Dr. Morris's sawdust seems quite dry. Mr. Jones keeps some, at least, of his in bins or barrels covered with burlap bags. He says that heartnut scions keep best not packed away but kept in the open cellar. I notice that Mr. Jones has been using some kind of mill planings in place of sphagnum moss. Branches and large scions will keep well in a medium that seems dry to the touch. Small scions, such as those cut from old parent trees, require careful handling to prevent shriveling, on the one hand, or bud starting on the other. A low temperature is probably desirable, but the right condition of moisture is essential to the proper keeping of scions for any length of time. I should

naturally prefer to keep them in darkness, but I am not sure that it is important. Undoubtedly the access of some air is necessary but it would be difficult to keep it altogether away. I do not know how long scions would keep if entirely covered with paraffine. One year I dipped all the cut ends of my scions in melted paraffine but I am not sure that it is worth the trouble. One year I packed away my scions in rather moist sphagnum moss. The first time I looked at them they were enmeshed in mold mycelium. Later many of the buds started to grow. As suggested by Mr. Jones, dipping either the scions or the moss in half strength Bordeaux mixture will remedy the mold trouble. Parenthetically, this should be of help in keeping chestnuts, chinkapins, and other nuts that spoil easily with mold, for planting in the spring. Packing scions tightly and heavily covered in boxes for any length of time has been, in my observation, disastrous. In shipping scions a method advised, and one that I have followed with satisfaction, is to wrap the scions, either separately or together, in paraffine paper without any packing next the scions but putting it, instead, outside the paraffine paper. This packing may be sphagnum moss or mill planings slightly moistened. This also is wrapped in a moisture impervious covering and then in ordinary wrapping paper. For shipping long distanced the moss or planings should be dipped in half strength Bordeaux mixture.

The surface of the bark of scions that are being kept should always be dry, never moist. But they should never be so dry as to look shrivelled. Until you know just what scions will do under the conditions you provide you should examine them frequently.

Equipment

The essentials are a knife, raffia and the wax heater with brush. A saw is necessary if stocks are to be cut back, and pruning shears are convenient for cutting scions into proper lengths and for trimming and pruning stocks. The

knife most used is the grafting knife of Maher & Gross, with a three inch straight blade and a round handle that gives a good grasp.

I used to suspect that the men who said that scions ought to be cut with two strokes of the knife were trying to establish an unattainable ideal. But after Mr. Jones and Dr. Morris had taught me how to sharpen my knife I found that I could cut one that way myself sometimes. Mr. Jones's method of sharpening is to hone the knife flat on the surface next the scion and with a bevel on the upper edge. I found that this made scion cutting so much easier that I thought it was the whole secret. But one day I saw another doubter come up to Mr. Jones and ask him if it was true that he could cut a scion with two strokes of the knife. Mr. Jones said he thought he could but he had no knife just then. The man pulled out his pocket knife and asked if that would do. Mr. Jones looked at it, took a stick and with two strokes cut a perfect scion. Since then I have felt that there is something to it besides the way you sharpen your knife.

A very important element in shaping scions is to give a drawing motion to the knife by keeping the handle well advanced before the blade. The cutting is done with a draw and not a push. This is one of the most important factors for success in shaping scions.

It seems hardly necessary to say that the stroke of the knife should be away from the grafter. Yet it is a common sight to see beginners cutting to the thumb.

Dr. Morris showed me that if, in sharpening your knife, you hold the little whetstone between the thumb and middle finger of the left hand you are less likely to put a feather edge on it. A feather edge is something to clip the sprouting wings of any budding saint of a grafter. When you get the right edge on your knife often you can use it the whole day without resharpening, or at most with simply a stropping on a piece of wood or leather. But

improper use of the knife, or the least nick, will spoil the edge and sometimes it will be quite difficult to get it back. Therefore the blade should always be protected by a sheath, never laid down or used for cutting raffia, or anything but the actual cutting of the graft. For this purpose a leather sheath worn on the front of the belt, as first used by Dr. Morris, is almost a necessity. This sheath may be made by any leather worker and should have at least two pockets, one for the grafting knife and one for another knife to be used for trimming, cutting raffia and other odd things. It is convenient to have a little pocket for a pencil also and one may provide places for other articles of equipment at fancy.

I do not know that there is much to be said here about raffia. But a great deal has been said, and will be said, elsewhere, when the raffia is rotten and breaks in the middle of tying a graft. It is the devil's own stuff to carry when you don't carry it right. The right way to carry it is to tuck one end of the bundle under one side of your belt, pass the bundle behind your back and the other end under the other side of your belt. Then the raffia never gets mixed up with scions, tools and profanity and the end of a strand is as handy as the knives in your belt. On the whole I do not know of any binding material as satisfactory as raffia. It is stronger and easier to use when it is damp.

One of the great advances in the art of grafting is the use of melted wax. I believe that we have to credit Mr. Jones for this. The use of paraffine for grafting wax we owe to Dr. Morris. To him also we owe the Merribrook melter which has added so much to the comfort and convenience of grafting that it can be recommended as an outdoor sport for ladies. I do not like the brush that Dr. Morris recommends but prefer a stiffer one such as can be bought for ten cents.

Equipments vary with the individual and with the difference in the work to be done. Mr. Slaughter carries into the nursery, when he is working for Mr. Jones in the semi-tropical sun of Lancaster, a stool with parasol attachment. Mr. Biederman of Arizona has the most elaborate equipment which includes a table, planes, curved knives and gouges. Dr. Morris carries a knapsack. I like an ordinary light market basket that Mother Earth holds up for me when I'm not moving from place to place. When in a tree I stuff my pockets with scions.

A saw is usually a necessity. For portability I prefer a curved one that has a draw cut. It has also an aesthetic element and doesn't look like a meat saw, which can't be said of Mr. Jones's saw that seduced Dr. Morris from church. For heavy and steady work I much prefer a carpenter's sharp hand saw. A two-edged saw is an abomination devised by conscienceless manufacturers for the seduction of innocent amateurs.

For pruning shears I have a personal fancy for the French, hand-made instrument, each one individual, a work of art and a potential legacy to one's horticultural heir, if one doesn't let the village blacksmith monkey with it, as I did with mine.

On some grafts it is desirable to use a bit of paper, either beneath or outside of the raffia, to make waxing easier. For this I have found scraps of Japanese paper napkin very adaptive to surfaces and absorptive of wax.

On very heavy grafts Dr. Morris uses the Spanish windlass, as devised by him, for which he carries sisal cord, wooden or metal meat skewers, small staples and a mallet. He uses a chisel to cut slots in very thick bark and planes for shaping heavy grafts.

I have tried fastening in grafts with a nail, using iron and brass nails and bank pins. Mr. Jones has suggested cement covered nails. My experience

with iron nails is that they damage the scions. The use of nails has not been fully worked out. They are almost essential in bridge grafting apple trees. I think that just the right kind of a staple might be a help with some kinds of grafts.

Paper bags, 2 pound size, are sometimes wanted, for protection from sun or insects or to make the grafts conspicuous. Mr. Jones shades grafts made close to the ground with a slip of paper.

For labels for immediate use the wooden ones, painted on one side and with copper wire fastening, are satisfactory. Attach them by the nurseryman's method, which it has taken me many years to recognize as the right one, by twisting the *doubled* wire around a convenient object. Do not separate the wires which will probably permit the label to flap in the wind and soon wear out the wires. I used to think that the nurseryman's method was the result of hurry or laziness.

Copper labels, to be written on with a stylus, cost 1-1/2 or 2 cents each, according to size. The smaller I consider preferable. I imagined that these would solve the label problem. Picture my disappointment when I found that many of them cracked, or broke off entirely near the eyelet, from flapping in the wind. If they are to be used they must be fastened so as not to move with the wind. Mr. Bixby has an excellent label made on an aluminum strip printing machine. It has a hole in each end and is fastened with a heavy copper wire. He uses two of these labels on each tree. Dr. Morris sometimes uses a heavy wire stake to which he fastens the labels. A good method of attaching labels, and one that does away with the risk of girdling the graft or tree, is to fasten the label to a staple driven into the tree. The matter of labels is a troublesome one for they will get lost no matter what you do.

Other conveniences of equipment are a small whetstone, a small hammer, matches, and some volatile oil, like citronella, lavender, wintergreen, or other black fly and mosquito repellant. It is almost suicidal to slap a mosquito on the back of your neck with a keen grafting knife in your hand. A supply of parowax and alcohol for the lantern's sake should be remembered.

Technique

If the stocks are vigorous and active, and the scions full of vitality, I doubt if the technique is of chief importance, provided it is ordinarily good. However, a good technique will increase the percentage of success. One should have a variety of methods at command for varying conditions of stocks and scions.

One may come as near 100% success in grafting hickories as one is able and willing to observe all the known factors of success. I think that we can say now that the factors of success in hickory grafting are known. They are a vigorous and active stock, a scion of abundant vitality, coadaptation of the freshly cut cambium layers and prevention of desiccation.

The stock and scion have already been considered. How is coadaptation best obtained? One of the best methods, one that can be used in all seasons and in most conditions of stock and scion, is the side graft, the one that Mr. Jones uses in his nursery work. That is the best argument for this graft. It is, perhaps, the simplest, and at the same time one of the most difficult, of all grafts. The scion is cut wedge shaped and pushed into a slanting incision in the side of the stock. Mr. Jones's modified cleft graft is only a side graft made in the top of the stock after cutting it off. The difficulty lies chiefly in cutting the scion and the incision in the stock so that the fit will be perfectly true. This requires practice.

The bark slot graft, as Dr. Morris calls it, I have used for several years. It can be used only during the growing season when the bark will slip. It is very successful, whether put in at the top of a cut off stock, or inserted in the side of a limb or the trunk. It is not convenient to use unless the scion is considerably smaller than the stock. The scion is cut with a scarf, or bevel, on one side only. When the slot is to be made in the top of a cut off stock two vertical cuts are made through the bark, as far apart as the scion is wide, the tongue of bark thus formed is raised slightly at the top, and the point of the scion is inserted, cut surface toward the center of the tree, and pushed down firmly into place. The superfluous part of the tongue of bark is then cut off. By slightly undercutting the edges of the slot, and slightly tapering it toward the bottom, the scion may be wedged, or dovetailed, in place so as to be very firm. It is even possible to dispense with tying, sometimes, but better not to do so.

When the slot is to be made in the side of a limb or trunk the same procedure is followed except that it is necessary before making the slot to remove a notch of bark, at right angles to the axis of the trunk, so as to free the upper end of the tongue of bark.

The bark slot graft is the easiest of all and readily mastered once the grafter learns to shape a true scion. It is much better than the old bark graft where the bark of the stock is forced away from the wood leaving considerable space to be filled or covered.

These two forms of graft, the side graft, of which Mr. Jones's modified cleft graft is only a variation, as before stated, and the bark slot, in its two variations as described, will meet all needs in topworking hickory trees.

Finally, prevention of desiccation of the graft is obtained by waxing. I have found Dr. Morris's method with melted paraffine satisfactory. The addition of raw pine gum, as advocated by Dr. Morris is undoubtedly an

advantage under certain conditions, described by him, but I have not yet used it. The melted parowax is applied to the whole graft and wrapping, leaving no cut surface exposed and the whole scion being covered. If the paraffine is at just the right temperature it will spread at a touch, covering the surfaces without danger of scalding. It is much more effective thus applied than if colder and daubed on. The thicker the waxing the more likely to crack and separate. If the paraffine smokes it is too hot. If it does not smoke, and is dexterously applied, I think we can feel safely that it cannot be too hot. But if applied with a heavy hand it may be too hot even at a temperature so low that it will not spread.

Season for Grafting

According to Dr. Morris nut trees can be grafted successfully in any month of the year. But practically I think that grafting will be limited to that part of the year during which the cambium layer of the stock is active. At other times of the year preservation of the vitality of the scion will be too problematical, it seems to me, even if it is very carefully waxed. However, I may be mistaken. At any rate grafting is not very pleasant work out of doors in very cold weather. The success of bench grafting would be an argument for the success of dormant season grafting out of doors.

After Care

Without thoughtful after care the labor of topworking will almost certainly be lost. There are many ways in which the grafts can be lost but the two commonest are by being choked, or inhibited, by growth from the stock, and by being blown out by the wind. All new growth from the stock must be rigorously prevented. Grafts often make so heavy a growth that, if not blown out by the wind, they will be dragged out by their own weight.

Consequently they must often be supported. When the grafts are in, or near, the trunk of the stock, and not too high, the handiest method of support is to cut a sapling of proper length, sharpen the butt, stick this into the ground at the base of the stock, and tie it in two places to the stock. When the grafts are too far out or too high for this method laths or slats or sticks may be tied or nailed to the branches. Support is likely to be even more necessary in the second season when the growth is often astonishing.

Bud worms will sometimes destroy your graft just as it is starting, but they are easily found if looked for. With my conditions the most harm by insects is done by the night feeding beetles, which are particularly exasperating as morning after morning you watch the progress of their destructive work without ever seeing them. Bagging is the only preventive and it pays to use bags when a particular graft is cherished.

Is Topworking Hickories Worth While?

Up to the present time it is the surest and easiest way, practically the only way, of getting good results with the hickories, excepting the pecan. The root systems of the native stocks are well established and push the grafts rapidly. I have had a Siers hybrid grow 11 feet Straight up in a season. A Taylor matured several nuts on the third season's growth. A Terpeny had a crop the fourth year, the Griffin bears annually since its fifth year, the Kirtland and Barnes since the sixth. The Kentucky is a little slower. None of the hybrids have yet borne with me but with others they have borne quite early. We can be sure that the hickories will bear when top worked as soon as the average apple tree. The size of the crop that any topworked hickory tree will bear will depend on the size to which you have been able to grow the tree and the habit of bearing of the particular variety. I think, also, that there is good evidence to show that the size of the tree, the size of the nuts

and the size of the crop will depend largely on the amount of care and the amount of plant food that is given the tree.

Two years ago I topworked a number of hickory trees for Mr. Patterson of Wilkes-Barre, one of our members, and Mr. Patterson's foreman put in a few grafts under my observation. This summer I went to Wilkes-Barre to inspect my work. The foreman took me out into a field where he had done a lot of grafting the year before and I found that he had had a little better percentage of success than I had had. He had used the bark slot graft for everything, even when the scions were almost as big as the stocks. Before this I had thought that long experience was necessary for successful grafting. Now I see that if you have good scions, a Morris melter and a half hour of instructions, you will have all the essentials for immediate success. Hickory grafting is easy now. But let no one be contemptuous, for this ease has come only after many years of experiment and countless failures by many men. The former difficulty in grafting the hickory seems now like a mystery. The history of its evolution would make a very pretty story for the nut grower.

NOTES ON MEDIATE AND IMMEDIATE GRAFTING AT ALL TIMES OF THE YEAR

By Dr. R. T. Morris, Connecticut

Any newly described fact which releases information on the subject of tree grafting opens vistas of the new frontier in world agriculture.

Time was when men went from one country to another in search of fresh top soil. That was when they did not know better. It was when their cogs of habit turned their cogs of thought. They were engaged in raising annual plants at a considerable expenditure of time, labor and expense. They committed wastage of soluble plant foods (a variety of sin).

Malthus formulated a famous over-population fear-thought. It had basis in his ignorance of the fact that steam was soon to become a factor in the spreading of food supplies. Furthermore, he seemingly did not know that when old top-soil frontiers had gone to the rear, new frontiers would appear in the sub-soil. The tree digs deeper than the farmer ever plowed.

After Malthus came hunger prophets who were ignorant of coming possibilities of fleet transportation through the air. The caterpillar tractor plunging into the tropical jungle will allow of the production of a

practically unlimited food supply. Famine in India, China, and Russia is a social matter and unnecessary. Trees cure famine.

Within the past decade a number of thinkers on one end of the see-saw have written heavily on the over-population question not knowing that they and their birth control ideas were to be tossed into the air by still heavier weight of fact on the other end of the see-saw.

The heavier weight of fact relates to the idea that famine does not belong to tree food regions. It relates to the fact that tree foods can supply all of the essentials of provender for men, livestock and fowls; proteins, starches, fats and vitamins in delicious form. It relates to the fact that tree foods come largely out of the sub-soil without apparent diminution of fertility of the ground. The tree allows top-soil bacteria and surface annual plants to manufacture plant food materials and then deep roots take these materials to the leaves for elaboration by sun chemistry.

Trees may be grown wherever crops of annual plants may be grown and where annual plants may not be grown profitably. They do not require the service of high cost labor for annual tillage of the soil. For example, four large pecan trees or black walnut trees on an acre of ground without tillage or fertilizer may average a thousand pounds of nut meats annually for a century. How often is the market value and food value of a thousand pounds of nut meats per acre equalled by crops from annual plants which would require from 100 to 200 plowings and harrowings during a hundred years of continuous cultivation leaving out the question of expensive fertilizers and labor. Large populations live upon dates, olives and figs. For trouble they have to look to religion.

Several centuries were required for the British farmers to raise the wheat crop from six bushels to thirty bushels per acre. Things move faster nowadays. It will not require so long a time to carry tree crops from the

seedling phase to the phase of grafted kinds with greater productivity and quality. In the past the successful tree grafter was a specially skilled man. Now almost anybody may graft almost any sort of tree at almost any time of the year.

Aside from grafting, the hybridizing of nut trees, like that of cereal grain plants, has become a scientific sport appealing to the play instinct of man. When work becomes play in any field of human activity progress goes by leaps and bounds. The recent advance in tree grafting has amounted almost to a revolution rather than an evolution process. Application of a few new grafting principles of great consequence is now the order of the day. Old established grafting methods frequently ran into failures when dealing with all but a few trees like the common fruit bearing kinds.

The two chief obstacles to successful grafting were desiccation of the graft and fungous or bacterial parasites which entered the land of milk and honey where sap collected in graft wounds. Both of these dangers have now been practically eliminated and it remains for us to extend the season of grafting, carrying it away from a hurried procedure in busy spring weeks.

The chief obstacle to this extension of the grafting season has been the difficulty in finding the right sort of grafting wax or protective material for covering the graft, buds and all, as well as the wound of the stock. For covering the entire graft in order to avoid desiccation grafting waxes had to be applied in melted form with a brush. They had to be applied in melted form for filling interstices of wounds in which sap might collect and ferment. These waxes had the effect of not retaining their quality under greatly varying conditions of heat, cold and moisture. The paraffin waxes which the author has preferred were inclined to crack and to become separated from the graft and stock in cold weather. Furthermore they would remelt and become useless in the very hot sun of southern latitudes.

Experimentation for several seasons has resulted in the finding that raw pine gum is miscible with the paraffins in almost all proportions because of physical or chemical affinity. This gives to the wax an elasticity and adhesiveness of such degree that we may now graft trees in cold weather. Cohesiveness of molecules of the mixture is such that remelting in the hot sun may not destroy the effectiveness of this protective coating in hot weather.

Since the author has depended upon this mixture he has grafted peaches, apples, hazels and hickories successfully in midwinter as well as in midsummer. Many other kinds of trees have been grafted successfully out of the so-called grafting season but these four kinds which represent two of the "easiest grafters" and two of the "hardest grafters" will suffice for purposes of illustration.

According to old-established idea trees may be grafted successfully only from scions that have been cut when dormant and stored in proper receptacles. This is what we may term "mediate grafting," a considerable length of time intervening between cutting the scions and inserting the grafts. On the other hand what we may call "immediate grafting" is the taking of a scion from one tree and grafting it at once in a tree that is to receive it. Mediate or immediate grafting may both be done at almost any time of the year, winter or summer, spring or autumn.

When preparing the scion for immediate grafting in the spring or early summer it is best to cut off all the leaves and herbaceous growth of the year. We then depend upon latent buds situated in the older wood of the scion. The latter may be one year or several years of age.

In midsummer when top buds have formed we may remove only the leaves, allowing the growth of the year to remain and to serve for grafting material.

In experiments with the apple for example it was found that mediate grafts inserted on July 10th in the latitude of Stamford, Conn., began to burst their buds five or six days later. Immediate grafts inserted at the same time began to burst their buds about fifteen days later from buds of the year and about twenty days later from latent buds in older scion wood.

New shoots from these mediate apple grafts continued to grow as they do in Spring grafting. Immediate apple grafts on the other hand put out about six leaves from each bud and then came to a state of rest with the formation of a new top bud. After about ten days of resting these new top buds again burst forth and grew shoots like those of the mediate grafts.

The philosophy of these phenomena would seem to include the idea that the mediate summer grafts had contained a full supply of pabulum stored up in the cambium layer. The immediate summer grafts, on the other hand, had contained only a partial supply of pabulum, enough to allow them to make six leaves and a top bud. After a few days of resting these shoots with meager larder could then go forward with new food furnished by the whole tree.

Mediate and immediate winter grafts were alike in their method of growth in the spring. This would seem to confirm the idea that character of new growth is dependent upon the relative quality of stored pabulum in the cambium layer.

In experimental work it was noted that both mediate and immediate winter grafts make a slower start in the spring than do the grafts inserted in springtime. This is perhaps due to the formation of a protective corky cell layer over wound surfaces. New granulation tissue would then find some degree of mechanical obstacle in the presence of a corky cell layer at first.

Herbaceous plants allow of grafting. We are familiar with the example of the tomato plant grafted upon the potato plant, furnishing a crop of tomatoes above and potatoes below.

It seemed to the author that the herbaceous growth of trees should be grafted quite as readily. This seems to be not the case. A number of experiments conducted with grafting of the herbaceous growth of trees in advance of lignification has resulted wholly in failure with both soft wood and hard wood trees.

The walnuts carried herbaceous bud grafts and scion grafts for a long time however. These grafts sometimes remained quite green and promising for a period of a month but lignification progressed in the stock without extending to the scion. Speculation would introduce the idea that lignification relates to a hormone influence proceeding from the leaves of a tree and that the leafless scion does not send forth hormones for stimulating the cells of the scion to the point of furnishing enzymes for wood building.

Perhaps the most interesting part of new tree work relates to experiments which are failures. Negative testimony is like the minor key in music. There are many men who care to do only things that "cannot be done." These are the ones who have made our progress in almost every field of human activity.

STOCKS FOR HICKORIES

Willard G. Bixby, Long Island

MR. BIXBY: The sheets which I am distributing to you contain tables to which I shall refer during this talk. But first I will give a little foreword regarding the trees. The trees enumerated in the tables shown were nearly all given me by Mr. Henry Hicks of Isaac Hicks & Son, Westbury, Long Island, and were taken to Baldwin and set out in the fall, practically the entire roots being saved and later the trees severely cut back. They were transplanted without loss except in the case of the shagbark, and those lost were all undersized trees. All of the hickories were of one age, but those lost were ones which had not made normal growth and had they been discarded in the beginning there would have been no loss whatever in the transplanting of 300 or 400 trees. Later, in the spring of 1924, I found some loose bark pignut (*Carya ovalis*) seedlings on a farm not far away from my place, and these were also transplanted; but they were too small to graft this year. These experiments in grafting, made during 1923 and 1924, have shown us some new things. With some of the walnuts we had 100 per cent success. With the hickories there was not 100 per cent success, but that was due to the fact that we were putting scions on stocks that were not congenial in many instances. You will notice the results as shown on the tables.

1923 GRAFTING

G—Grafts Set C—Successful Catches

	Shagbark	Mockernut	Pignut	Pecan	Bitternut	Total	
	G	C	G	C	G	C	%
Barnes	6	6	3	3	3	3	18 18 100.0%
Brooks	5	0	4	2	5	1	5 2 19 5 21.0%
Clark	5	1	5	0	5	2	5 1 5 2 25 6 24.0%
Fairbanks	27	17	27	17			59.3%
Gobble	1	0	1	1	1	1	1 1 1 1 5 4 80.0%
Griffin	1	1	1	1	1	0	1 0 1 1 5 3 60.0%
Hales	5	3	4	1	5	4	5 5 19 13 52.5%
Kentucky	5	4	3	1	5	4	5 4 5 1 23 14 61.0%
Kirtland	3	1	3	2	3	2	3 2 12 7 58.4%
Laney	6	4	6	4			66.7%
Long Beach	4	3	3	2	4	1	4 2 3 1 18 9 50.0%
Manahan	5	1	5	1	6	2	5 1 5 1 26 6 24.2%
Siers	5	5	5	5			100.0%
Stanley	3	3	3	2	3	3	9 8 89.0%
Taylor	4	3	3	3	4	3	4 3 15 12 80.0%
Vest	5	1	5	0	5	1	5 2 5 1 25 5 20.0%
Weiker	5	1	5	2	5	1	15 4 26.8%
	32	17	51	20	52	26	46 24 91 53 272 140
	53.1%	29.2%	50.0%	47.0%	59.3%	51.5%	

An inspection of the 1923 grafts made August 21, 1924 showed the following number growing: on shagbark 14, on mockernut 6, on pignut 26, on pecan 24, and on bitternut 16, the only place where there was any material difference being in the case of the mockernut where nearly three-quarters of the number of grafts growing last summer failed to grow this

spring, in fact all varieties failed to grow excepting three, the Barnes, Gobble and Long Beach, all three of which I suspect from other evidence, have mockernut parentage. In the case of those on pignut and pecan stocks there was no loss from 1923 and in some instances at least of those on shagbark and bitternut stocks the loss was due to outside causes, such as being broken off.

1924 GRAFTING

G—Grafts Set C—Successful Catches

	Shagbark	Mockernut	Pignut	Pecan	Bitternut	Total	
	G	C	G	C	G	C	%
Barnes	8	7	10	4	18	11	61.0%
Beaver	5	1	5	1			20.0%
Brooks	11	8	10	5	21	13	61.9%
Clark	6	0	8	0	5	0	5 1 24 1 4.6%
Fairbanks	5	3	5	3			60.0%
Greenbay	5	0	5	0			0.0%
Hales	5	1	5	1			20.0%
Kentucky	5	2	4	2	9	4	44.5%
Kirtland	5	5	4	3	9	8	88.8%
Laney	5	3	5	2	10	5	50.0%
Manahan	6	2	6	2			33.3%
Mosnat No. 5.	7	1	7	1			14.7%
Mosnat No. 6.	10	6	10	6			60.0%
Siers	5	4	5	4			80.0%
Stanley	12	1	12	1			8.3%
Vest	10	3	15	5	16	5	10 3 12 3 63 19 34.2%
Weiker	5	3	5	3			60.0%

16 3 122 52 54 21 15 4 12 3 219 83

18.7% 42.6% 38.9% 26.7% 25.0% 37.9%

In 1923, it was very evident that the Barnes was the only variety showing 100 per cent success on every stock. That was not repeated in 1924, but it still showed a high percentage of success.

From the comparatively modest percentage of catches, 51.5% on the average in 1923 and 37.9% in 1924, one might hastily conclude that the grafting was not skillfully done or that the grafts did not have proper attention afterward, but as noted above the grafting was done by Dr. Deming, whom I regard as one of the most skillful men that we have, and as the work on walnuts done at the same time showed 100% success with a number of varieties, I think any question as to the skill with which the work was done and the care the grafted trees had afterwards can be dismissed.

It is to be regretted that the number of scions at hand was not sufficient to repeat exactly the experiments of 1923 as well as to follow out the points suggested by the 1923 work, but as there was not enough for both, the latter was done.

The 100% success of catches of the Barnes in 1923 was not repeated in 1924; but the high per cent of catches on the mockernut, (7 out of 8 in 1924), is gratifying in view of the few varieties that we have that have shown adaptability to that stock. As the Barnes is one of our good varieties and there is such a wide section of the country where the mockernut is the prevailing hickory, it is believed this behavior of the Barnes will prove a valuable addition to our knowledge in top-working the hickory.

No variety as strikingly adapted for use on the pignut has appeared, but there are a number that have shown fair adaptability.

The varieties most desirable for top-working various species of hickories as suggested by this work supplemented by other observations of the writer, would be as follows:

Shagbark—Most varieties.

Mockernut—Barnes.

Pignut—Brooks, Kentucky, Taylor, Kirtland.

Bitternut—Beaver, Fairbanks, Laney, Siers.

It is useful to know that the Barnes is the only one especially successful on the Mockernut. By the spring of 1924, all grafts on mockernut had died except the Barnes, the Gobble and the Long Beach, and each of these is thought to have mockernut parentage.

In the cases of the pignut and the pecan stocks, all of the grafts successful in 1923 were still living in 1924. With the shagbark and bitternut most lived. As to pecans there is not much to be said; pecan varieties would usually be used for the topworking here.

The results of a few grafts set in 1924 on *Carya ovalis* and on shellbark seedlings which were 100% failures, are not noted, as the shellbarks were, in the judgment of the writer, too small for the purpose, and the *Carya ovalis* had been set out in the spring of 1924 but a few weeks before the grafting was done. In other words the latter had not become sufficiently established to make good stocks, and the former were not large enough. In each case there was not sufficient vitality available to expect success.

This brings out one point which has impressed me strongly; that is the need of having vigorous stocks if they are to be grafted or transplanted successfully. I feel that this point cannot be too strongly emphasized. If a stock either from youthfulness or inherent lack of vigor is not rapid growing

it is almost useless to try to graft it or transplant it until it does show the needed vigor.

As to stocks to grow in the nursery with the idea of grafting them later, the two commonly used, the bitternut for the bitternut hybrids and the pecan for others, there is little further to be recommended at this time, although for some varieties, notably the Vest, a stock better adapted to it than any we now have is earnestly to be desired.

* * * * *

THE PRESIDENT: Are there any questions on these three papers on hickory grafting?

MR. REED: There are two points in regard to propagation which I believe should be mentioned; one is that these various methods that have been discussed make it possible to propagate successfully during a great portion of the year. By beginning early in spring with the dormant graft, and continuing throughout the summer, these methods can be made to follow one another so that if one fails still another can be used. These methods greatly prolong the season, and when it is not convenient to propagate at one period by the method proper to use at that time another can be employed at a different season.

The other point is that we are constantly learning more in regard to the influence of stock upon scions. For example, hickories on pecans seem satisfactory while the reverse is at least doubtful. Mr. Jones finds that *sieboldiana* is not a good stock for *regia*. We all find *nigra* apparently satisfactory as a stock for any species of *Juglans*. These conspicuous differences of influence of various species upon scions suggest the possibility of less, but perhaps quite as important, difference of varieties. It

is one of the newer phases of study and experimentation which should be considered by all and reported upon to this association.

THE SECRETARY: At my place the Vest, used in top-working large shagbark hickories, has been very successful. I do not know any that have been more successful or that grow more rapidly than it does on the shagbark hickory.

DR. MORRIS: The Marquardt is successful at my place.

MR. O'CONNOR: I do not know why we have not had success with paraffine in a single instance. In grafting fruit trees I had excellent results. I thought that if this could be done on fruit trees why not on nut trees? But I am going to try with the hickory again. I am going to be more careful in selecting good, strong stock for that purpose, and I think in that way we should have better success.

DR. MORRIS: Did you not perhaps cover the buds of your hickory grafts too thickly with melted grafting wax? Might not that account for your failure? Hickory buds will burst their way through almost any thickness of grafting wax, but when the paraffines are used without pine gum admixture the paraffine over the buds is particularly apt to crack and to allow the graft to dry out.

MR. O'CONNOR: I did not cover the hickory grafts with melted grafting wax at all; I simply put them in like apple grafts with ordinary grafting wax.

DR. MORRIS: Practically all hickory grafts will fail under such circumstances, but practically all hickory grafts will catch if they are covered with melted grafting wax of the right sort, provided that the scions and stock are also of the right sort.

THE SECRETARY: May we now have the President's address?

THE PRESIDENT: Before I begin I wish to call to your attention this pamphlet regarding the fifth Mid-West Horticultural Exposition, to be held in the Hippodrome, Waterloo, Iowa, November 11 to 16, 1924. It will be under the auspices of the Iowa State Horticultural Society, co-operating with its affiliated societies and the Greater Waterloo Association. The exposition will cover the Mid-West territory, from Pittsburgh to Denver. I wish especially to mention the printed list of premiums on page 27. Mr. S. W. Snyder, Center Point, is superintendent of this department. Cash premiums in Department b-Nuts, amount to \$289. In addition there will be a grand sweepstakes, a trophy cup, donated by a member of the Northern Nut Growers' Association, for the exhibitor winning the greatest number of points. Anyone interested could write to the secretary, Mr. R. S. Herrick, State House, Des Moines, for a printed premium list. If any members of our Association have pet nuts of a variety which they would like pushed to the front now is the chance. Snyder Brothers are offering special premiums for new nuts unnamed and unpropagated.

The object of this association, as defined in its constitution, is "the promotion of interest in nut-bearing plants, their products and their culture," and as its name implies, in the northern part of this country. Without going into detail it seems to me that we have achieved the object of our association, at least to the extent of making practical use of our accumulated knowledge. Public interest has been aroused, which may become stale. Articles have appeared in magazines and newspapers from time to time on subjects relating to nut culture. We are also on a continual lookout for new varieties, and those of our members skilled in the art are constantly improving and working out new methods of grafting and budding, particularly as evidenced by Dr. Morris' work entitled "Nut Growing." We know approximately how soon a grafted nut tree, especially the black

walnut, will begin to bear. At Mr. Jones' Nursery, Lancaster, Pa., an Ohio black walnut tree in the nursery row bore a cluster of seven nuts 17 months after the graft was placed. Mr. J. W. Wilkinson, of Rockport, Ind., has demonstrated that grafted northern pecan trees bear early and abundantly for their size.

We have given advice conservatively in reply to all inquiries relative to nut-bearing plants, perhaps too much so. Much honor and credit is due to certain members of our association for their untiring work and efforts in its behalf. It is not necessary to mention names as I am sure most of you present know to whom I refer. Our annual reports testify to their splendid work.

From this time forward I believe we should adopt the policy of boldly advocating the planting of orchards of nut trees. The intending planter will decide for himself what variety he will plant, and as a guide he should judge from the wild varieties growing in his vicinity. By so doing he cannot go very far astray in what will be to him a new venture. Of course certain varieties will be restricted to certain limited areas. This applies particularly to the introduced varieties, as distinguished from the native nut-bearing trees.

The black walnut has a wider range than any of the other nut trees. Travel wheresoever you will about the country and you will observe wild black walnut trees growing almost on every farm. The planting of the Persian, or English walnut, as it is more generally known, has had more of a popular appeal, perhaps from the fact that we are accustomed to seeing clean, smooth nuts of uniform size of that variety in almost every grocery store, the kernels of which may be extracted without great effort. The black walnut, on the other hand, has been tolerated as a sort of poor relation, and has been given no particular attention, because we have been used to seeing

it around. It has not been made to do its share of contributing towards its keep. Our earliest recollections of it bring to mind bruised fingers as a result of our endeavors to crack the nuts and the tedious work of manipulating a darning needle to extract the kernels, which we usually picked to pieces in the process. We now know that we simply did not have the right kind of black walnuts. We should put our accumulated knowledge to practical use to urge on every occasion the planting of nut orchards, especially of approved varieties of the black walnut. This I understand is what the United States Department of Agriculture is advocating, and we should co-operate all we can with the department in that recommendation.

It will, no doubt, be urged that sufficient grafted black walnut trees are not available for orchard planting on a large scale. This, no doubt, is true, but on many farms there are wild black walnut trees of a size suitable for grafting or top-working. Grafting wood may be obtained in larger quantities than the grafted trees. Those of our members skilled in the art have not been selfish in imparting their knowledge to others and are always ready and willing to instruct others in the art. Most owners of these trees would only be too glad to substitute profitable tops for their trees in lieu of their unprofitable ones.

I believe that at all our meetings we should have practical demonstrations in budding and grafting, as this will tend to arouse the interest of the uninitiated and will spur the initiated to greater perfection.

During the past year there has been a discussion relative to the calling of the black walnut by some other name. Personally I believe we should not attempt the change. The public will not understand and it will take them a long time to become educated to the change. Valuable time will be consumed in picking out a new name. Let us take the name as we find it. Properly handled, after the husks are removed, the walnuts will not be as

black as they are painted, and besides, we do not eat the shell anyhow. The quality of the kernel will make its appeal. The trouble with all of us has been that too much attention has been given to the looks, rather than the quality, of our food stuffs. Quality has been sacrificed for looks. Various illustrations of this come to mind with all of us.

I believe success will attend the planting of black walnut orchards. This will encourage others to follow with orchards of other nut-bearing trees. Orchards of all kinds of fruit trees are being planted each year and the planters are content to wait until the trees are large enough in order to reap the benefits thereof. But somehow the impression prevails in the minds of many people that a nut tree should show results and yield profits soon after it is planted. In recommending to a lady of means that she should plant, as shade trees, northern pecans she promptly wanted to know how many bushels of nuts she would get off of the trees the next year.

Perhaps we place too much importance on selecting just the right spot and soil in which to plant a nut tree and thus cause the intending planter to be too timid in making a start. Those who know anything about trees know pretty well where it is not advisable to plant trees, especially those with a long tap-root. They can judge fairly well from the wild trees of the same variety growing round about.

As evidence of what a nut tree will do, those of you who have visited Devil's Den in Gettysburg Battle Field, have perhaps noticed a butternut tree, now quite old, growing out of the top of the cleft in a huge rock, having sent its roots down to the adjoining soil for nourishment. This tree has borne nuts even in its adverse situation.

For the benefit of those interested in the northern pecan, I wish to record the fact that a seedling pecan tree is growing in Clermont County, Ohio, on upland, not far from the eastern boundary line of Hamilton County, about

five miles north of the Ohio River. The nut from which the tree grew was brought from Rockport, Indiana, and planted about forty-one years ago. The tree is quite large and bears nuts comparable with the wild seedling nuts that may be obtained from the Rockport district. If a seedling does this, you may readily see what a grafted tree will do.

THE PRESIDENT: We will now ask Prof. Collins for his address.

THE SEARCH FOR BLIGHT-RESISTING CHESTNUT SPROUTS[A]

Prof. J. Franklin Collins, Rhode Island

The chestnut blight has now been with us for more than twenty years and has destroyed practically all the chestnut trees of the northeastern part of the country. It has spread in all directions from its original center in the immediate vicinity of New York City until it has reached the limits of the native chestnut growth in the northeast and north, and is steadily approaching its limits in the west and south. The disease, a native of China and apparently imported into this country on some Japanese or other oriental chestnut, found a more susceptible host in our native chestnut and so became a virulent parasite on this new host. It was not until 1904 that general attention was attracted to the disease. By that time it had obtained a strong foothold on the chestnuts of southeastern New York (particularly the western end of Long Island), in southwestern Connecticut, and in northern New Jersey.

All of you are more or less familiar with the efforts made in Pennsylvania, New York, and elsewhere in the northeast, in co-operation with the federal government, to control the disease. These efforts are now an old story to most of you and there is no need of repeating it at this time.

Early in the fight against the blight the attention of many of us was directed to locating possible immune or resistant species, varieties, or individuals. The search for resistant native individuals and the accompanying experiments in crossing and grafting various species and varieties has been kept up ever since. Foreign explorers have constantly been on the lookout, with more or less success, for chestnuts in other countries that might be resistant to the blight. It has long been known that most forms of the Japanese chestnut (*C. crenata*) were in general highly resistant to the blight. Later it was found that the more recently introduced Chinese chestnut (*C. mollissima*) was also quite resistant, although both the Japanese and the Chinese were far from being immune. Quite recently Mr. Rock, explorer for the U. S. Department of Agriculture, has brought a new chestnut from southern China for experimental purposes. Notwithstanding newspaper reports to the contrary the possibilities of this chestnut in this country apparently are unknown at the present time. Nobody seems to know if it will stand our climate, resist the blight, produce worthwhile timber or fruit; nor is its name known, according to late advices that have reached me.

Some years ago the late Dr. Van Fleet made numerous crosses between the Japanese and the American chestnuts, the Chinquapin, and other species and varieties. Personally, I have not been in very close touch with Dr. Van Fleet's experiments. Doubtless some of you know more about them than I do. Regarding these I will only say at this time that the work begun by Dr. Van Fleet is being continued by the Federal Bureau of Plant Industry, with Mr. G. F. Gravatt in direct charge of the work so far as the Office of Investigations in Forest Pathology is concerned. Mr. Gravatt is also testing out the value of scions taken from seemingly resistant native trees when grafted on resistant stocks.

Some years after the blight had destroyed most of the chestnut trees in the northeastern states we kept getting reports from various localities to the

effect that the blight was apparently dying out. Many of these reports came from sources that made us doubt their value, but others came from more reliable sources. We have had opportunity to investigate a number of these reports and have usually found that the statement that the blight was dying out was, in a sense, strictly true, the reason being that the chestnut trees were entirely dead, except for sprouts. This fact naturally prevented the disease from showing us as much as in former years.

Some twelve years ago I noticed in Pennsylvania a sprout of an American chestnut about an inch in diameter which had a typical hypertrophy of the disease, apparently completely girdling the sprout at its base; also a girdling lesion farther up on the stem. The hypertrophy was such a pronounced one and in other respects such a typical example of the disease that I photographed it. A few years later I was surprised to observe that this sprout had increased to more than three times its former diameter and that the two diseased areas just mentioned apparently had disappeared—at least they were no longer in evidence except as rough-barked areas. To make a long story short this sprout is still alive and has increased in size and height each year. Although now (1924) it is considerably branched and makes a small bushy tree it is badly diseased in numerous places and is only partially alive, but the dead portions have not resulted from some half dozen of the original disease lesions (apparently girdles), but from later infections. The very fact that a sprout should have lived for more than twelve years in the center of one of the most badly diseased areas known to the writer seems at least to suggest the possibility that future chestnut sprouts may yet grow in spite of the disease and persist—at least in a shrubby form if not as a tree.

The sprout to which I have just called attention is not an isolated case, but merely one of the most pronounced that I know about. In a careful survey in July (1924) of the region immediately surrounding the sprout just mentioned two or three other notable, but less pronounced, cases of a

similar sort were discovered. In two cases fine looking branched sprouts some twenty feet high with healthy-looking foliage were noted. Both were diseased but the disease seemed not to be very conspicuous or virulent. In a recent survey of woodland in Rhode Island (July, 1924) much healthy foliage was observed and several large sprouts were found on which the disease (although present) seemed to be doing little damage when compared with its former virulence in the same general region.

I call attention to these cases primarily to acquaint you with the results of our latest observations on what seems to me to be cases of gradually developing resistance in some of the remaining sprouts. In all my intensive work on the blight between 1907 and 1913 I cannot now recall a single instance where a chestnut sprout in a disease-ridden area ever reached a diameter of an inch or thereabouts before its existence was cut short by the blight; and yet today—a dozen years later—we are finding quite a number of living sprouts over two inches in diameter, and a few that are three, four, and even up to seven inches in diameter. Last Friday, August 29, I heard of a small chestnut tree in New Jersey that bore a few burs last year and which has a dozen or more this year. If the nuts mature we hope to get some of them to propagate. Last Sunday, August 31, I saw a three inch sprout in Connecticut that had had a few burs on it. I would be glad to learn of any cases of this sort that may come to your attention.

You are all thinking men and women and all of you have had experiences with diseased trees of some sort, many of you with very serious diseases, and some of you I know have had a wide experience with the chestnut blight, so you can draw your own conclusions as to the significance of the facts that I have stated.

As to the state laws for transporting material from one state to another I am not posted, but I believe that we can be advised by writing to the

government at Washington.

DR. MORRIS: We do not know whether the Washington government will sterilize those scions and send them out for us, but there should be some way of sending from one state to another.[B]

It seems to me that in all probability, the vital energy of the protoplasm of the endothia is diminishing. Quality, flavor, or anything you please, is bound up with certain vitality, and that diminishes and finally will cease. That is the reason for the endothia growing less now.

PROF. COLLINS: My point was perhaps not exactly that. I meant that the result is that, with the average cases, we are now getting chestnuts not so quickly destroyed. The explanation may be exactly what you have stated.

DR. MORRIS: There are two factors to be considered. First, the running down of the vital energy of the protoplasm; and second, in the factors which affect the vital energy of the plant.

PROF. COLLINS: In the paper I have just read there was mentioned the apparent number of trees in various parts of the country which are very slowly dying from the blight, and some which have resisted it entirely, so far; but that was not the point I desired to emphasize. There are some around New York City which are still growing, and Dr. Graves could tell us of this.

MR. O'CONNOR: Would it be desirable to take out an old tree where there are new sprouts? One tree on Mr. Littlepage's place in Maryland has a number of sprouts coming up. I suggested that if we could get people together and clean the woods up we could dig up the old trees and only leave the blight-resistant ones.

PROF. COLLINS: That is near Bell Station where we do our experimental work. We found one place infected. I cleaned it out and we have not seen anything of the disease since.

MR. BIXBY: Some five or six years ago I sent a number of chestnuts to Warren, New Hampshire, which is outside of the blight district. I did not know then much about the blight. They grew for several years and it was not until one year ago that the trees were found with blight. I got the party to cut them down. How long must I wait before it is safe to send other trees there? I believe they will grow there and bear, but we do not want to get them affected with the blight.

PROF. COLLINS: I do not know that anybody could answer that. Apparently we have waited 20 years and are still unsafe. It is a case of experimentation.

MR. KAINS: As to the hybrids of Dr. Van Fleet and Dr. Morris, in the spring of 1923 I planted 10 and there are only four alive now. They were affected by blight and killed. They were rather large trees when planted, and I think for that reason more susceptible. We had the idea from the nursery that they would be more likely to withstand the disease than would the American sweet chestnut. Have you any reports as to the way these hybrids behave?

MR. REED: As to Dr. Van Fleet's hybrids, so far as we know they are all going with the blight. The collection in Washington is practically gone. We are still caring for them and doing what we can but the prospect is not at all good. We get reports of these distributed around the country, but in no case have we had a report indicating that the Van Fleet hybrids were at all resistant.

[Footnote A: Note—"Blight-resisting" as used in this paper should be interpreted as a slower death of the host than in former years, whether or not the result of increased resistance to the parasite on the part of the host, or to decreased virulence of the parasite, or to both factors combined.]

[Footnote B: Decision From the U. S. Department of Agriculture.
Washington, D. C.]

In a letter of later date, addressed to Mr. C. A. Reed, Dr. B. T. Galloway, of the U. S. Dept. of Agr., wrote regarding the matter of distributing Merribrooke chestnut scions, as follows:

"I have talked with Mr. Stevenson, of the Federal Horticultural Board, regarding this matter, and he says that, while there is no federal quarantine covering the chestnuts, as a matter of policy we have not been letting any chestnuts or scions go through our hands into the non-blight regions. Mr. Stevenson says that Dr. Morris himself might be able to carry out the plan he suggests by dealing direct with some of the state institutions in non-blight regions, selecting states that have no quarantine against chestnuts."]

PROF. COLLINS: I will now read my paper on

PROTECTION OF WOUNDS IN NUT TREES

I have been asked to discuss briefly the handling of wood decay in top-worked nut trees. I am not sure that I know very much about the latest methods employed in this type of work. Personally I have had no practical experience with it. I understand, however, that nut trees are top-worked by cutting off limbs and inserting one or more scions. I am informed that limbs as large as six inches or more in diameter have been cut for this purpose, particularly on pecan trees in the South, and that decay has started at the top of these stubs after the scions have become established, resulting in a pocket of decay. I assume that it is about such places as these that you want me to say something. Such conditions, whatever their origin, call for straight tree surgery methods. My work on tree surgery has been almost entirely with shade trees and chestnuts, and only to a very limited extent on other nut trees.

The general methods of handling decay are essentially the same on all trees, as also are the fundamental principles underlying the same, whether on nut or shade trees. I must admit I do not know just what methods are being employed by nut growers at the present time to counteract such decay in top-worked trees, so my suggestions may include nothing with which you are unfamiliar. Again, they may include some methods that you have already tried and found wanting so far as nut trees are concerned.

As a *prevention* of decay my suggestions, based on my own shade tree experience, would be:

(1) Avoid cutting large limbs when smaller ones are available and will serve the purpose just as well or better.

(2) Keep the scars thoroughly and continuously covered with some good waterproof and antiseptic material so as to prevent infection of any part of the cut surfaces.

(3) Always make the cut somewhat slanting so that rain water will readily run off, and insert the scions preferably at the upper extremity of the cut. Such an oblique cut normally heals quicker and better on shade trees than a transverse cut, particularly if a vigorous young sprout is left at the peak of the cut. I am quite certain the same statement will hold true with scions of nut trees placed at the peak of the oblique cut.

After decay *has started*, I would suggest—

(1) Cut out all the decayed woody matter, preferably from one side, so that a free and easy drainage of the wound may result. If necessary, when several scions have been placed around the stub, sacrifice one of the grafts and make a rather long oblique cut or groove from which all decayed matter has been removed. Use shellac, liquid grafting wax or melted paraffine over the cut bark, cambium and adjoining sapwood immediately after the final cut is made.

(2) Cover the entire wound with some good preparation to keep out disease germs and water. Preferably use for a covering such materials as will be more or less permanent and which have been found by practical experience to be least injurious and most effective on the particular nut tree that you are treating.

(3) Keep the wound thoroughly painted or covered at all times until it is completely sealed over by a new growth of callus.

(4) If the top-working was originally done in such a manner that the removal of all the decay results in a cavity that cannot be properly drained, it is advisable to fill the cavity with some waterproofing and antiseptic material in order to prevent it holding water and also to assist the cambium in covering the wound. The cavity must first be treated in accordance with approved tree surgery practices. In shade tree work, quite a variety of substances have been used to fill cavities with more or less success; e. g., wood blocks and strips, asphalt and sawdust, asphalt and sand, clear coal tar, clear asphalt, elastic cement, magnesian cement, Roman (or Portland) cement, etc. Of these only two—wooden blocks and Portland cement, have been in general use more than a few years. Blocks of wood were used in France to fill cavities more than 60 years ago, and in this country to some extent about 50 years ago. Later, Portland cement was used in preference to wood for fillings, probably mainly because it was more easily handled. To us of the present generation, Portland cement in combination with sand is the one material that seems to have been in general use sufficiently long to allow us to draw any seemingly reliable conclusion as to its real merits.

For the personal use of the average orchardist, Portland cement is one of the last in the list mentioned above that I would recommend. According to a few reports that have reached me, wooden blocks and tar proved to be fairly satisfactory half a century ago, and strips of wood embedded in some flexible and antiseptic material, are proving very satisfactory today. An excellent preparation to use between the strips of wood, containing asphalt and asbestos, can be readily bought on the market, and it has the advantage of being mixed ready for use. For cavities with horizontal openings that will hold semi-fluid substances, clear asphalt or gas-house (coal) tar may answer all purposes. For cavities with oblique or vertical openings, or for those on

the underside of a limb, probably some of the magnesian cements, which readily adhere to wood, will be found more satisfactory when properly mixed and applied.

Although I have said more about filling cavities than of other phases of the work, I do not wish the impression to go forth that I recommend such work except as a last resort, so to speak. The one thing that I do most emphatically recommend above all others is the prevention of decay so far as possible by practices that are less likely to allow decay-producing organisms to gain entrance in the first place, or at any other time.

THE PRESIDENT: Does anyone care to discuss this paper?

MR. KAINS: Mr. President: During the last five years, I have planted several hundred nut trees, including the English walnut, black walnut, the heartnut, pecan (northern ones) and some hybrid hickories. I have noticed that in this nursery stock there has been a good deal of dying-out of the original stock where the trees had been grafted, and where the scion had not covered over. In some of those cases decay has set in, and the trees have died before they could be attended to or have been broken down by the wind. The point is, I think it a mistake for nurserymen to use as large stocks as they have been using in many of these cases, because the stump of the stock is too large for the slowly growing scions to cover over quickly enough. My experience in the planting of fruit trees has been uniformly successful with smaller stocks (that is, trees smaller than I have been able to buy for nut trees) with peaches one year from the bud and with apples not more than two years; with berries and stone fruits, not more than two years. In every case, with the fruit trees, one year stocks have given me better results. First, because they healed over more quickly, and second, because I could cut to better advantage in the trees. In no case have I been able to get nut trees as small as I can apples and peaches. I believe that with the

smaller trees amateurs will have better success. I bring this matter to the attention of those men who are devoting their lives to the propagation of nut trees.

THE SECRETARY: The subject of transplanting nut trees was treated fully by Mr. Bixby in his paper this morning and will be treated by Mr. Hicks this afternoon in his address on the subject. Mr. Hicks will give a lecture, illustrated with slides, showing how the larger nut trees may be successfully transplanted.

DR. MORRIS: Mr. Kains' thought was that there was a good deal of difficulty from using stocks that were too large. Paraffine will keep them safe from microbes.

MR. KAINS: We had difficulty from the drying of the scions.

DR. MORRIS: I find that if raw pine gum is put in it prevents the paraffine from cracking.

MR. O'CONNOR: In regard to wounds on the trees I find that creosote makes a very good antiseptic. I use coal tar and creosote, mixed to a consistency of cream. I have used Portland cement but I treated with creosote first. In some cases I used bichloride of mercury.

MR. REED: It seems to be the experience in the South that, so far as the amateur is concerned, he gets better results with the pecans by planting trees of from three to five feet. Trees smaller than that are regarded as dwarfed; but the man who is in a position to exercise greater care could get quicker results from buying the large-sized trees. Yet it requires more care in transplanting, more fertilizer, and more attention.

MR. REED: I wish to make the motion that the chair name a nominating committee at this time.

THE PRESIDENT: Is that agreed? All right; then I name Mr. O'Connor for chairman, Mr. Reed, Mr. Olcott, Mr. Bartlett and Mr. Hershey on the committee. Are those names acceptable? (Vote shows unanimous acceptance).

THE PRESIDENT: The convention will adjourn until two o'clock.

FIRST DAY—AFTERNOON SESSION

Meeting called to order by the President.

THE SECRETARY: I will read a communication from Mr. Snyder, of Center Point, Iowa. But first I would like to explain that when the President in mentioning the Horticultural Exposition at Waterloo, spoke of a sweepstakes cup from a member of the N. N. G. A. for the greatest number of points won in the nut exhibition of which Mr. Snyder has charge he did not state that he himself was the member who gave the cup.

A HARANGUE ON THE NUT SITUATION IN IOWA

By S. W. Snyder, Iowa

Previous to the organization of the Mid-West Horticultural Exposition the Iowa State Horticultural Society had given but little attention to the nut question. But along with the exposition came a demand for a nut department, which resulted in the writer being appointed superintendent and given authority to prepare a limited premium list.

This resulted in bringing out a number of new and unnamed varieties of nuts and created some enthusiasm. When it came time to prepare for the second exposition, authority was given to greatly increase the premium list, which resulted in bringing out more new varieties and created a wonderful lot of enthusiasm.

When it came time to prepare for the third exposition a list was adopted calling for \$138.00 in cash premiums, which resulted in bringing out such a large exhibit of choice nuts that when we came to make preparation for the fourth exposition the premium list was increased to a total of \$181.50. This brought out so many fine nuts that it became a common thing to hear the remark, among the visitors that it was the most important department in the exposition.

For the coming exposition, to be held next November, the premium list as adopted calls for \$280.00 in cash premiums, and while I am no prophet I am going to predict that it will result in bringing together the largest nut exhibit ever collected under one roof in the United States.

At our last exposition held in Council Bluffs, some of the directors of our state fair observed that the nut department was attracting much attention and was bringing a good many visitors to the exposition. They decided that they must have a nut premium list for the state fair and requested me to make up a list covering the nut subject as strictly applied to the State of Iowa. This I did and am attaching the list hereto. Although our state fair comes off in the month of August, and no nuts are available for exhibit, except such as happen to be kept over from the previous year's crop, yet it brought out at our 1923 fair the largest and best exhibit of nuts that has ever been shown within this state, not excepting the exhibits of the exposition. The board of directors were so well pleased with the interest manifested in the nut department that they are continuing the list for this year's fair and doubtless it will become a permanent feature of future fairs of this state.

So much publicity and attention has been given the nut question within our state that it has resulted in bringing to light several new varieties that we think should be propagated before the original trees may have been destroyed.

The horticultural department of our Iowa State Agricultural College is now taking an active interest in the nut question and has assigned one of the professors to the job of collecting information about and taking pictures of, the best known nut trees within the state.

If they follow up the nut subject with as much vim and energy as they have other phases of horticulture we may look for something in the nut line in the next few years that will be worth while.

The native nut situation might well be summed up by saying that we have so many good walnuts, butternuts, hazels, pecans, hickories, and hybrids of the two last named species, that we could banish all foreigners and still have plenty left to supply every need.

The crop of nuts for this season is fairly good; some trees have none, others a light crop, and some varieties are carrying a heavy load.

Of introduced nuts some are proving to be hardy and fruitful, but in my judgment they are all lacking in eating quality as compared with our own native nuts, unless I should except the filbert which has not yet proven that it will bear profitable crops in this climate.

In closing I want to give just one instance of the great interest that has been aroused for nut growing within this state.

A certain little city of less than two thousand inhabitants happens to own thirty acres of land that is suitable for the growth of timber. The citizens propose to plant the entire tract to nut bearing trees and bushes, and eventually make it a free park in which the children of the village may be turned loose to gather the nuts.

Just imagine, if you can, how the enthusiasm of the boys who may be fortunate enough to live in that little city, will more than bubble over as the nut gathering season approaches. I hope to be able to assist those people in their laudible enterprise and wish I may live to see it develop into the greatest thing of its kind in the United States.

THE PRESIDENT: Dr. Brooks, will you kindly give the Vice-President's report from West Virginia, preceding your paper?

DR. BROOKS: I have no special report to give as Vice-President of the association from West Virginia. I might say, perhaps, that the West Virginia station is in a land of hills and dales. Our latitude is from 200 to 5,000 feet above the level of the sea, and our average elevation is 1,500 feet. From our excellent position we can look down 600 feet or so upon the Ohio. Our land contains many species of trees, including nut trees. Among these there is one species of beech, two of hazel, two of chestnut, six of hickory, two of walnuts and fifteen of oaks. Fortunately, the chestnut blight has not swept the entire state. The chestnut has been in the past and is still our most popular tree. There are areas where tons of chestnuts are still put on the market every year. The people are still thinking more and more of some plant that might take its place; they are considering the shagbark hickory and the black walnut. I predict that in the future there will be more planting of hazel nuts, black walnuts and shagbark hickories in this state. The prospect there is promising.

SOME OF THE MORE IMPORTANT INSECTS ATTACKING NORTHERN NUTS

By Fred E. Brooks

Associate Entomologist U. S. Department of Agriculture

The prevalence of insect pests need not be regarded as an alarming obstacle to nut growing in the North, and yet there are numerous species of insects which are capable of destroying our nut crops. On the whole I presume there are fewer insects that attack nuts in this country than commonly attack apples, but apple growers are not limited in planting nor prevented from making profits on account of insect depredations. Neither should the probability of more or less insect injury discourage the would-be planter of nut trees.

The presence of an insect in any locality may mean, among other considerations, that the soil, and climatic conditions of that locality are favorable to the plant upon which the insect feeds. We may be sure that wherever the Baltimore butterfly is abundant, nearby is a congenial spot where the turtle's-head, the food plant of the butterfly, flourishes. Just so, in localities where there are many chestnut weevils we may expect to find chestnut trees thriving and fruiting generously. The same is true of the associations of many other insects and plants.

Theoretically speaking, one would not care to risk the expenditure of much time or money in propagating a plant in a region that was destitute of insects that might attack that plant. The absence of such insects would possibly indicate a lack of natural conditions favoring the growth of the plant in question. Thus the presence in any locality of insects that feed on nuts may mean that nuts thrive naturally in that locality and that insects are there because of the abundance of a favorite food.

May I hasten to add, however, that this fact should not lead to an under-estimation of the possibilities of insect destructiveness, nor encourage lax methods in dealing with injurious species. In the beginning of any nut-growing enterprise we should anticipate the coming of insect pests and be ready to meet them. The planting of pure stands of native nut trees sets up a condition under which insects coming from the forest may increase more safely and rapidly than under the more hazardous environment of a scattered forest growth. This applies to cultivated plants generally. It is true of an orange grove, a cornfield or a potato patch. The mass planting of any crop is quite sure to call sooner or later for measures to offset the stimulus which such plantings offer to insect increase.

Reference may be made to a familiar nut plantation which illustrates a natural result of neglecting one of the insect factors. This plantation is the government's chestnut orchard at Bell, Maryland, which was planted for scientific purpose some years ago by Dr. Van Fleet. This orchard of around one thousand trees contains numerous species and varieties of chestnut, some of which bear fruit every year. The various scientific projects carried on in this orchard in the past have all been of such a nature that they called for no consideration of weevil increase. Many nuts have been allowed to lie under the trees until the weevil larvae issued and entered the soil. This has resulted in a constant increase of weevils until infestation of the nuts became practically one-hundred per cent. All nuts of the crop of 1922 were

so wormy that when planted they failed to germinate. Injury to the crop of 1923 seemed somewhat less severe, but its extent may be indicated by the fact that 3080 nuts from this orchard which were kept by the speaker in rearing jars yielded 11,085 worms. In the woods adjacent to the orchard the native chestnut trees are disappearing on account of the blight, and presumably weevils are on the decrease. Within the small area of the orchard, however, the increase has been abnormal, due, as has been indicated, to the peculiarly favorable and man-made conditions. If, from the time the trees of the orchard began to bear, the investigations being carried on had called for close gathering of the nuts at maturity and the destruction of all the worms that issued from them, there is little doubt that infestation would have been kept within reasonable bounds. At present, after two years of attention to the collection of ripening nuts, there is an apparent decrease in the number of weevils. Strong emphasis should be placed upon the importance of gathering chestnuts as soon as they are ripe and prevention of the worms from reaching the soil. This is especially true of districts where woods surrounding chestnut orchards do not contain bearing native chestnut trees.

The Nut Weevils

Now that the subject of nut weevils has been introduced, let us consider in more detail these grotesque, long-snouted insects whose larvae, or grubs, play havoc with so many of our nuts. Most of us have had the experience of gathering in autumn rich stores of our delicious native chestnuts. But how often our anticipations of boiled and roasted feasts have been blighted. We have found that the chestnuts were like the manna which fed the children of Israel in the wilderness, "When we left of them until the morning they bred worms and became foul." There are numerous cases in this country where chestnuts in shipment have been seized and condemned under the

provisions of the Food and Drugs Act. Usually the phraseology of the libel has been "because the shipment consisted in part of filthy animal substances, to wit, worms, worm excreta, worm-eaten chestnuts and decayed chestnuts." Altogether the loss to chestnuts from weevil injury is beyond computation.

The beetles which are the parents of the familiar worms in chestnuts are not commonly seen, or, if observed, they are not associated with the disgusting inhabitants of the nut kernels. These beetles represent in their structure a very interesting adaptation to a special end. The mouth is located at the tip of an enormously long snout, or proboscis, and the drill-like instrument is used for puncturing the thick covering of various kinds of nuts so as to admit the egg into the kernel upon which the young will feed. In some cases the mouth is situated at a greater distance from the eyes and other head appendages than is the anal extremity of the insect. There are in the northern part of this country two species which attack chestnuts, one which attacks hickory-nuts, one which attacks hazel-nuts and about a dozen which attack acorns. And here may be mentioned an interesting peculiarity of the feeding habit which is decidedly to the advantage of the nut-grower. Each species adheres closely to its own food plant. The hickory-nut weevil does not attack hazel-nuts nor the hazel-nut weevil hickory-nuts. None of the acorn-infesting species will seek for food in the nuts of chestnut, hickory or hazel. Once the chestnut weevils are absent in a locality, there is no chance that oak trees will serve as a means of spreading the weevils back into the locality. So closely confined are these weevils to their particular food plants that many of them distinguish between the different species of oak and will oviposit only in certain kinds of acorns.

All the different species resemble one another in both the adult and larval stages. There is also a general similarity in their behavior. I have recently discovered, however, a marked difference in the life cycles of certain

species. For example, the larger chestnut weevil and the smaller chestnut weevil look alike, but they are decidedly unlike in their development. The grubs of the larger weevil begin to leave the nuts at about the time the nuts drop. They enter the soil to a depth of several inches and fashion smooth-walled cells in which they remain unchanged until the following summer. During June and July they transform to pupae, and soon afterward to adults. In August they issue from the ground and seek the trees where they collect around the burs and begin to deposit eggs soon after the nut kernels start to form. This life cycle is continued year after year. To forestall starvation of the race in case of entire failure for a year of the chestnut crop, a few individuals carry over the second winter in the ground and then issue as beetles along with the one-year-old specimens. It is probable that a small per cent of the insects may remain in the soil over three winters. Thus does nature by unique arrangements safeguard the lives of even the very small creatures.

The life cycle of the lesser weevil is quite different. The larvae of this species leave the nuts somewhat later in the autumn than do those of the larger weevil. Like them, they enter the ground and pass the first winter unchanged. The grub stage is continued throughout the summer, but late in autumn, after the beetles of the larger species have been on the trees for some weeks and deposited most of their eggs, the larvae of the smaller species transform to adults. Instead of coming from the ground, however, they remain in their earthen cells throughout the winter. The next spring, prior to the blooming of the chestnut-trees, they emerge from the ground and soon thereafter collect in large numbers on the male catkins of the chestnuts. At this time very little feeding is done and the sex instinct does not manifest itself. As the time approaches for the nuts to mature, however, the beetles begin to feed and pair and soon thereafter to lay their eggs in the ripening nuts. Most of the eggs are deposited directly into the nuts after the burs begin to open. In the case of the larger weevils the beetles are present

only about three months of the year. Those of the lesser species, however, are perpetually present, those of the younger generation reaching the adult stage in the ground before those of the previous generation have finished laying their eggs in the ripening nuts. As with the larger species, a few of the smaller weevils carry as larvae for several years to tide over possible failures of the chestnut crop. The life cycle of the hickory-nut weevil is similar to that of the larger chestnut-weevil, and that of the hazel-nut weevil is like that of the lesser chestnut weevil. Both cycles are represented among the acorn-infesting species.

Any intelligent warfare against the nut weevils calls for a knowledge of these distinctive life histories. Thus, an abundance of maturing larvae of the larger species this autumn will insure an abundance of beetles to deposit eggs in the nuts next autumn. With the lesser weevil, however, maturing larvae this autumn will not affect the number of beetles on the trees the succeeding autumn but will provide beetles for the crop two years hence. Large numbers of beetles of the lesser species may be destroyed by collecting them from the blossoms of chestnut, but, at that season of the year there are no beetles of the larger species abroad.

These weevils are to be made the subject of a bulletin by the Bureau of Entomology in the near future, in which it is hoped to go more fully into a discussion of control measures.

Walnut Husk Maggot

Although none of the weevils of the group just discussed attacks walnuts, the fruit of this tree has a serious enemy in the walnut husk maggot. This insect is most familiar in the form of multitudes of dirty-white maggots inhabiting the blackened, slimy husk of ripening walnuts. Originally, the black walnut furnished the favorite food of this insect, although the husk of

butternuts was sometimes attacked. More recently the pest has turned its attention to the Persian walnuts which are fruiting in many places in the east. The watery, dark-colored pulp into which the husk of the nut is converted when the maggots begin to feed penetrates the shell of the nut and injures the kernel by staining it and imparting a strong flavor. The operation of hulling is also made doubly disagreeable, the nut coming out of the husk discolored and dirty.

These maggots hatch from eggs inserted into the husk of nuts by a light-colored fly about the size of our common housefly. Although easily overlooked, these flies may be seen on the nuts at almost any time in August and September. They have strong ovipositors with which they puncture the surface of nuts and insert into the openings masses of white eggs from which the maggots hatch.

As to the control of this pest, the speaker obtained very promising results in spraying Persian walnut trees belonging to our friend, J. G. Rush, at West Willow, Pa., with a solution of 1-1/2 pounds of lead arsenate to 50 gallons of water with 10 pounds of glucose sugar added to impart a sweet taste. The flies were observed feeding on the sweet coating given to the leaves and the nuts that ripened later were comparatively free from maggots. It was obvious that the flies died from the poison before depositing many eggs in the nuts.

Twig Girdlers

During the past two seasons the speaker has made special studies of several species of beetles which cut or girdle young hickory trees, or the branches of larger trees, causing the severed part to break off or die. Not fewer than four distinct species of beetles in the east have this habit. Three of the insects do their damage in the larval stage. One of these, *Elaphidion*

villosum, has been called the twig-pruner. It is a well known species and its work in pruning the branches of hickory and various other trees has often been referred to. The other two species which sever the wood in their larval stage are *Pseudobidion unicolor* and *Agrilus arcuatus*. Thus far, these two have no common names. In certain localities they are proving to be very troublesome to both young and bearing trees. In one block of a nursery in Virginia I estimated that the *Agrilus* larvae had ruined one-hundred dollars worth of young hickory trees. Fortunately, the adult of this species feeds freely on hickory foliage and can be killed readily under nursery conditions by spraying with arsenical poisons.

The fourth girdler referred to is our familiar hickory twig-girdler, *Oncideres cingulatus*. In this case the adult insect cuts a ring-like furrow around the wood and the portion above dies. The purpose of the girdle is to provide dead wood in which the young may feed. After the girdle is made, a process which occupies several hours, and, sometimes several days, the eggs are laid in the bark above. In central West Virginia and northward the grubs which hatch from these eggs require two years in which to reach maturity. In the vicinity of Richmond and southward, however, the larvae mature in one year. This more rapid development in the south probably accounts in part for the recent serious outbreak of this insect in Virginia and the Carolinas.

Each female beetle is capable of girdling several twigs. One female of about a dozen kept in confinement last autumn made eleven girdles and deposited 55 eggs. Several of the beetles continued their interesting operations until after several snows and severe frosts had occurred.

The twig girdler in the beetle stage feeds rather freely on the bark of twigs. Enough of the surface is eaten to justify the belief that the beetles may be killed by spraying with arsenical poisons. This treatment is being

tested at the present time. In the cases of all these insects which sever the branches the wood is killed for the safety and comfort of the insect as it undergoes further development above the severed point. There is a period of at least several weeks in each case after the twig dies during which the insect in one stage or another remains in it to complete its growth. This affords an opportunity to gather the twigs and burn them with the assurance that the insects are being destroyed thereby.

At least some progress has been made in discovering the habits and the methods of controlling these and various other insects that may be expected to give nut growers in the north more or less trouble. The remedies that can be offered at the present time are not in all cases entirely satisfactory. There is much yet to be learned, but there are control measures within the reach of most of the nut growers which are well worth consideration and adoption.

THE SECRETARY: Dr. Zimmerman, will you read to us now?

DR. ZIMMERMAN: Perhaps some of the members will not be so glad to hear what I have to say, but I feel that there is a need for something along the line I will refer to.

DEVELOPING A NUT INDUSTRY IN THE NORTHEAST

BY DR. G. A. ZIMMERMAN, PENNSYLVANIA

We have all heard of the pecan. No doubt most of us have traveled through the South at some time or other and have entertained a wish for a pecan grove. A personal friend of mine, a minister, told me recently that the only time he was ever tempted to invest in a commercial proposition was when a real estate agent laid a picture of a pecan grove before him. I had entertained the thought that some day I might possess an orchard. Therefore, a couple of winters ago, when I found it necessary to go south for my health, I silently hoped I could kill two birds with one stone, by getting some undeveloped land and starting a pecan grove, which at the same time would keep me in the open air and give me exercise. Consequently, my eyes were always open and I was on the constant lookout for pecans. After miles of travel they appeared. They were very interesting and I went into the subject pretty thoroughly. I was informed that no cheap land was available any more that was desirable for pecans. I am not so sure of that. I was also informed that most of the people who had planted groves had made a mistake, that the pecan business was just beginning under new ideas, and that most of the work would have to be done over. From the amount of trees that are being top-worked I am inclined to believe this is true.

But I didn't kill the two birds with one stone. I did not attempt to build up a pecan grove, but instead I came back with the idea firmly impressed that we have a better proposition for the future right here, that we have right here in the North the building material in the shagbark hickory and the black walnut for a nut industry that will rival or even surpass the enviable position the pecan holds today. Was I correct or was I wrong? A second trip last winter has served only to imbed that idea into a firm conviction.

What ground have I for drawing this conclusion? Some of you, my friends, may disagree with me in some of my remarks, and no doubt insist that I am uninformed. Perhaps I am, but I am giving my convictions nevertheless, and I ask you to withhold judgment for twenty years before deciding against me.

Why has the pecan forged to the front as it has? Because the pecan is a good food, easily available, of pleasant taste and presents a fine appearance. From a commercial standpoint, after 20 years or more on the pecan, there is only one really desirable variety available, namely the Schley, and the fact that it readily sold last fall for 80 cents per pound wholesale, while the choice of the other varieties brought 60 and 65 cents per pound, bears me out in this. I am not referring to the greater productivity and other qualities of some of the other varieties. Many of them are tolerated for various reasons.

How about the shagbark in the North? It is my belief that we do not have at present a shagbark that will anything like meet the pecan of the South, yet the consensus of opinion of the people I know who have eaten both, decides in favor of the shagbark. The quality of a very ordinary shagbark is better than the best of pecans. What then, is lacking? Size, shape, thinness of shell, cracking qualities, color, everything but flavor is lacking in most shagbarks. Don't misunderstand me. I am not condemning what we have,

for I believe that if as many years are spent by as many people in finding or developing a shagbark, we will have one that will surpass the pecan. But as the matter stands I am constrained to say that I do not know of a really good nut today that will stand the test of building an industry that will compete with the pecan. We must find or develop a couple of really good nuts that will compete, nuts that are large, smooth, shell thin enough to crack with the fingers, a white kernel that is plump and easily extracted. I do not believe that any thick shell nut will ever meet the favor it should or become extremely popular. The Weiker, one of our best, is of good size, looks fairly well, but the shell is thick and it is poorly filled. It will never fill the place for a real industry, and yet they sell for a good money-making price today.

If we build our groves after this standard we will be in the same place in a few years that many of the pecan growers are now, namely, with a lot of trees on hand that must be top-worked later on. But they are the best we have and, like the old adage that it is better to love and lose than not to love at all, it is better to go ahead with these than not to go at all.

How about the black walnut? This nut will come to the front and be popular for baking purposes and candy-making, for it is the only one that holds its flavor after heating. But its competition will be against the thin-shelled English walnut. It will not be extremely popular until we get one with a shell equally thin. At present we do not have one.

How then can we anticipate a great future industry after meting out this doleful outlook? Are we going to discard everything we have and start again? By no means. The price of nuts, even of the ordinary class, is sufficient even now to well repay any man for his effort, if producing them on a large scale, and what must be done is to encourage more people to become interested.

If we could arrange to have nice exhibits of named varieties of nuts at the various county fairs, and have someone there to explain them, a good deal of interest could be created. I frequently see native nuts displayed, but not named varieties.

I shall not refer to the hazel, chestnut, pecan nor butternut, all of which I believe can be developed into a more or less successful industry but only repeat in closing that I am convinced, after pretty thorough investigation, that the shagbark hickory and the black walnut can be developed into an industry in the Northeast in a much shorter time than it has taken to develop the pecan, to a point that will equal or surpass the enviable position that nut holds today. But, and let me impress this point, we must develop a few new and better nuts to do it. On account of the colder climate, which goes for the developing of fine flavor in all products, I do not believe the pecan will ever equal the shagbark in quality. This is our great natural advantage.

DR. MORRIS: I accept all of the statements by Dr. Zimmerman with one exception. The pecan is tremendously prolific and so productive that there are records of 30 bushels to a tree. I do not know that any of the shagbarks or shellbark hybrids ever will rival that in production. From the marketman's point of view production is of prime importance. In this the pecan out-rivals the black walnut.

TRANSPLANTING NUT TREES

Willard G. Bixby, Baldwin, N. Y.

When I set out the first nut trees which now are growing at my place at Baldwin, I was very particular to follow the best advice obtainable. What this was is to be found in Bulletin No. 5, published by the association, pages 8 and 9, under Planting Directions. I will not take time here to read them but will refer those interested to that publication.

Much that is to be found there is unquestionably the best practice that we know today. The importance of preventing the roots from drying out, digging holes of sufficient size and filling with good top soil, firming the soil well about the roots, severely cutting back after planting and staking newly set trees if they are of appreciable size above ground, are of the utmost importance and should be emphasized, but others of these directions have been modified in my practice and I will relate the unfortunate experiences which caused these changes to be made.

From the start there has been trouble in transplanting hickories, difficulties with other trees being small in comparison. Out of a number of fine looking little grafted hickories set out in the fall or spring some would be sure to die. They mostly came from Mr. Jones, who, as a rule, has furnished the finest looking hickories that I have received, and were finely

packed and seemingly ought to have lived, but only part of them did. After losing a number out of one lot, I watched the lot purchased next year with particular care. Three out of a lot of six, which had put out leaves well in the spring, by the middle of July began to show signs of distress, the edges of the leaves beginning to turn brown which the year previous had been the beginning of the end. I knew what had happened the year previous, felt that the trees would die if something was not done, and did something. That something was to dig about six quarts of chicken manure and two trowels of nitrate of soda around the three trees that looked sick and saw that they were watered plentifully till a heavy rain came. At first nothing was noticed, but after a while the brown disappeared on the leaves that were only slightly brown, while in other cases new leaves put out and finally a second growth of shoots, very small to be sure, but the trees had been saved. This was diametrically opposed to previous practice of putting no manure or strong fertilizer in holes when planting the trees, but the result was so satisfactory that I have continued to dig in about 1/4 of a wheelbarrow of well rotted stable manure around each tree when planting and two trowels of nitrate of soda in May when the growth should start in the spring.

The above treatment seemed almost entirely to solve the difficulties of transplanting and for about two years practically no hickories were lost. Twenty-four Hales trees, 10 years from grafting brought here from Monticello, Florida, all lived through the first year and 23 of them through the second and now seemingly have a long life ahead of them. Inasmuch as Mr. Jones expressed his doubts as to how successful this experiment would be I regarded it as somewhat of a triumph. On the other hand out of the finest looking lot of young Iowa hickories grafted a year ago this spring and shipped in the fall and set out just as carefully as I knew how, with well rotted stable manure in the holes and seemingly having every prospect of a long life before them, all have died now, excepting four, two of which I am making desperate efforts to save.

The reason for this failure has not yet been proved, but I have an idea what it is. With two exceptions the stocks were not large, unusually small in fact, and the growth of the grafts was small, but, except for their small size of stock and graft they were fine looking little hickories as one often sees. The two that are in good condition today were bitternuts on bitternut stocks and both the stocks and grafts were notably larger than others. One of these bitternuts by the way, is bearing this year. Evidently there was not as much vitality stored in the smaller trees as in the larger ones. I am inclined to believe that the real trouble was because the grafts, excepting the bitternuts, had not become sufficiently established before having to stand the shock of digging, shipping and transplanting. I have noticed in experiments made to determine the adaptability of a number of species of hickory as stocks that it was not unusual to find that a graft would do reasonably well the first summer and die the second. If this happens occasionally when hickories have not been transplanted it is undoubtedly very much more likely to happen when they are transplanted. I have had practically no losses in transplanting hickories when the graft had grown two seasons before being transplanted. The safe plan, then, would seem to be to let a graft grow two seasons before transplanting. Unfortunately this will add to the cost of grafted hickories which even now are so expensive to produce that almost no nurserymen grow them.

Another one of the commonly accepted principles that I do not now follow is that of not planting trees any deeper than they grew in the nursery. I prefer to plant them a little deeper, say two inches or so. I do not recall losing any trees seemingly from this slightly deeper planting, while I did lose a considerable number of seedlings last year that were inadvertently planted two inches or so too shallow.

Outside of the hickory I have had little trouble in transplanting any trees excepting some of the hazels. Unless hazels, particularly American hazels,

are very well rooted, they will need more care the first year than most nut trees, particularly protection from the hot sun and drought. If I get poorly rooted hazels I now plant them in a shady place for a year or two if they have not grown well the first year, and then move them where they are to stay.

THE PRESIDENT: Mr. Snyder of Center Point advocates planting trees two to four inches deeper.

DR. MORRIS: In Dr. Brooks' paper he spoke of some of the twig girdlers in the beetle stage which feed upon the bark of twigs before ovipositing, and he said that gives a weak point where we may attack them. On my place at Stamford, where there are forests, that would be impossible. I have had a good many hazels partially destroyed this year by girdlers. A great many of the branches have the larvae in them. I find also a large number of small hazels on which the leaves and branches are dying, though there is no apparent injury to the bark. Suddenly, however, a little twig will drop off and yet, in cutting into them, I did not find any larvae.

DR. BROOKS: That happens to be the work of an insect which I am just beginning to study, one of the flat-headed borers, and the reason you have not seen the larva is that it is very small. It is not half an inch long. In the second year it comes out as an adult. I judge that control measures should be used in the spring, when I think without doubt that it would feed on the poisoned spray.

DR. MORRIS: I find a great many larvae in dead twigs on the ground. If we are going to get this pest out of the way, we should not only look at the twigs on the tree, but at those on the ground as well.

DR. BROOKS: That is true of all of these curculios. Dr. Morris' statement is true. The ground should be gone over and the dead and dying

branches and twigs of the trees should be collected. The insects mature in them.

DR. COLLINS: Would you advocate pruning often?

DR. BROOKS: No.

Adjournment to lecture hall. Mr. Henry Hicks of Westbury, Long Island, gave a talk on the transplanting of large trees by his methods, illustrated with lantern slides. This was followed by a talk with lantern slides, on

HEREDITY IN TREES AND PLANTS

By Dr. A. F. Blakeslee, New York

Dr. Blakeslee said in part:

One of the first things we notice as we go out into the open is diversity in the habits of trees and plants. It is through the details thus presented that we are able to distinguish one species from another. You can see this diversity the year round in nut trees, and in the nuts.

If you arrange nuts, or any other objects for that matter, in a curve according to size, you will find that the most numerous of them are of about the average size. This is equally true when applied to mankind. What is the reason?

There are a number of factors affecting this, but, in general, there are two main causes—environment and heredity. We do not know which is the more important but both are absolutely necessary.

In the picture being shown we see the influence of the black walnut upon plants around it. It creates an environment which influences the ability of other plants to grow near the roots.

It must be remembered, however, that what the animate plant transmits is not the actual character in question, but the ability of the animate plant to develop characteristics. By placing the plant near a black walnut tree we do not affect anything but the capacity of the plant to develop in certain directions.

I have shown here a diagram to illustrate a certain stock fertilization. Here we have the plant with its stamen and pistils, the egg cells and the pollen. There are two types of pollenization, one where the pistil is fertilized by insects carrying sticky pollen; the other by movement of the wind carrying the pollen. If I should believe my records, in attempts to cross trees, I might have a cross between a birch and an alder, in which the pollen is carried by the wind. I tried once to hybridize pines. I put some pitch pine pollen on the female flower of another species and seed resulted. I did this the second year and again I got seed. The third year I put bags on the female flowers before I could see them developing. Then I got no seeds. I believe that the pollen which had caused the seed to set in the preceding instances had come from the south for perhaps hundreds of miles.

There are times when the pollen of the staminate plant is all shed before the pistillate gets ready. Sometimes we have a plant that is self sterile. I have experimented with pollen from several different nut trees and also with the Norway spruce. Then again, there are abnormal cases; sometimes there is parthenogenesis. The jimson weed is the first plant which has ever been reproduced by parthenogenesis. Since that was discovered, an investigator in California has found a similar case in fruit developed without pollination.

One of the most important conceptions in heredity is its effect upon characters and factors. Take the Japanese bean here shown for example, one dark bean and one mottled. In the next hybrid generation we find three

mottled and one dark. That is the familiar "three to one" ratio of Mendel's law. We believe now, that all, or at least a very large proportion of the heredity characters in plants of all kinds may be due to a series of factors; but the habit of growth of the plant is due to a single factor. We have the case here of a second generation of the weeping mulberry that I crossed with the white mulberry. As a result there was an average of three erects to one weeping one. Certain characteristics may be made up of the inter-action of a large number of factors. This will give a little idea as to the complexity of Mendel's law.

How do we get new characters in nature? New types are due to the rearrangement of previously existing characters, just as with the old-fashioned kaleidoscope, where you turn the crank and get new pictures. Another way is by the sudden appearance of new factors.

I wish to speak about one effect of hybridization, which is really connected with heredity factors, the vigor which occurs when we cross different varieties, species, or even races. In my experience certain types that have been naturally contrasted finally lose vigor, and after two or three generations the plant disappears. Then again I could show you cases where yields are greatly increased due to hybridity. These are established facts, not only as regards species of plants and trees but also as regards the human race. Hemy, in Dublin, who has done the best work in this line of endeavor, believes that many of our more rapid-growing trees are rapid-growing because they are hybrids.

To summarize, I have tried to point out the fact that diversity which we see in nature is real, and that it is brought about by two causes, namely, environment, and heredity. And that heredity is brought about by factors in the bodies of the chromosomes which are shuffled around like cards in a pack; they reappear in the same way that the cards will reappear. We have

no means, as yet, of controlling the appearance of the factors, but we have two methods of getting new factors, as follows:

One—The finding of new things in nature; that, probably, is the very best method that can be used. The work of the theoretically planned project points out the tremendous importance of the exceptional individual.

Two—By taking the exceptional individuals, and by crossing them, you can recombine, although the results may be very complex, and obtain characters that are very desirable.

As ministers sometimes say to clinch the moral, I would say, "Seek earnestly that which is best and hold fast to that which is good."

THE PRESIDENT: Has anyone a question he would like to ask?

DR. MORRIS: In attempting to make crosses between *juglans* and *carya* we find often that the pollen of *carya* will excite the cell of the *juglans* but without making a fusion. What is the element of the male cell of the hickory which starts the female cell of the walnut into action?

THE SECRETARY: I would like to ask Dr. Blakeslee one thing; he showed the influence of the black walnut on the growth of the hedge, and he showed that something other than the effect from the black walnut had caused these plants to be dwarfed. Is that understood to be a fact?

DR. BLAKESLEE: No; some of the effect was due to the black walnut.

MR. HICKS: In some cases the trees get sick and die. I have observed many plants and trees growing close to walnuts and I can point out peach trees and other fruits planted close to black walnut trees which have been injured. I should like to see the question determined.

MR. O'CONNOR: On Mr. Littlepage's place it seems that some blackberries thrive better in the shade of the walnut tree than anywhere else.

DR. BROOKS: In West Virginia there is a locality where blackberries grow wild, and it is a matter of common knowledge that black berries will grow under the black walnut but that apple trees will not grow there. I have noticed that the best place to plant jimson seed is under the black walnut trees. I have no definite information about this but there is something in the influence of the black walnut trees.

MR. BIXBY: I have noticed at my place that cabbages planted under black walnut trees were somewhat stunted. I believe that it was the effect of the walnut trees growing so speedily that there was not enough nourishment for both.

THE PRESIDENT: The next lantern slide lecture will be by Mr. Reed.

MR. REED: (This lecture was delivered in a darkened hall where it was not possible for the reporter to take notes. However, the gist of the talk is here given).

The slides illustrated various methods of nut tree propagation, and that it is possible successfully to graft or bud nut trees at almost any time from February until the very end of the growing period. In working over large trees the first method in the season to be employed was shown to be that of the cleft graft. Following this, with large stocks, would be the slip-bark graft, or with smaller stocks, the chip-bud. The slip-bark graft has the advantage of being feasible at any time when the bark slips. Dormant scions are more often used with this form of propagation, although by no means necessary, as Dr. Morris has demonstrated that by applying a coat of paraffin over the entire scion and the cut surfaces of the stock, it is possible to use growing scions at almost any time when they can be obtained. The

chip-bud is most successful during a relatively short period, beginning about ten days before the buds begin to swell and continuing until after the trees are practically in full leaf. From this time on the patch, or some other modification of the annular bud, is most commonly used.

In top-working, when the cleft-graft has failed, the patch-bud may be used late in summer, by inserting buds of the current season's growth in the base of the new shoots springing up from below where the cut was made in the stock for the graft, thus affording two opportunities for propagation during the same season.

The slides showed various methods of propagating the filbert by layering, and of propagating more difficult species by inarching. They were from a collection soon to be placed in the hands of the extension Service of the U. S. Department of Agriculture and of the various state colleges of agriculture.

THE PRESIDENT: We will now adjourn, and will meet in the room upstairs in this building at 10 o'clock tomorrow morning.

SECOND DAY—MORNING SESSION

Meeting called to order by the President, at 10 a. m.

THE PRESIDENT: I have the great pleasure of introducing to you Dr. Howe,
Assistant Director of the Botanical Gardens.

DR. HOWE: I shall only take a minute to say that we are delighted to have you here, and that if we can do anything to assist you, or to perpetuate your success, I hope you will please let us know. As the Spaniards say, "The house is yours."

I hope that your visit will be so pleasant that you may find it convenient to come here again.

THE SECRETARY: Mr. Jones will you tell us something about the handling of seeds for planting?

MR. JONES: I did not give the subject any thought before coming here but I might say that the nuts should be gathered promptly and dried, placing them in a shady spot, for they can be injured where the sun is too warm. We stratify them in sand. Then in the spring you can sift the sand through a sieve, take out the nuts and plant them.

In stratifying chestnuts we keep them between layers of wire mesh, for mice are very fond of these nuts. We cover the nuts with sand and leaves. Chinkapins we usually keep in cold storage.

THE SECRETARY: When you stratify these nuts where do you keep them?

MR. JONES: Right out in the open on top of the ground. A frame may be made with wire nailed on the bottom. This may be set out anywhere in the garden, but down a little into the dirt. Put in the nuts between layers of sand and leaves.

THE SECRETARY: Mr. Kelsey told me that the best way he had found to keep nuts was to bury them in a deep hole, perhaps two feet deep. Have you had experience with that way?

MR. JONES: The way I described is the usual way to keep seed and we get very fine results. We do that in order to keep the seed cool and so that they will not dry out. But we always have to watch out for mice. It might be a good idea, in stratifying chestnuts in the box with wire mesh on the bottom, to place the box at an angle that would drain off at least part of the water.

THE SECRETARY: Dr. Zimmerman, have you anything to say?

DR. ZIMMERMAN: I discovered by accident that black walnuts and hickories could be kept very nicely in the dry state until spring; then put water on them and they will sprout very nicely. But my chestnuts get moldy that way.

MR. BIXBY: We cover the nuts with at least a sprinkle of earth, may be four or five inches.

THE SECRETARY: Mr. Jones would keep them with practically no dirt but with sand and leaves.

MR. JONES: I would use a little sand over them, two parts of sand to one part of nuts. We put in six inches of nuts and alternating layers of sand.

DR. BROOKS: I know of a man who puts a layer of chestnuts and one of moss and says that in the spring the nuts are in splendid condition.

MR. BIXBY: I have had the nuts sprout very much better when they were stratified as soon as gathered.

MR. O'CONNOR: I bought about 5 bushels of black walnuts, paying 75 cents a bushel for them. I simply dumped them out on the ground, not bothering about the shucks at all, and covered them over with dirt. I paid no more attention to them until spring. Then I put the nuts in trenches with dirt about 5 inches over the top. The mice did not bother them, and I think they did well that way.

THE PRESIDENT: Did the frost affect them?

MR. O'CONNOR: No, not at all.

THE PRESIDENT: I have a black walnut tree at home that started to grow in a neighbor's cellar. It had grown a foot and a half and was rather white in color. I cut off the top and planted it out in the open. Today the tree is still growing and is all right.

We will now have an address by Prof. Neilson, of Canada.

PROF. NEILSON: Mr. President, Ladies and Gentlemen: It is a real pleasure for me to get back to this convention once more. I tried to come last year but owing to certain difficulties I was not able to do so.

Before I give you my report on nut culture in Canada, I want to tell you some of my troubles. Two or three years ago, when I began to express my interest in nut culture, I thought it would be a good idea to get some nuts from China. I wrote to several missionaries in Northwestern China at about our latitude, and I finally secured five bushels of Persian walnuts and one bushel of Chinese chestnuts. The nuts were a long time on the road and very few were in fit condition to use when they arrived. I stored some of the Persian walnuts in our cellar at the Ontario College. The rest of the nuts I distributed to others.

The nuts at the college did not fare very well. When I left there I gave directions to the members of the Department to look after them carefully. This is how they did it. Someone broke into the cellar where the nuts were stratified in the sand, and ran off with about one bushel. The Chinese chestnuts arrived in about the same condition as the Chinese walnuts. Of these I managed to save about a peck. We divided the nuts into three equal lots. Some we kept at the Guelph Experiment Station, some at Vineland, and some in the Southwestern Station. Of those at Guelph, out of the whole lot, 35 nuts germinated, and of these the mice ate all but five. These five were taken outside and carefully placed in a flat; but someone came along and ran into the flat and smashed those five plants all to pieces.

In addition to this some of my friends tried to tell me that I was chasing wild geese; that nut trees would not ever be important commercially in Canada; that 99 per cent of the value of the nut tree was for shade anyhow (as if he meant shade for pigs and cows); and that they were not even ornamental.

Before I read my paper, however, I will say that the work I am now doing is somewhat different from that I had when I was last here, when I was Prof. of Horticulture. I am now doing extension work for the government.

PROGRESS REPORT ON NUT CULTURE IN CANADA

Jas. A. Neilson, M. S., Extension Horticulturist, Horticultural Experiment Station, Vineland, Ontario

During the season of 1923-24 there has been a marked increase in the interest shown in the culture of nut bearing trees in all parts of Canada where nut trees can be grown. This is indicated by the numerous letters of enquiry and personal requests for information on nut culture which have been received by our Station. A total of 450 letters were received or sent out by our office during the past year besides numerous enquiries answered by a personal visit.

The search for good nut trees has resulted in some interesting additions to the data presented in the paper published in the last report. One of the most gratifying features of this phase of the work has been the discovery of several new localities where the European filbert is growing successfully. It has been located or reported at twenty widely separate points in Ontario, the northernmost of which is on Wolf Island at the mouth of the St. Lawrence River in approximately 44,100 N. Lat. This plantation is said to have been established before 1840 and would therefore be nearly 90 years old. Another interesting point in connection with filberts is the amazing way in which they thrive under conditions of absolute neglect. Several of the

plantations observed during the past year were not given the slightest attention and yet were doing very nicely. Obviously this is not good practice but it would seem to indicate that excellent results could be secured in Southern Ontario by the proper choice of varieties and the best cultural methods. This survey also showed that the sweet chestnut grew as far north as Georgian Bay.

The prize nut contest staged by our office last autumn resulted in the discovery of some very good black walnuts and a fine Japanese heartnut. Samples of these are shown in some of the plates on the table.

The Persian walnut was found to have a wider distribution and is more abundant in Ontario than was expected when our nut survey began. About 150 bearing trees have been located in that part of Ontario extending from Toronto on Lake Ontario to Goderich on Lake Huron. This number of course will seem insignificant in comparison to the numbers of trees in some sections of the northern United States, but it must not be forgotten that Ontario is on the northern margin of the Persian walnut territory, and therefore the results are rather encouraging.

Several fine Paragon chestnut trees have been located which bear good crops and which appear to be resistant to chestnut blight. This disease has unfortunately appeared at several places in Ontario and will undoubtedly destroy the majority of our chestnut trees.

The members of this association will be interested to learn that Gellatly Brothers of Gellatly, B. C., prepared and sent to the British Empire Exhibition at Wembley a large collection of nuts that has attracted a great deal of attention and favorable comment. This should do a great deal toward advertising the nut cultural possibilities of that province and of Canada generally.

The trial plantations on the experiment station grounds are doing very well indeed. The black walnuts are making a fine growth and one variety the McCoy, has a good crop of nuts at two years from planting. The Ten Eyck is making an extremely rapid growth, in some cases, producing new shoots over four feet in length.

The English walnuts are also making a good growth and two varieties, Mayette and Hall, have borne nuts in the third season.

I am pleased to state that we now have about 100 seedlings of the Chinese walnut growing on the station grounds and at various other points in Ontario. These little trees seem to be making a more rapid growth than our seedlings of the "Ontario," a Persian walnut which is a native of St. Catharines.

We also have about 60 seedlings of the Persian walnut from the Northern slopes of the Carpathian Mountains in the Ukranian region of what used to be the old Austrian-Hungarian Empire. These nuts were obtained from Rev. Paul Crath, of Toronto, who informs me that the winter temperatures in that part of Europe often go lower than in Toronto. We hope for some interesting developments from the growth of these trees because of the rigorous climatic condition of their native land.

During the latter part of the past winter an experiment was conducted in propagating the walnut under greenhouse conditions. For this purpose 100 well grown one year black walnut seedlings were obtained from our forestry station at St. Williams in the late autumn and heeled in out of doors until about February 1st. These were then brought inside, planted in 8 inch pots and placed in the greenhouse where they were allowed to remain until a good leaf growth had been produced. The young trees were then side cleft grafted with scions of the best English walnuts in the district. While engaged in this work one of the trees was inadvertently cut off a few inches

above the ground. The stub was then whip grafted and to my surprise it made a better growth than the others which had a part of the top left on. The results of our experiment were much better than I expected. About 40% of the scions grew which was quite satisfactory considering that I was a mere novice in the art of grafting nut trees and that my method was an experiment. I believe I could get 70 to 75% to grow with greater care in the selection and handling of scions. The object in doing the work in the greenhouse was to obtain better control conditions of moisture and temperature and thus reduce the mortality of scions due to these factors.

I also outlined an experiment in propagating nut trees by cuttings as a thesis subject for one of our fourth year horticultural students at the O. A. C. In this experiment ten cuttings each of English walnut, butternut, Japanese walnut, hickory, chestnut and black walnut were planted in sand and watered at intervals with a 1 to 10,000 solution of potassium permanganate. In the course of time the majority of cuttings came out in leaf, but none formed roots, and hence soon died. It is admitted that this experiment may have been improperly planned and conducted, but it showed at any rate that it is not an easy matter to propagate most nut plants by root or stem cuttings.

In 1923 I purchased with my own funds another lot, 1-1/2 bushels, of good heartnuts and sent them in lots of about two dozen to the secretaries of 125 horticultural societies, and to about 30 other parties for trial planting. I found that this little contribution was gratefully received and in many cases brought forth inquiries for the names of people from whom good trees might be purchased. I do not propose to carry on much more of this free distribution of nuts as that would not be fair to the individuals themselves or to those engaged in the propagation of nut trees. My chief reason for distributing these nuts was to stimulate interest, and now that my objective has been attained I will refer inquiring parties to reputable nut nurserymen.

Numerous requests for addresses on nut culture have been received from horticultural societies, women's institutes and other organizations. I have always endeavored to comply with these requests and have invariably found keen interest shown in the subject. American members of this association will likely be interested to learn that the Ontario Horticultural Society is the largest of its kind in the world, having a membership of over 60,000 while the Women's Institute is an almost equally large and influential organization.

These powerful and widespread organizations can be and are of great assistance in carrying on the propaganda for the planting of nut trees.

The Ontario Horticultural Association, the Ontario Horticultural Council and the Canadian Horticultural Council have each passed resolutions expressing approval of our work in nut culture and asking the Dominion Minister of Agriculture to appoint a man to fully investigate the nut cultural possibilities of Canada. I regret to state that no action has as yet been taken to meet the desires of these organizations. Because of many other urgent duties and lack of departmental support, I have not been able to devote as much of my time to nut culture as I would like, and therefore have had to make the very best use of the little time I have had at my disposal. I am looking forward to the time when those in authority will have a greater appreciation of the value of nut trees and will see their way clear to appoint someone to devote his whole time and energy toward increasing the productiveness and adding to the beauty of our country by means of more and better nut trees.

To sum up briefly, my objective is as follows:

1. To carry on the nut tree survey of Canada until we have located the very best natural and exotic species.

2. To propagate these best strains, provided they are as good or better than the best so far discovered.

3. To introduce the best hardy species from the northern United States and northeastern Asia, on a more extensive scale for test purposes and breeding work.

* * * * *

THE SECRETARY: Prof. Neilson has placed on the table in the hall, very modestly, a very interesting collection of nuts from Canada and I hope that you will all look at them.

THE PRESIDENT: Are there any present who would like to ask Prof. Neilson questions?

DR. MORRIS: It seems to me that the Ontario walnut is the best in quality of any I have tried. What did you think of them Mr. Jones?

MR. JONES: I do not think there is any better.

PROF. NEILSON: I am in favor of another one which I think you will agree is still better. It is larger and betterlooking and the flavor is just as good. (Displays walnut).

The interesting feature is that although the tree is a third generation tree, now about 15 years old, it has produced more nuts than the older trees.

DR. MORRIS: If I remember correctly the Ontario is a milder type.

PROF. NEILSON: I think that this is just as good as the Ontario. I have several trees of this.

THE PRESIDENT: From what I gathered from your remarks, Prof. Neilson, possibly some moral support would be of assistance to you in your work. Would it be out of order?

PROF. NEILSON: I think it would be a very good idea. The trouble I am having is perhaps very localized; it is with but one or two individuals. I think that a resolution by this association would have some effect. It would at least present to the authorities the fact that we were being recognized. I hope so at least. Our present Minister of Agriculture has openly expressed himself in sympathy with the idea of planting more nut trees; also Mr. Martin, our specialist in poultry keeping and I think if I can get them lined up it would be all right. The resolution might help to do this.

THE PRESIDENT: Dr. Morris the Chair appoints you to that committee; also Mr. Jones and Mr. Ellis. It wishes you to draw up a suitable resolution for that work.

PROF. NEILSON: I may say that the public in Canada is behind our work. About 97% of my time is spent on the road and I go long distances. The rest of my time I am writing letters, about 1,200 of them, and about 450 of these are on nut culture.

DR. MORRIS: I have the following resolution to offer: That a letter be written to the Dominion Department of Agriculture, along the following lines: "The Dominion Department of Agriculture has officially stated that the nut growing industry of British Columbia has become an important one. The Dominion nevertheless is importing \$5,000,000 worth of nuts annually from other countries.

In view of these facts, the Northern Nut Growers' Association in assembly at its 15th Annual Meeting, in New York, commends the work of Prof. J. A. Neilson of the Horticultural Experiment Station at Vineland,

Ontario, and expresses the hope that the Canadian Government and private support will further his work in such a way as to make it a matter of large public service. Service of the sort relates not only to eastern Canada but to the commerce of this entire continent."

(Signed) ROBERT T. MORRIS,
J. F. JONES
Z. H. ELLIS.

THE PRESIDENT: The secretary will accordingly transmit this message to the Canadian Government.

NOTES BY PROFESSOR A. S. COLBY

Purdue University, Illinois

Friends: I believe an apology is due you. I was away on my vacation at the time the invitation came to me to make an address at this meeting and I have come here without one. But I shall be glad to give you some sort of an idea of the past, present and future of nut culture in Illinois.

I became actively interested in nut growing about a year ago. Our work started partly in response to public demand. We have been receiving an increasing number of letters of inquiry from people interested in the subject but who know little about it. We are attempting to secure such information as will be of value regarding the best species and varieties of nuts to plant, where to plant them, and how to care for them. There are a number of members of the N. N. G. A. in Illinois and they are very kindly helping me in this work. The Illinois State Horticultural Society, founded in 1856, has also been interested to some extent in nut growing.

Illinois has had three grand old men in the nut industry, Mr. George W. Endicott of Villa Ridge, Mr. E. A. Riehl of Alton, and Mr. Benjamin Buckman of Farmingdale. Mr. Riehl is eighty-seven years young now and is the only one of the three men living.

Mr. Endicott was interested, not only in the commercial side of horticulture but was a pioneer in scientific work. He originated the Endicott plum and other valuable fruits and, since he was interested in plant improvement, naturally turned to hybridization of the chestnut, a tree which grows readily in southern Illinois. In 1899 he crossed the Japanese chestnut (*Castanea japonica*) with pollen from the American Sweet (*C. americana*). He must have had some difficulty in crossing the species because they did not bloom at exactly the same time. He was, however, successful in securing five hybrid seeds, raising three trees from them, naming them the Blair, the Boone and the Riehl. Naturally there were differences in the characteristics of these trees though they were all vigorous and produced nuts of commercial value. The Blair and Riehl began to bear at four and five years respectively, while the Boone bore its first crop at seventeen months of age. The Boone is the most valuable since it matures fruit of good quality about two days earlier than the Blair and two weeks before the Riehl. It also retains the burr and drops the nuts free at the beginning of the season so that about half the nuts can be picked up before the burrs fall.

Mr. Endicott was so pleased with the results of the cross that he raised over 175 seedlings from the Boone tree. From these second generation hybrids he secured trees very uneven in growth and size with a great range in time of coming into bearing. The nuts differed widely in size, quality, and season of ripening. The character of the burr showed all gradations between the extremes of thickness, length, rigidity of spines, etc. These striking variations in the second generation trees show that many hereditary factors had been segregated and recombined and offer a most interesting opportunity for scientific study. I have visited the orchard several times.

Mr. Endicott died in 1914 but his son Robert has since cared for the trees which have brought him considerable revenue. He tells me that he secures about 160 pounds of nuts per year from each of the three original trees. At

an average price of thirty-five cents a pound wholesale the crop from each tree is worth \$56.05 per year. Since the chestnut blooms late it is pretty certain to escape spring frosts. The Blair, for example, has had a crop failure once only since beginning to bear.

(Displays photographs of the Japanese and American chestnuts and the Boone tree).

Mr. Endicott is now working some of the worthless second generation trees with wood from the Boone tree.

(Displays photographs showing method of grafting).

I have had the good fortune to visit Mr. Riehl several times and have secured many representative nuts from his collection. While he has grown a large number of nut species and varieties he believes that the chestnut pays the best in southern Illinois. He plants them on rough and hilly land, difficult to cultivate, pasturing with sheep, and has had very good success. He does not worry about the chestnut blight, since the chestnut is not native here and there is such a great distance between the blight ridden East and Illinois.

Mr. Buckman was an amateur horticulturist, in the work for the love of it. On his land he had nearly two thousand varieties of apples and hundreds of varieties of peaches, plums, pears, cherries, grapes, small fruits, and nuts collected from all over the world. I was much interested to study the fine pecan and chestnut trees growing and producing good crops as well as the persimmon and papaw trees, of which he had a number of rare varieties. I was able last spring to secure cuttings of a number of rather rare papaw varieties which I sent to Doctor Zimmerman for propagation at the request of Doctor Fairchild.

Mr. Buckman recently died and there is now a movement on foot to secure, either through the University or the Horticultural Society, as far as possible, all the valuable data which he had been collecting for years.

There are several other men interested in nuts as a commercial proposition in Illinois, such as O. H. Casper of Anna and Judge W. O. Potter of Marion. I recently visited these orchards. Mr. Casper has mostly pecans and walnuts growing in sod. They are from six to eight years old and would have borne this season if weather conditions had been favorable.

Judge Potter has over twenty acres of pecans interplanted with chestnuts and filberts. For part of the orchard this is the fifth growing season. The trees are growing vigorously and make a very impressive showing. I counted thirty-nine nuts on a representative Thomas black walnut tree. The filberts look especially promising. Although the weather at blooming time was unfavorable a fair crop of nearly a peck was gathered from four or five bushes of a late blooming imported variety. Judge Potter is also growing another orchard using apples as fillers between black walnut trees. This experiment will be watched with great interest since it will be of great value in showing future possibilities in nut growing in Illinois.

Now as to some of the things we are trying to do at the experiment station at Urbana. This will be necessarily a progress report. I am making a survey of the state to find promising individuals of the different species and varieties and marking them for future use. We have our state fair at Springfield next week and as I speak to the boys and girls attending the state fair school I hope to interest them to tell me of any trees in their neighborhoods of particular value.

Some of the agricultural leaders in the various counties, that is the farm advisers, are awake to the value of the nut industry and we have a number of these men co-operating with us. From Gallatin County, in the Wabash

and Ohio river bottoms, around \$100,000 worth of native pecans are sold in some seasons. In the southern counties and over north of St. Louis in the western part of Illinois there are also native pecan groves which are quite profitable. We hope to find valuable northern pecans, adaptable to our conditions. We, of course, know that the English walnut is very difficult to grow in Illinois and we are not recommending it as a commercial proposition. We believe that the black walnut, all things considered, has the most promise and we hope to have something worth while in a few years as propagating material. The Thomas, Stabler, and Miller are especially to be recommended for Illinois at this time.

We hope soon to have a complete collection of hardy nut trees on our experimental trial grounds. Here we shall study not only the varietal characteristics but try out new methods of propagating, pruning, fertilizing, etc. There is very likely some connection between winter injury and hardening up of the wood in autumn and we hope to learn something about that problem through the use of various cover crops, for example. We have at the station a complete experimental cold storage plant in operation where we may be able to learn more about the effects of extremes of temperature on the roots and trunks of certain species.

In such new but important work we must make haste slowly. We have some things to unlearn and many things to learn. We hope to be able in a few years to make a worthwhile contribution to such an interesting and important subject as nut growing in the middle west.

I shall be glad to have you ask me any questions which occur to you.

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THE PRESIDENT: DO you happen to know Mr. Spencer?

PROF. COLBY: No, I wrote Mr. Spencer but I did not get any reply from him. I hope to visit him this fall.

MR. REED: DO you know anything about the top-working of black walnuts from Missouri at the university?

PROF. COLBY: No, I do not know about them.

MR. GREEN: In regard to those Gallatin County nuts; has any survey ever been made by the U. S. Department of Agriculture of the nut trees in Illinois?

Prof. Colby: Not that I know of.

Question: At what age are they planting those walnuts in Williamson County with apples and how far apart?

PROF. COLBY: The walnuts are from 50 to 80 feet apart interplanted with apples. The walnut trees are about two years old; the apples four and five.

A SPEAKER: I believe those apple trees will die.

PROF. COLBY: That's what I want to find out. There is a great difference of opinion as to the compatibility of walnuts and other fruit trees.

MR. BIXBY: You will see at Baldwin, this afternoon, peach trees planted between nut trees. It is too soon to say what will happen but so far, it is all right.

DR. SMITH: As a matter of very great importance, how will you "round up" the forces in Illinois?

PROF. COLBY: We have a number of interesting suggestions brought out in Professor Neilson's paper. He would use every way possible, including questionnaires sent out judiciously, as well as the boys' and girls' clubs, and the Boy Scouts, of which Dr. Morris speaks. The horticultural society can be of very great help. In Illinois where we have over one hundred counties, almost all of which are very efficiently covered by farm bureaus, the farm advisers are of considerable assistance. The local horticultural societies, as for instance the one with which Mr. Riehl has been so prominently connected in Alton, have helped very much in the past. The Smith-Hughes teachers in charge of agricultural teaching in the high schools can easily get in touch with promising native trees through their students. I know most of these teachers and know they will be glad to help me. I recently had a request from the Associated Press representative in Springfield to write an article on nut growing in Illinois. There is a wonderful field for development along such lines as this.

THE PRESIDENT: It seems to me that if the agricultural colleges were asked to hand in information that might bring results, and particularly the students' work in isolated sections which would not be reached by Boy Scouts.

PROF. NEILSON: For the benefit of those who did not hear my address in 1922, I may say that I have circularized the whole county and the college stations; I have sent about 125 circular letters to the horticultural society and to its officers, high school inspectors, and to anyone I thought might be glad to get the information. I wanted to carry this further but could not. I wanted to send letters to every school teacher in the Province of Ontario and ask them to bring the matter to the attention of the boys and girls, and to offer them a substantial prize for the location of the best tree in their locality. I will say, however, that I got a great deal of encouragement from the horticultural society, the public school and the high schools.

THE SECRETARY: I will read again a sentence from Mr. Howard Spence's letter:

"The Minister of Agriculture has agreed to instruct all their inspectors over the country to make a collection of all walnuts of merit and to forward them to me for classification and identification of varieties which may be worth perpetuating."

If we could do something of that kind in the United States to enlist the extension agents, we should get some valuable information.

MR. OLCOTT: I think that a very important thing would be to send that message not only to the state experiment stations, but also to the government authorities. Why should not the Department of Agriculture make a systematic survey of that kind? Why should it be left to the small societies like this one, when the federal Department of Agriculture is so thoroughly equipped to get this? The department at Washington has expressed interest; I wonder if it would not be appropriate for this association to take some formal action, suggesting federal government action in that matter, in co-operation with the extension service, Boy Scouts, etc.

THE PRESIDENT: Will you put that in a resolution?

MR. OLCOTT: I submit the following resolution:

WHEREAS, The investigational and experimental work of the Northern Nut Growers' Association during the last fourteen years has been signally successful in improving native nuts of the northern United States, based upon discovery and propagation of superior specimens; and

WHEREAS, This work could be greatly extended with the facilities at the command of the United States Department of Agriculture, as compared with the efforts of the small number of members of this association; therefore be it

RESOLVED: That it is the sense of the Northern Nut Growers' Association, in fifteenth annual convention in New York City this fourth day of September, 1924, that the U. S. Department of Agriculture be asked to take up systematically the work of discovery and investigation of promising native nuts in the northern states and of testing selected specimens at government stations in co-operation with the authorities of the state experiment stations; such discovery to be brought about by enlisting the aid of boy scouts, school children and others, in connection with the activities of county farm agents, inspectors and other attaches of the department.

THE PRESIDENT: Prof. MacDaniels, of Cornell University will now address us.

L. H. MacDaniels, Professor of Pomology, Cornell University

It gives me great pleasure to bring you greetings from the Agricultural College at Cornell University and to express my appreciation for your invitation to address this convention concerning what the college is doing along the line of nut growing. I have a very real interest in nut growing and in this association. I like to think of it as comparable with the American Pomological Society when it started more than one hundred years ago. All of you men who are spending your time and energy in finding new facts regarding the propagation and culture of nut trees are doing pioneer work,

and your names will go down in the history of nut growing in the same way as those of Wilder, Downing, and Prince have come to us linked with the early development of fruit growing in the United States. I feel confident that the work of the association will stand the test of time.

Interest in nut growing at Cornell, as you probably know, was started by John Craig who died about a dozen years ago. He was greatly interested in northern nut growing and also in southern pecans. As a result of his work we are still receiving inquiries about southern pecans addressed to Professor Craig. While at Cornell he established a course of study in nut growing which was a part of the regular curriculum. At the time, however, the actual known facts about the growth of nuts in the northern states were so few, and reliable information so scarce, that after Professor Craig's death, when there was a general consolidation of courses in the department, nut growing was combined with another course in economic fruits. Since that time, as our knowledge of nut growing has increased, more and more attention has been given to the subject. Our aim is, in fact, to give all of the up-to-date information that we have regarding the propagation and culture of nut trees.

The nut tree plantings in the experimental orchards at Cornell have not been particularly successful. About ten years ago Professor Chandler set out about one-half acre of named varieties of pecans, Persian walnuts, black walnuts, hickories, hazel nuts, chestnuts and Japanese walnuts. These have received good care, both as to cultivation and fertilization but to date the only trees which have borne are the Japanese walnuts and these have not had good crops. Apple trees of the same age in adjacent land have been bearing commercial crops for a number of years, especially such varieties as the McIntosh, Wealthy and R. I. Greening. The climate at Ithaca is apparently rather too rigorous for most of the nut trees. Persian walnuts, hazel nuts and frequently Japanese walnuts suffer from winter injury. In the case of the chestnut, blight has practically killed all of the trees. The pecans

are perfectly hardy but as yet have not borne, probably because our seasons are not sufficiently long or warm enough to grow this nut to advantage. Hickories have been very slow to become established and in fact have never made really good growth. This experience, of course, makes us feel that nut growing is really not as easy as some enthusiasts would have us believe.

In addition to this variety planting there are four or five acres of recently cleared woodland in which there are hundreds of hickory seedlings which can be top-worked. We are aiming also in this area to establish seedlings of all of the hardy nut trees to use as stocks and eventually to get a collection of all named varieties of nut trees. Grafting so far has not been particularly satisfactory due in some cases to failure of the grafts to set; in other cases to the winter killing of grafts which have made fairly good growth. Injury by bud moths and wind storms have also been detrimental factors. Our own experience together with observations upon the results of nut grafting elsewhere by experts lead us to believe that grafting of nut trees is a very difficult undertaking as compared with that of other fruit trees. It involves a knack which must be acquired by very considerable experience. I realize, of course, that new facts regarding nut grafting are being discovered almost daily and in the future we may look for better results.

The attitude of the Department of Pomology at the College with regard to nut growing is of necessity conservative. First of all, the men in the department are trained in scientific methods and have a somewhat critical attitude when it comes to statements regarding marked success in any line. The tendency is in each case to try to find the data or the experience upon which statements are based. Unfortunately, in nut growing there are very little data upon which statements can be based. Mr. Bixby's experiments with stocks are a very good start in the right direction, and it is upon such experiments as he is carrying out that real knowledge regarding nut growing will be gained.

We have heard enthusiastic statements as to the profits which may be derived from the planting of nuts in the northern states, but I must confess that I have looked in vain both for the facts upon which such statements might be based and also for orchards which actually are profitable. If such exist in New York state I have not been able to find them even after considerable travel.

In order to be profitable, an orchard must pay all the expenses involved, including interest on the initial cost of land; the cost of labor and materials and depreciation on tools, etc. We have cost accounts covering these items on many crops such as apples and wheat, but not on nuts. It seems to me we must recognize that nut culture is in its experimental stage only. This is in fact one thing that makes it particularly attractive for the amateur.

Another reason for our conservatism is that we feel it our duty to the growers to give out statements which are based upon facts only. If a man in a northern state wants to plant ten acres of nuts what shall we tell him? Shall we tell him to go ahead and assure him that if he takes care of his trees a profitable plantation is certain? On the basis of what we know I think surely not. A hundred and one unanswered questions come up. What kinds of nuts will succeed under his climatic and soil conditions? What stocks should be used? What varieties will succeed under his conditions? Will the meats of the nuts fill out in the average season? Are the seasons long enough, etc. The fact is in most cases we do not know. In most parts of New York state we are extending a natural range of many of the nut trees and they have not been grown long enough under the new conditions to make it possible to answer these questions with certainty. On the other hand, we can tell the prospective nut grower that nut growing is in its experimental stages and under certain conditions has great commercial promise. On the basis of our present knowledge we cannot recommend large plantations but would encourage the planting of nuts in an

experimental way, especially for home use. It should be borne in mind that in the early days of fruit growing in America it was the amateur planting of varieties that laid the foundations for the present industry. If shade trees are to be planted let them be nut trees. Plant nut trees as a hobby but do not go into nut culture on a large scale for profit unless you can afford to lose.

I have great hopes for the future of nut growing in the northern states and also for this society. I am confident that new and better varieties of nuts will be found and better methods of propagation and transplanting originated so that in the future there may be a commercial industry in the north. For the present, however, I believe that conservatism is advisable, and that great harm may be done by misrepresentation. Sound growth of a northern nut industry will be built upon facts and honest experience and not on conjecture, hearsay, or even on enthusiasm, however necessary this may be. I believe that we should encourage people to plant nuts for pleasure, plant nuts as a hobby, plant them for shade and for posterity, but under present conditions not for financial profit.

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THE SECRETARY: We must adjourn at once to the lecture room, that we may hear Dr. J. Russell Smith's talk on "Nut Tree Crops as a Part of Permanent Agriculture without Plowing." He will have some interesting slides to show during his talk.

Dr. Britton has asked that we have lunch today at noon instead of one o'clock. Everyone present is invited to take luncheon at that time as a guest of the Botanical Society and of Dr. Britton, it makes no difference whether they be members or guests.

MR. REED: May I make the motion to extend a rising vote of thanks to Dr. Britton and his associates for the cordial and generous way in which

they have entertained us?

(Motion seconded, passed, and acknowledged by rising vote).

THE PRESIDENT: Dr. Britton, you are officially notified.

DR. BRITTON: I would like to have that vote of thanks mentioned in the official record of this convention, and in the record of the Botanical Society.

THE SECRETARY: We will see to that.

DR. BRITTON: You will be interested in knowing that we have with us the very distinguished Curator of the British Botanical Herbarium of the Royal Society. Dr. Stapf has been traveling in Canada, attending the meetings of the Royal Society there.

THE PRESIDENT: We shall very much appreciate the opportunity of meeting him.

We will now adjourn to the lecture hall, to hear Dr. J. Russell Smith.

NUT TREE CROPS AS A PART OF PERMANENT AGRICULTURE WITHOUT PLOWING

Dr. J. Russell Smith, Professor of Economic Geography, Columbia University, New York

My first experience with nut culture was gained on the farm of a man I knew more than 30 years ago. It was a truck farm not far from Philadelphia near a boarding school which I infested and the farmer complained that I infested the farm. The farm had its fence rows and driveways lined with grafted chestnut trees bearing abundantly of large fine nuts of European origin. It was remarkable how quickly they filled my pockets. I usually succeeded in gathering them on the hundred per cent basis.

I am interested in this subject today because of an innate love of trees and because the development of a tree crop agriculture offers a way to stop soil erosion. To me the worst of all economic sins is the destruction of resources, and the worst of all resource destructions is the destruction of the soil, our one great and ultimate resource. "After man the desert" has been truly said too often of many old lands.

Soil cover is after all about the only thing that man has as a basis for the support of his life on earth. All of our food depends directly or indirectly

upon plants.

In hilly countries there is usually but a thin layer of earth and rotten rock between the surface of the field and the bed rock. It is a very difficult problem to maintain this cover of earth and it is very easy to lose it. Sometimes it is lost through over-pasturing and destruction of turf; but more largely through plowing.

The nut tree is particularly effective as a part of a plowless agriculture which can use the soil permanently where annual crops ruin it quickly because the plow prepares the land for erosion.

The speed of soil destruction, with its erosion after plowing, is particularly noticeable with the great American crops, cotton, corn and tobacco, which require clean cultivation. Many orchards are also cultivated for the double purpose of keeping down rival plants and preserving moisture, but we pay high in soil loss for the moisture that we get by that means on hilly lands. The plow is one of the greatest enemies of the future. As a matter of fact we have already destroyed enough land in the United States to support many millions of people; and therefore the tree is the more important because it permits an agriculture that will keep the soil indefinitely, and in permanent production, without plowing.

I have accidentally discovered a better way of conserving moisture than by plowing, and I have found it going on in widely scattered places and in widely different climates.

Primitive peoples in many parts of the world have long since obtained the advantage of cultivation, mainly, increasing the available moisture for the tree or plant, without cultivation of the soil and the loss which follows the washing of cultivated soils. As an example I cite the Indians of Arizona, who have grown corn crops for centuries in a country with but from six to

fifteen inches of rain. They do this by planting in little patches at the mouth of a gully where at the time of rain the flood water is led away into furrows and depressions so that it thoroughly soaks the ground in which the corn is planted.

My attention was first called to this practice by observing a good patch of barley in the edge of the Sahara in Southern Tunis, where the gully flow resulting from a winter rain had spread itself out fan-*like and soaked the triangular alluvial area of sand, which bore a fine crop of barley in the midst of the desert.

For centuries the olive growers of parts of Tunis have led gully water to the olive trees where it was retained, in areas that resembled a tennis court, with a 12 inch bank of dirt around it and two or three olive trees within this area thus watered by impounding.

A practice somewhat similar to this is shown in F. H. King's classic book on Chinese agriculture, "Farmers of Forty Centuries;" but the most extreme case that has come to my attention is furnished by the Berber tribe of the Matmatas, of Tunis. These people live on the edge of a hilly, limestone plateau, where the rainfall is less than 10 inches and in some years as low as five.

An important part of the food supply of these people is furnished by date and olive trees which they grow in the gulches of their limestone plateau. They built a dry rock dam behind which earth-wash lodges. In this the trees are planted and every rain sends more earth and soaks that which has collected. The plan can certainly not be called an experiment for the people have lived there for centuries. They have olive trees that are several centuries old and I have never seen such fine olive trees, not in California, or the plains of Spain, Portugal, France, Italy, or in Algeria or Tunis, and I have seen a good many olive trees in those countries. The olive tree is

usually open, light and feathery. These in the Matmatas gulches are thick and black and rank.

For automatic cultivation and fertilization the plan of these primitive agriculturists is hard to beat. You put up your stone dam, and every time the gulley runs with water your crop is irrigated and fertilized. Can you beat it?

Three Americans of my acquaintance have independently experimented and discovered along similar lines.

The late Freeman Thorpe of Hubert, Minnesota, did it with much enthusiasm. So did the late Dr. Meyer, a friend of J. F. Jones, near Lancaster. He discovered it accidentally. He put a brush dam across a gully. Water stood behind it for days after every rain. The apple tree near it grew much more than the others. That started the Doctor. He began to dig small field reservoirs and collect water near trees and he found that it paid even with the very expensive process of hoe and shovel.

The idea has been modernized and brought to the machine stage which characterizes our present-day agriculture, by Mr. Lawrence Lee, a civil engineer-farmer of Leesburg, Va. Mr. Lee runs a level line across the face of the clay hills, and then with a Martin ditcher scoops out a terrace on this horizontal line. It makes the terrace so that the water will hold and will not run away. Mr. Lee is sure that nine-tenths of the heavy thunder shower runs off of the hills, in normal conditions of non-plowing, and that if he plows, most of the water and much of the soil go off together. He is also sure that the water pockets hold both water and soil.

Rows of apple trees planted below these waterholding terraces thrive without cultivation as well as do other trees across the row with cultivation, but with this difference, ordinary cultivation impoverishes the soil and this enriches it by keeping all mineral and organic matter in the field.

The combination of principles worked out by many primitive peoples and also by Messrs. Thorpe, Meyer and Lee makes it possible for the farmer to arrange his rough land in tree crops so that every rain will water his crops, even though the land may be rough and in sod. If he cannot run horizontal terraces he can dig holes near the trees and lead the water to these holes by two furrows with the turning plow. This is really an automatic kind of irrigation. By this means a farmer can use his odd time whenever he can work the ground, and thus do the cultivation for a whole year or two and at the same time preserve the soil and establish a permanent agriculture.

This gives the hill land the same chance as the level lands to grow fat sods. It offers a very interesting combination of blue grass pasture along with crops of black walnuts, Persian (English) walnuts, pecans, grafted hickories, mulberries (for pigs and chickens), persimmons (for pigs and sheep), oaks (which make more carbohydrate food than corn in many situations), honey locust (which has a bean as rich as bran and good for the same purpose) and many other crop trees that will be available if good brains keep developing the idea.

In this connection it may be pointed out that France exports millions of dollars worth of Persian walnuts and most of them are grown on isolated trees scattered about the fields and along roadsides.

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THE PRESIDENT: We will now adjourn to Sormani's for luncheon and then we will immediately start for Mr. Bixby's place on Long Island.

(Adjournment).

NOTES AT MR. BIXBY'S NUT ORCHARDS AND NURSERIES BALDWIN, NASSAU CO., N. Y.

September 4, 1924

Japan walnuts (seedlings) on street set out in 1918 or 1919. All except the tree on the south have borne, 1924 being the third year for one. One of them is a heartnut.

Chinkapins raised from seed outdoors.

Black walnuts grown in pots and transplanted with a ball of earth and the entire root. Set out without cutting back and sod and vines allowed to grow around them. While they grew rapidly before transplanting they have scarcely grown since.

Beaver Hickory seedlings. These illustrate well the information to be obtained frequently as to parentage by raising seedlings. The history of the Beaver tree was ascertained four or five years ago and from this and the appearance of the tree and its nuts, it was decided to be a shagbark x bitternut hybrid. The seedlings bear this out, for they vary from seemingly pure shagbark to pure bitternut with several in between looking somewhat

like the parent tree. It may be that some of these will bear nuts that will be found valuable.

Japan walnut tree killed with butternut blight.

Chestnut trees killed with chestnut blight.

Main experimental orchard. This comprises about four acres and is laid out in rows running north and south, starting at an east and west road. There are 29 trees in each row running north and south, the trees being about 15 feet apart. A nut tree is put every 30 feet and a peach or apple or some other tree that is intended to be taken out later, is put in between.

Row 1 South—(1) Niblack Pecan (5) Warrick Pecan (7) Warrick Pecan (9) Greenriver Pecan (11) Greenriver Pecan (13) Mahan Hickory (15) Marquardt (?) Pecan (17) Siers Hickory (19) Wilkinson (?) Pecan (21) Kirtland Hickory (23) Greenbay Pecan (25) Weiker Hickory (27) Burlington Pecan (29) Kentucky Hickory. This Kentucky Hickory blossomed full and some two dozen nuts set which grew to about 5/8 inches long then they dropped off. Probably it will bear next year.

Row 2 South—(4) Moneymaker Pecan (10) Pleas Hickory (24) Dennis bitternut, bearing (26) Hatch Bitternut (?).

Row 3 South—(3) Stanley Hickory (5) Ridenhauer Almond (9) Burkett Pecan (11) Hales Hickory on shagbark (13) Hales Hickory on bitternut (21) Cedarapids Hickory on shagbark (23) Cedarapids Hickory on bitternut (25) Dennis Hickory (27) Fairbanks Hickory.

Row 3A South—Seedling Black Walnuts.

Row 3B South—Seedling Chinese Chestnuts.

Row 3C South—Seedling Chinese Chestnuts.

Row 4 South—(2) Rush Chinkapin (3) Miracle Chestnut (4) Chinkapin (7) Chinkapin (8) Chinkapin (9) Champion Chestnut (10) Paragon Chestnut (13) Riehl Chestnut (15) Paragon Chestnut (16) Paragon Chestnut (17) Miracle Chestnut (22) Champion Chestnut (29) Boone Chestnut. The above trees are all that remain of a row of 29 Chestnut and Chinkapin trees most of which were bearing two years ago, from which a good many quarts of Chestnuts were gathered. Some of them died in 1922 and more in 1923.

Row 5 South—(1) Beaver Hickory (2) Hacheye (?) Persimmon (3) McCallister Pecan (4) Hayakuma Persimmon (5) McCallister Pecan (6) Kawakami Persimmon (7) Busseron Pecan (9) Busseron Pecan (10) Lambert Persimmon (11) Butterick Pecan (12) Josephine Persimmon (13) Butterick Pecan (15) Kentucky Pecan (17) Kentucky Pecan (18) Golden Gem Persimmon (bearing) (19) Indiana Pecan (20) Rush Chinkapin (21) Indiana Pecan (23) Posey Pecan (25) Posey Pecan (27) Major Pecan (28) Parry Chestnut (29) Major Pecan.

Row 5A South—Pecan seedlings.

Row 5B South—Shellbark seedlings.

Row 6 South—(1), (2), (3), (4), (5), (6),-(7), (8), (9), (10), (11), (12), (13), (14), (15), (17), (18), (19), (20), (21), (22), (23), (24) Hales Hickory, transplanted some years ago, brought from Monticello, Florida (25) Kentucky Hickory.

Row 6A North—Butternut seedlings.

Row 6B North—Butternut seedlings.

Row 7 South—Vest Hickory seedlings, Hales Hickory seedlings, *Juglans cathayensis* seedlings, Chinese Persian walnut seedlings, Papershell Chinese Persian walnut seedlings, Hybrid hazels (native Long Island x Italian Red 1923).

Row 7A South—Mockernut seedlings.

Row 7B South—Mockernut seedlings.

Row 7C South—Close bark pignut *carya glabra* seedlings. Loose bark pignut *carya ovalis* seedlings, Japan walnut seedlings, Adams Black Walnut seedlings.

Row 7D South—Persian walnut seedlings, Stabler Black Walnut, perfect form seedlings, Stabler Black Walnut, one lobe seedlings.

Row 7A North—Japan Walnut seedlings.

Row 7B North—Japan Walnut seedlings.

Row 7C North—Japan Walnut seedlings.

Row 8 South—8A South—8B South—8C South—Seedling Japan Walnut x butternut hybrids.

Row 8A North—Japan Walnut seedlings.

Row 8B North—Japan Walnut seedlings.

Row 8C North—Persimmon seedlings.

Row 9 South—(1) Miller Black Walnut (3) Thomas Black Walnut (4) Purple Hazel (5) Thomas Black Walnut (6) Fruhe Lange Hazel (7) Stabler Black Walnut (9) Kinder Black Walnut (11) Allen Black Walnut (13) Wasson Black Walnut (15) Peanut Black Walnut (17) Ten Eyck Black Walnut (19) Mattingly Black Walnut (21) McCoy Black Walnut (bearing) (23) Paradox Walnut (25) Ohio Black Walnut (bearing) (27) Herman Black Walnut (29) Stabler Black Walnut.

Row 10 South—(2) Stranger Heartnut, bearing (4) California Black Walnut (6) Seedling Allen Black Walnut (8) Seedling Allen Black Walnut (10) Seedling Allen Black Walnut (12) Casper Hickory (14) Casper Hickory (16) Reike Hickory (18) Vest Hickory (20) Swaim Hickory (22) Swaim Hickory (23) Jordan Almond (24) Wampler Hickory (25) Jordan Almond (26) Wampler Hickory (27) Texas Prolific Almond (29) Texas Prolific Almond.

Row 10C North—Hickory Seedlings. Here may be seen the melancholy results of not planting hickory seedlings deep enough.

Row 11 South—(1) Aiken butternut, bearing (3) Stranger Heartnut, bearing, (5) Ritchie Heartnut, bearing (7), (9), (11), (13), (15), (17), (19), (21), (23), (25), (27), (29) Lancaster Heartnut bearing.

Row 11A South—Grafted and budded black walnuts.

Row 11B South—Grafted and budded black walnuts.

Row 11C—South—Grafted and budded butternuts and Japan Walnuts.

Row 11 North—(1), (2), (3), (4), Aiken butternut (6) Juglans mandshurica (8), (10) Deming butternut.

Row 11A North—Seedling Japan walnut x butternut hybrids.

Row 11B North—Seedling Japan Walnut x butternut hybrids.

Row 11C North—Seedling Japan Walnut x butternut hybrids.

Row 12—(2) Faust heartnut, bearing (4) Deming butternut, bearing (8) Burlington Pecan (10) Rockville Pecan (20) Snyder Hickory (27) Early Golden Persimmon (28) Rockville Pecan (29) Ruby Persimmon.

Row 12A South—Grafted and budded Black Walnuts, Stabler, Ohio, Thomas & Adams.

Row 12B South—Grafted and budded Black Walnuts, Wasson, McCoy, Ten Eyck, O'Connor hybrid Witte Persian Walnut.

Row 12C South—Grafted and budded butternut & Japan Walnut, Aiken butternut, Lancaster Heartnut.

Row 13 South—(1) Franquette Persian Walnut (3) Eureka Persian Walnut (4) Early Golden Persimmon (5) Holden Persian Walnut (7) Eureka Persian Walnut (8) Grosse Kugelnuss filbert, bearing (9) Holden Persian Walnut, bearing (10) White Lambert hazel (11) Alpine Persian Walnut, bearing (12) Italian Red Hazel (13) Lancaster Persian Walnut (14) McFarland Chestnut (15) Meylan Black Persian Walnut (16) Hale Persimmon (17) Rush Persian Walnut, bearing (18) Imperial Hazel (19) Cording Walnut, bearing (J cordiformis x regia) (20) Early Golden Persimmon (21) Hall Persian Walnut (22) Yemon Persimmon (23) Paradox walnut (24) Yemon Persimmon (25)

Mayette Persian Walnut (26) Floreams Almond (27) Holden Persian Walnut (28) Floreams Almond (29) Mayette Persian Walnut.

Row 13 North—Chinese Almond so-called, 3 years old, really an apricot with edible kernels. Has proved perfectly hardy so far.

Row 14—Grafted and budded black walnuts, Boston Persian Walnut. O'Connor hybrid Walnut, Adams Black Walnut, Alley Black Walnut, Mosnat butternut.

Row 15—Grafted and budded Black Walnuts, O'Connor hybrid, Thomas, Stabler. Ohio Persian Walnut. Minnas Zeller Italian Red Hazel, bearing.

Row 16—American Hazels from West Virginia and Ohio.

Row 17—Landesberger Lange Zeller, Buettners Zeller, Hempels Zeller, Barnes No. 6, Hazel bearing hybrid nuts, Barnes No. 5 Hazel bearing hybrid nuts, Kentish Cob, Noce Lunghe filbert, Daviana Hazels, both bearing.

Row 18—Merveille de Bollwiller filbert bearing, Medium long filbert. Like Merveille de Bollwiller, Althaldestenbener Zeller.

Row 19—*Corylus californica*, White Lambert filbert, Vest hazel, Grosse Kugelnuss, Hallersche Riesen filbert. Barcelona filbert, Italian Red filbert, Du Chilly filbert.

Row 20—Long Island Hazel, bearing Blueberries. 8 plants of selected varieties, Jujube, Tree hazel, *corylus colurna*, Vest hazel bearing hybrid nuts, Daviana hazel bearing, White Aveline hazel, tree hazel, *corylus colurna*. Long Island hazel bearing, Red Aveline hazel bearing.

Row 21—*Corylus californica*, tree hazel *corylus columnata*. On the southern end of these rows will be found the grafted hickories.

Row 21—Grafted Shagbark hickories.

Row 22—Grafted Mockernut hickories.

Row 23—Grafted Mockernut hickories.

Row 24—Grafted Pignut hickories.

Row 25—Grafted Pignut hickories.

Row 27—Grafted Pecan hickories.

Row 28—Grafted Pecan hickories.

Row 30—Grafted Bitternut hickories.

Row 31—Grafted Bitternut hickories.

Row 32—Grafted Bitternut hickories.

Row 33—Grafted Bitternut hickories.

Row 31—Grafted Bitternut hickory.

Additional Notes by Stenographer

This is a Royal Burbank walnut brought from California, in 1911. It stood in a yard in Brooklyn until 1917. It did not grow well there but since we have brought it out here it is growing and bearing, as you see. It is a hybrid of the California black and the Eastern black. The nut itself has not

much value. The leaves are rather smaller than others. It would not compare with the propagated varieties. It is only considered as a rapid growing tree.

Here is a row of Beaver seedlings. This one is a typical shagbark. This one is like a bitternut. Every once in a while you will find a tall one with buds like the old tree. They are all Beaver seedlings from nuts gathered at the same time from the same tree.

Here are chinkapin seedlings grown out of doors. I simply threw them on the ground and covered them with leaves.

Here is a dead Japanese walnut tree. It died of a fungus, melanconium. You can see the fungus all the way down the trunk. It is a weak fungus and sometimes if the tree is nourished properly it will disappear.

This is a Lancaster heartnut. And so is this. One is much more prolific than the other. Both grafted on Japanese stock. It is bearing pretty well. It was put out in 1918.

Here is a Kentucky hickory. It had about 24 nuts, but they have fallen off.

This is a Moneymaker pecan. It is growing finely. I bought this tree from J. B. Wight, of Cairo, Ga. I also have a Burkett from Texas.

There is a Paragon chestnut which has escaped the blight. Fungus is beginning on the end of the branch, however.

Two years ago we had a whole row of these Boone chestnuts. This is the only one left. They were all in bearing then and a good many quarts of chestnuts were gathered. Some of them died in 1922 and more in 1923.

From here up, the trees are hickory (Hales) on pecans. They are ten years from the graft, and planted here from Monticello, Fla., two years ago. 23

out of the 24 trees living.

There are 12 varieties of Japanese persimmons, bought from Texas. This one shows winter-killing but will apparently live. (Hayakuma persimmon).

Here is a Jap. persimmon (Kawakami). It has not borne yet. Here is a McCallister pecan; originated from between the Wabash and Ohio Rivers.

Those are Thomas black walnuts; they have been out five years, and have not yet borne.

This is a Ten Eyck; it has made good growth this year and is a heavy bearer. This is a McCoy black walnut. This tree is bearing heavily this year, and bore one nut last year. It is about five or six years from the nursery. The parent tree is from near Rockport, Ind., and is a very large one.

Here is an Ohio; it came from Mr. Jones, I think. These trees are bearing heavily; they have been set out 5 or 6 years.

These trees are Lancaster heartnuts. They will probably bear heavily one year and less the next.

(Here catkins and nuts were found on the same branch, and a photograph was made).

MR. REED: There will probably not be any Lancaster here next spring; the late growth has devitalized the tree.

Here is a California black walnut but it has not grown very successfully.

Here is a Stranger heartnut from South Carolina, bearing.

Here is an O'Connor hybrid walnut on black walnut. The whole tree is 3-1/2 feet high; splendid growth for one year. The parent tree is in Maryland,

about two miles from Mr. Littlepage's place.

Here is a Lancaster heartnut which has borne every year, without a stop; you see it is planted in a chicken yard.

EXHIBITS AT THE HOUSE OF WILLARD G. BIXBY, BALDWIN, N. Y.

September 4, 1924

BLACK WALNUTS

Varieties:

Adams
Alley
Herman
McCoy
Miller
Ohio
Stabler, Perfect Form
One Lobe
Ten Eyck
Thomas
Wasson

Species:

Juglans major, Arizona rupestris,
Texas boliviensis, Bolivia
insularis, Cuba
The extremes of black walnut

shape. Adams, long and
narrow, Corsan, short and
broad

Varieties: Butternuts

Aiken

Deming

BUTTERNUTS AND JAPAN WALNUTS

Varieties: Japan Walnuts

Heartnuts

Lancaster

Ritchie

Stranger

Species:

Juglans cinerea

manshurica

cathayensis

sieboldiana

cordiformis

Rough shell Japan walnut

Juglans sieboldiana x

cinerea

Juglans sieboldiana x

nigra

Cording, *Juglans cordiformis* x

regia

Nuts from 4 trees on Grand Ave.

Baldwin

CHESTNUTS

Varieties:

Boone

Paragon

Rochester

Morris No. 2

Morris No. 3

Species:

Chinkapin

Castanopsis

HAZELS AND FILBERTS

Varieties:

Althaldensleben

Barcelona

Daviana

Du Chilly

Emperor

Grosse Kugelnuss

Imperial

Italian Red

Merveille de Bollwiller

Montebello

Noce Lunghe

Red Aveline

Red Lambert

Rush (American)

Vest (American)

White Aveline

White Lambert

Species:

Chinese tree Hazel (*Corylus*
 chinensis)
Constantinople Hazel (tree
 corylus columna)
Thibet Hazel (*Corylus tibetica*)
Hazel Blight (Specimen)

HICKORIES

Varieties:

Beaver
Brooks
Dennis
Fairbanks, Parent tree
 Grafted tree
Galloway
Glover
Griffin
Hales
Kirtland
Laney
Milford
Pleas
Siers, Parent tree
 Grafted tree
Vest
Weiker, Parent tree
 Grafted tree

It will be noticed that nuts
from young grafted trees are
generally larger than those

from the parent trees

Species and Hybrid:

Arkansas Hickory, *carya buckleyi*

Arkansana

Bitternut, *carya cordiformis*,

Dennis, Hatch

Buckley Hickory, *carya Buckleyi*

Chinese Hickory, *carya cathayensis*

Pallid Hickory, *carya pallida*

Shellbark, *carya laciniosa*, from

3 locations

Water Hickory, *carya aquatica*

Zorn, the largest hickory yet

found, *carya buckleyi* Arkansana

x alba

PECANS

Northern Varieties:

Burlington

Busseron

Butterick

Campbell

Greenriver

Indiana

Koontz

Major

McCallister

Niblack

Norton

Posey

Witte

Species and curiosities:

Seedling Pecan from Adams,
Ill. The most northern native
growing pecan yet seen
by Willard G. Bixby
Curtis Pecan, without inner
shell partition
Schley Pecan, one grown in
Georgia, the other in southern
Pennsylvania. This
shows how the nuts are
dwarfed by lack of sufficient
summer heat

PERSIAN WALNUTS

Varieties:

Alpine
Boston
Colona
Franquette
Hall
Holden
Hutchinson
Lancaster
Mayette
Milbank
Ontario
Pomeroy
Rush
Sayre
Witte

Seedlings and Hybrids

Chinese Paper Shell

Juglans regia x cinerea from
2 locations

Allen, juglans regia x rupestris

MISCELLANEOUS

Almond, Ridenhauer

Chinese (edible apricot)

Beechnuts, American (2 locations)

European

Queensland Nut Macadamia
ternifolia

Water Chestnuts:

Nelumbium Luteum

Nelumbium Speciosum

NOTES TAKEN AT MERRIBROOKE, DR. MORRIS' ESTATE NEAR STAMFORD, CONNECTICUT

Excursion of Friday, September 5, 1924

Arriving at Stamford, all guests and members were met at the station by cars from Dr. Morris' place. After coming together at the house, the members followed Dr. Morris to the main gateway, where the following program commenced:

DR. MORRIS: If you will all follow me here inside the gateway we will take the trees as they come in the order of the mimeographed sheet which you hold.

I will first say that the abnormalities at Merribrooke this year were three in number. First, a destructive invasion of the tent caterpillar which attacked nearly all kinds of trees during its traveling stage. Then came a canker worm invasion with partial or complete defoliation of even the forest trees. Almost all of the whole leaves on any tree represent the second set for the season. Then came a drought said to have been the most severe since 1871. As a result of these three influences most of the fruit trees and nut trees dropped their crops this year.

Among the many introduced and grafted trees at Merribrooke only about one hundred typical forms have been tagged for this occasion. The large tags on the trees represent types, the smaller tags represent different variations of the type. Numbers on the tags correspond to numbers on this list.

We will begin with No. 1—Original Taylor Shagbark hickory. Nut large, thin shelled, good cleavage and high quality. This is practically an annual bearer. The weevil likes it because it is very thin-shelled. Consequently we seldom get a good crop. Most of the trees were defoliated. This is the best all-around hickory that I have found. I gave prizes for years and got seedlings from all over the country, and this is the best one that I obtained growing right here at my gate. It is defoliated by both the tent caterpillar and the canker worm.

2. Buckley Hickory from Texas. Nut large, round, thick-shelled, peculiar flavor and fragrance. This hickory was first described in 1872 in Texas and then it was forgotten. Dr. Sargent was quite surprised when I told him that I had one for the variety really passed out of history among the botanists until the past two years. The bark is deeply ridged in the older trees. The tree has been crippled by the twig girdler this year.

3. Carolina Hickory Seedling (scaly bark hickory). Nut small, thin shelled, sweet. I think this is one of the most beautiful hickories we have. It has been crippled this year but not enough to hurt. It has a small, thin-shelled nut with sweet flavor. The older trees have the scale on the bark.

4. Carolina Hickory grafted upon other local wild stock, and I do not know whether it is macrocarpa or pignut.

5. Shagbark top-worked to Vest variety of shagbark from Virginia that Mr. Bixby described yesterday as having a shell so thin that it could be

cracked with the hand.

6. Shagbark top-worked to Carolina and Kentucky varieties. Note the different foliage, and smaller leaves. Here is a graft of three hickories on one stock.

7. Shagbark top-worked to Vest shagbark above and to McCallister pecan below. The foliage of this McCallister would justify putting the tree in any grounds; but here on the shagbark stock the leaves are not so large. The foliage on Mr. Bixby's was large and beautiful.

8. Shagbark top-worked to Brooks shagbark. That tree prolongs the name of one of our audience into history.

9. Asiatic Winged Walnut (*Pterocarya fraxinifolia*). I think this would be valuable for hybridizing.

10. Grafted Woodall American (black) walnut. Nut small, thin shelled. Tree very prolific. This tree has not yet borne, but it should next year. I got that from a man near Milford, Del. The nut is thin-shelled and cracks very easily.

11. Grafted Lutz American Walnut from North Carolina. This tree is about six years from the graft. The nut is large.

QUESTION: When do you have frosts here at Stamford?

DR. MORRIS: The frosts are from about the middle of September until sometime in May. Sometimes we miss the September frosts.

12. Korean Nut Pine. Furnishes important food supply in northern Asia.

13. Grafted Papaw. Larger part Ketter variety. Prize fruits have weighed about one pound each. Smaller part Osborn variety No. 3, a choice kind.

14. Seedling Papaw.

15. Seedling Papaw, christened "Merribrooke prolific" with clusters of fruit of the first year's bearing. Five bunches on the tree and it is the first year out from the nursery. It is a very beautiful tree for the lawn.

The growing season of pawpaws is so long that a hard September frost may catch the fruit before it is ripe in this locality. Fruit will stand a light frost only.

16. Chinese Pistache seedling. Tree beautiful but nut too small for the market. May serve for hybridizing purposes. The autumn foliage of this tree is very wonderful.

17. Grafted Wolfe persimmon. Ripens fruit in July or August. This is an ordinary size fruit but the peculiarity is that it ripens before the others do.

18. Grafted Cannaday seedless persimmon. You see another member of our party has gone down to fame with this Cannaday seedless persimmon.

19. Stanley shellbark hickory grafted on shagbark stock.

20. Stock grafted to Kentucky shagbark.

21. Jeffrey Blue Bull Nut Pine. Nuts small, thin-shelled, rich. Eaten shell and all by the natives. This is one of the most beautiful of pines. In the top of the tree is placed one of the large gourds which I raise here on the place. I place these gourds in the tree-tops for bird-houses. All kinds of birds nest in them, from the chickadee to the barred duck. A squash may be used for this purpose as well as a gourd.

I raise the pines from seed.

22. Torrey nut pine from southern California. Nut is large, and has a fine flavor. I get my seeds from Bartner Brothers. Pines do not do so well near cities. The sulphites in the air are picked up by the pines and this kills them. This particular pine is a surprise to all botanists who have seen it; it is native in California and is one of the disappearing pines. I have had five of them and I raised them all from seed.

23. Chinese hazel. Grafted on common hazel and outgrowing it, The Chinese hazel makes a tree from 80 to 100 feet in height. This is the first year this tree has borne. It is grafted on common stock, and is beginning to bear earlier than it would have done on its own roots.

24. Butternut parthenogens. Some are large and some small but all are grown under the same conditions. That one was defoliated by the canker worm and then by the tent caterpillar and this is the fourth set of leaves it has put forth this year.

25. Hybrid walnut (Siebold x butternut) four years old.

26. Grafted American walnut. Peanut variety. Only one chubby half of kernel to each shell. The scions were sent here from Washington, D. C.

27. Mediate shagbark grafts (Cook variety). Grafted July 10 in midst of great drought. Compare this with the trees you will see farther on in the walk, grafted near the end of the drought. I do not have much trouble with the plain splice graft and I expect it to start ten days after I put it in.

Here is the way I treat a borer, although I have two or three ways of doing this. First I find a hole on the tree, like this one. Then I follow down

to where the borers work. I cut that part away, inject chloroform and fill up the opening with common kitchen soap.

28. American Chestnut. Merribrooke variety, root-grafted on Japanese chestnut. I grafted that very low, below the ground. It is the best chestnut I have among several thousands that I planted. This tree was one of the first to go down with the blight, but I have grafted on other scions and have kept it going ever since.

29. Drescher chestnut (European origin) grafted on Japanese chestnut. The graft is about three years old. It has borne since the first year. There are several nuts on it now.

(Now we must be careful of the sharp stubs in the woods. These are newly cut brush paths, and all guests wearing low shoes should step carefully).

30. Stanley shellbark hickory, grafted on pignut hickory. Mr. Jones introduced this hickory.

31. Kentucky shagbark grafted on shagbark stock, with bark slot graft. I let another twig grow from the same lead for nourishment. I put in three grafts here two of which are dead. I do not quite approve of that method. I prefer now to go up to the small branches and then splice-graft on small branches.

32. Marquardt pecan grafted on stock of pignut. It does well on this hickory.

33. Hardy, hard-shell almond.

34. Woodall American walnut. This shows that the Woodall black walnut grows fairly well on butternut stock.

35. Shagbark hickory top-worked to Marquardt pecan.
36. Staminate persimmon trees.
37. Bony Bush filbert, grafted on common hazel. (Bush badly cut up by girdler beetle. Elaphidion. Five nuts on the bush).
38. Purple hazel. Look sharp to find the 20 nuts on this bush. This tree is about 5 years old.
39. Four large bitternut-hickory trees, top-worked to Beaver hybrid. Beaver branches distinguished by larger leaves and fewer leaflets. Stock shoots will be cut out gradually, allowing Beaver to have entire tree finally.
40. Bitternut hickory top-worked to Marquardt pecan.
41. Hybrid walnut. (Siebold x Persian). Tree riddled by walnut weevil every year hopelessly.
42. Taylor shagbark hickory grafted on shagbark stock. I fill the cavities with paraffin and turpentine. There are three or four nuts left in the top of the tree. The tree has borne nuts for three years.
43. *Pinus edulis*.
44. Marquardt pecan on bitternut.
45. Dead hybrid hickory, grafted to Beaver hybrid. Grafts made enormous growth in first year—10 feet for some grafts. All blew out in one minute of hurricane in advance of thunder storm.
46. Bartlett hazel grafted on common hazel. There are a number of dead ends, caused by a small worm you can hardly see.

47. Chinese chestnut. Blighted at foot of trunk but the tree continues to bear.

48. Garritson persimmon. Best of all varieties called seedless, but the large staminate tree nearby spoils that feature. It is about five years old, and bears very regularly and heavily. The stock came from Mr. Jones.

49. Early Golden persimmon. Carries one graft of Everhart seedless variety on lowest large branch.

50. Hybrid walnut. *Juglans nigra*. I do not remember which parent I used.

51. Pignolia nut pine. *Pignolia pinea*. It is a seedling. You can buy pignolia nuts in Europe for food everywhere.

52. Hardy soft-shelled almond. I do not know the variety as the label is lost; but the tree was put there about 3 or 4 years ago. It came from the Government.

58. Deming purple walnut. I think Dr. Deming can best tell you about this.

DR. DEMING: It grows on the side of the road between Norwalk and Danbury, where the very large black walnut tree is, 15 feet in circumference, said to be the largest in Connecticut. This purple variety has nuts with a brownish red involucre showing sharply against the green leaves. The young foliage is purplish red, and the cambium and the pellicle of the kernels are purple. It is a very fair nut and the tree is very striking when it starts in spring with the beautiful tufts of leaves.

DR. MORRIS: It may be a valuable wood for cabinet-makers. Every part of the wood is purple. There are two purple trees. The smaller tree is evidently a seedling of the larger.

54. Young Major pecan.

55. Webb Persian walnut on American walnut stock. The nuts are enormous and of Alpine type of good quality. You saw some of these yesterday among those brought in by Prof. Neilson. You sometimes see these in the French market where they are called "Argonne." I picked this up in Greenwood. It has many nuts this year and this is the second crop of leaves.

56. Busseron pecan. This had a full crop of flowers this year, both staminate and pistillate.

57. Appomattox pecan, from the James River in Virginia. This and four other kinds of pecans would have borne nuts this year excepting for defoliation. It is a handsome tree and will bear next year.

58. Seedling filbert. About six years old.

59. Daviana filbert from Europe. Many people call them "hazels," but I think we should call them "filberts."

60. Josephine persimmon. It has borne heavily every year except this year. It still has some leaves left. Some people are very fond of the fruit. I do not like that as well as the Garretson. It is a big persimmon and a very good one. The fruit stays on until late November and December. I think the Garretson is the best persimmon I have ever had.

61. Lambert persimmon. Largest fruited American kind.

62. Japanese persimmon, planted between the rocks for protection from wind in winter, and from heat in summer. Hardy now for two years but of slow growth.

63. Beaver grafted on bitternut.

64. Weiker hybrid hickory on shagbark stock.

65. European filbert grafted upon common hazel stock. The squirrels have lived on it. I can count 7 nuts left. I made grafts more than a foot long. It was planted three years ago. I could show you several hundred trees bearing heavily this year, and on all of them we lost the first crop of leaves.

66. Beaver grafted Nov. 5, 1922, on bitternut.

DR. ZIMMERMAN: Will they live when grafted at any time throughout the year?

DR. MORRIS: I would not be afraid to graft anything at any time of the year.

67. Taylor shagbark grafted July 21, 1924. Probably mockernut stock. Growth slow but sure.

68. Wild beak hazel. Nuts not so good as those of common hazel.

69. Bitternut top-worked to Beaver.

70. Hazel, patch-grafted here and there with Bony Bush filbert. The larger and darker leaves are Bony Bush.

71. Leonard shagbark grafted on stock probably shagbark. Nut very small, thin shelled, highest quality and keeps for four years without becoming rancid.

72. Shagbark top-worked to Taylor variety, but only a few grafts. Too much work for a tree of this size.

73. Pleas hybrid pecan on butternut stock.

74. Bitternut top-worked to Beaver.

75. Here is a very interesting object lesson. No. 74 is a bitternut top-worked to Beaver, and all doing well. The same day, with the same graft, I top-worked this pignut. The pignut refused the graft and died insulted. But another stock from the same root accepted Marquardt.

76. Bitternut stock accepting Marquardt pecan tardily.

77. Here is another form of borer. I treat them in this way: Cut away a little of the hole, pour in the chloroform and stop up the hole with soap. That will kill all of the borers in the tree.

78. Grafts of Laney hybrid hickory on bitternut.

79. Group of four filberts—not blighting, but not thriving this year or last. Reason unknown. Soil is heavy clay hardpan near top. Top swampy in spring.

80. Taylor shagbark on bitternut.

81. Taylor shagbark on shagbark stock.

82. Bitternut grafted to Lucado pecan. Grafts grew well for two summers, but died in second winter.

83. One poor graft of pecan on bitternut.

84. Pleas hybrid pecan.

85. Merribrooke chestnut grafted upon Chinese chestnut sprouts.

DR. ZIMMERMAN: Have you been able to bud chestnuts successfully?

DR. MORRIS: Yes.

86. Daviana filbert.

87. Hybrid hazel. (*Colurna x Americana*).

88. Avellana hazel. Variety *Contorta*.

89. Siebold walnut. *Parthenogen*.

90. Hybrid chinkapin. (*C. pumila x C. dentata*). Grafted to another hybrid, but stock now blighting.

91. One of a series of chinkapins, natural or hybrids, grafted over to other hybrids or to the Merribrooke variety of American sweet chestnut. Some are blighting.

92. Original Bony-Bush hazel. Blighting moderately. Treatment for blight not followed because of wish to note the degree of resistance.

That bush was named by Dr. J. Russell Smith. The nut is remarkably thin shelled, very long and curious in form.

93. Chinkapin, not grafted. These bear heavily every year notwithstanding the blight. From the same root common chinkapin will keep on bearing year after year. When one stock blights another takes its place so that heavy continuous bearing is the rule.

94. Original No. 1 Morris hybrid chinkapin. (*C. pumila x C. dentata*). Nuts of size and quality of American sweet chestnut. Tree blighted in its 13th year after bearing crops for 8 or 9 years. New stump sprouts now growing.

(Note: At this time, the guests were called to the lawn back of the house, where a luncheon was served by Mrs. Morris. The tables were laid sumptuously, and all enjoyed it the more because of the surroundings, where trees on one side bent over a clear trout-stream, and elsewhere old-fashioned gardens splashed colors over the green background.)

BUSINESS SESSION

Held on Third Day

(Note: It was planned that this session should be held during the afternoon of the third day, after the trip through Dr. Morris's estate. However, while the members were exploring deep in a wooded portion of Merribrooke, a sudden downpour of rain occurred. The nearest shelter was found to be the barn, where the members agreed that the following session should be held, since it was not possible to reach the main house. All members were standing during the session, including the reporter who wrote with the notebook resting against one of Dr. Morris's cars.)

Session called to order by President Weber.

DR. SMITH: There should be added to the by-laws the following amendment:

ARTICLE V. Members all be sent a notification of annual dues at the time they are due, and if not paid within two months thereafter they shall be sent a *second notice*, telling them that they are not in good standing on account of non-payment of dues, and are not entitled to receive the annual report.

At the end of thirty days from the sending of the second notice, a *third notice* shall be sent, notifying such members that unless dues are paid within ten days from receipt of this notice, their names will be dropped from the rolls for non-payment of dues.

The President: The motion has been seconded; all in favor please signify by saying "Aye."

(Vote carried unanimously).

The Secretary: The association should have a fiscal year. Shall we discuss this or will the president authorize the secretary and the treasurer to agree upon a date most convenient to them for the beginning of the fiscal year?

MR. REED: I move that we leave this to the discretion of the secretary and the treasurer.

THE PRESIDENT: All in favor of the motion, please signify.

(Voted as presented).

THE SECRETARY: I move that combination membership in the Association with subscription to the American Nut Journal be \$4.50, a deduction of 25 cents each by the Association and the Journal.

THE PRESIDENT: All in favor of the motion please so indicate.

(Motion carried).

THE SECRETARY: The next thing is to elect new officers.

THE PRESIDENT: Mr. Reed will please read the report of the Committee on Nominations.

MR. REED: The making of this report was one of both great pleasure and of extreme regret. Since Dr. Deming has found that it will not be possible for him to continue as secretary, the following names are offered:

President—Harry R. Weber.

Vice-President—Mrs. W. D. Ellwanger.

Secretary—Mrs. B. W. Gahn.

Treasurer—H. J. Hilliard.

THE PRESIDENT: Are there any exceptions to this? Will those who are in favor please so state?

(Election carried unanimously).

DR. SMITH: Dr. Deming's retiring from the secretaryship is a matter which all old-timers will regret, and I want to move that this association record in its proceedings the fullest appreciation of his great and faithful service in helping to carry the organization through so many years. I do not know what we would have done without his service and it is with great regret that we see him step aside.

(Motion seconded and unanimously carried).

DR. DEMING: I wish to express my gratitude to the members for their kindness, but I also wish to say that although I have stepped aside, I have not entirely passed away. I am still with you and I shall always give the association the best of my efforts in whatever way they may be needed; its interests shall always be dear to me.

DR. MORRIS: It seems to me that we have an object lesson here. Excepting for Dr. Deming's efforts I doubt whether this organization could have held together and worked harmoniously during its years of existence.

He has been the key-note of the work with which others have helped, and we have been successful because of concerted work on the part of a number of men who are looking forward to the great future of this new agriculture, this new source of agriculture for the entire world, wherein we are going to be able to depend upon the sub-soil for our sustenance. It is through untiring work and self sacrifice that those who are so interested in this work have been able to work as a mass unit. I do not know of anything more that I could say.

THE PRESIDENT: I am sure that we all regret to see Dr. Deming step aside, but we will still have him with us and I am very sure that he will do all possible for the good of the association always.

DR. DEMING: I stated a few moments ago that although I had stepped aside I had not passed away; but since then I have changed my mind. I believe that I have entirely passed away.

DR. SMITH: I move a resolution of great appreciation for Dr. Morris's and Mrs. Morris's hospitality to us, and for enabling us to enjoy the beautiful day we have had here.

(Motion seconded and unanimously passed).

THE PRESIDENT: Dr. Morris, you now have notice of the official action of the association in their desire to thank you.

DR. MORRIS: I thank you, but I must say that I have had very little to do with it; I may have made the suggestion, but the women always do all of the work and in this case my wife and daughters have done it all.

THE PRESIDENT: We have not yet decided on the place for our next convention. I would like to have your ideas.

DR. MORRIS: I had three ideas as to that; one is to go to Mr. Riehl's place next year. Prof. Colby said that if we should, he would assume the responsibility of the committee on arrangements. We are first to ask Mr. Riehl whether it would be in accordance with his ideas and wishes.

The second idea is this. We saw yesterday only a small part of Mr. Bixby's exhibit, one of the finest collections in the world. We should have to spend more than a day there to see it satisfactorily. In connection with a visit to the Hick's nurseries, and others in the vicinity, it would take more than a day.

The third idea is to go again to Lancaster to see Mr. Jones' nursery and other things in that vicinity. It seems to me that we must make a choice between these three.

MR. JONES: I would be very glad to have you come to Lancaster.

DR. MORRIS: The objection to that is that Mr. Riehl is now 86 years of age. In view of that our first choice ought to be Mr. Riehl's place.

DR. SMITH: I move that, if it prove acceptable to Mr. Riehl, we meet in western Illinois.

MR. JONES: Why not add, "If that is not satisfactory, to go to Lancaster?"

DR. MORRIS: We should go back to Long Island next year and complete what we did not see this year, if we do not go to Mr. Riehl's.

THE SECRETARY: The Secretary has received from the St. Louis Chamber of Commerce an invitation worded with rather more cordiality than usual to hold our next convention in St. Louis. They offer to provide a meeting place, speakers, publicity, to do all except give the cash prizes and

entertainment. I do not know exactly how far St. Louis is from Alton, but I understand it is one hour's ride by rail.

MR. REED: We could also see the Botanical Garden and the collection of large trees.

THE PRESIDENT: I think the sentiment is in favor of the western meeting.

We can easily get to Mr. Riehl's place from St. Louis.

MR. REED: It is 22 miles from St. Louis to Alton, and there you can change and go to Mr. Riehl's. I think it best to go to St. Louis for the convention and to take a day at Mr. Riehl's place.

THE SECRETARY: As to the date we would not be able to decide upon that without first consulting Mr. Riehl and learning the time convenient for him. However, we should express our opinion as to the best time, approximately.

MR. REED: I believe it would be to the advantage of the organization to go there at a time when the nuts are on the trees. We have seen the species and varieties in bearing, but we have not seen a paying orchard ready for harvest. I believe we should have the meeting about September 10, or a little later.

THE PRESIDENT: Then we will move that the convention next year be held at St. Louis on September 10, or a little later as may be decided by the Executive Committee after consultation with Mr. Riehl.

(Motion put, voted and carried).

DR. MORRIS: Another important matter is in regard to publicity. For this meeting I have sent notes to about 15 different publications, expecting that

they would give us notices. Not a single one of them gave us notices. This morning one of the reporters called me and said he was sorry he could not be here as he had an important meeting to attend. He wanted to know what the Northern Nut Growers' Association was like, if it was something like the Tree Planting Association. The fact is that people do not understand, as yet, the meaning of this association or its purpose. They do not realize that California sends 25,000 tons of walnuts to market, worth millions of dollars, and 10,000 tons of almonds this year. They don't realize that down in Georgia, in the poor, puny pinewoods where men had a hard time to make a living at one time, they are now riding around in limousines because they are growing nuts. They do not realize the enormous social and economic importance and consequence of work of the nut growers of today in the part that they play in the agriculture of the world for tomorrow. The newspapers would rather send some representative to see a prince fall down with his horse. But I know from mutual acquaintances that the Prince would rather be with us here today at this meeting than to be listening to a thousand and one nonentities and taking part in conversations with no future meaning. I believe that if I had thought about inviting him in time I should have had him out here. I have had experience with members of royalty before and I know what serious-minded people they are.

The next subject discussed was that of dropping members who are not in general good standing. After the discussion the decision stood that no action could be taken unless specific charges against the member were presented and proven true.

Another matter discussed was that of compensation to Mrs. Gahn for doing secretarial work for the association. It was voted by those present that she should be compensated, but the amount of compensation should be left to the decision of the Executive Committee.

The President adjourned the session sine die, at 4 p. m.

Because of lack of time, several papers were not read. These are included herewith:

NUTS

By Hon. Royal S. Copeland, U. S. Senator from N. Y.

Whenever there is a peculiar individual in the community, he is apt to be called a "nut." As ordinarily used this is a term of derision, but the more one studies the value of the nut the more he is impressed with the idea that this isn't a good word to apply to an abnormal individual, unless he happens to be abnormally good. The nut is one of the best of the products of nature. It is one of the oldest of foods, and among certain animals it is almost the only food depended upon for health and growth.

If Mr. Bryan is mistaken about the origin of man, and if his antagonists are right, the natural ancestors of the human race were all nut eaters. At least the gorillas and chimpanzees are fond of the nut. When we go back to the early history of the Greeks and the early inhabitants of Great Britain, we find that they depended largely upon the acorn for food.

When measured by the caloric method it is surprising how much richer in nourishment the nut is than almost every other food substance. Nuts average about ten times as many calories per pound as the richest vegetables.

It makes you hungry to hear the names of the nuts. In this country we have the walnut, butternut, hazel nut and the hickory nut, the chestnut and the beechnut. These are native to our land. Then there are cultivated orchards of Persian walnuts, pecans, almonds and peanuts.

Christmas and Thanksgiving would be a failure without nuts; they are a part of the hospitable fare and no stocking is well filled at Christmas time unless a handful of nuts is added to the surprises.

Isn't it amazing what popular ideas there are in existence about the digestibility of foods. Many of these are fallacious. For instance, it is common belief that nuts are difficult to digest. This is not well founded. Of course nuts like all foods which are used as a part of the dessert are considered merely as an addition to the meal, and not a part of the meal structure. You finish your meal, having eaten everything you need and having filled your stomach, then you are given a dish of ice cream and, perhaps, after that the nuts are passed. They taste so good that you are tempted to take one more about ten times. You fail to chew the nut thoroughly and you crowd it into an already overfilled stomach. Because it happens to be the first thing to come up in case of disaster you jump at the illogical conclusion that your indigestion is due to the nuts. I need not tell you how unscientific is your conviction.

Several varieties of nuts are used for the making of nut butter, and this food is a very excellent substitute for meat.

Certainly nuts have material advantage over a good many foods. They keep indefinitely. They never putrefy. They are not infested with harmful bacteria. You can never get tape-worm or any other parasitic trouble, which occasionally follows the eating of infected food.

I am glad there are societies organized to propagate the nut. A prominent concern of New York City is very active in promulgating the value of the nut, and is encouraging the planting of nut trees.

Somebody has estimated that there are three million miles of country roads, and that if nut trees were planted alongside these roads there would be enough protein food for the entire population.

Nuts are rich in protein, lime, iron and vitamins.

Many dishes may be made from the nut which have the appearance and flavoring of meat, without the objectionable effects of flesh diet.

Last year we imported twenty-five million pounds of almonds, forty million pounds of Brazil nuts, eighteen million pounds of filberts, and forty-four million pounds of walnuts,—about twenty million dollars worth of these nuts were brought into the country.

This shows that there is some appreciation certainly of an article of food which deserves to be even more commonly used than it is at present.

HARDINESS IN NUT TREES

By C. A. Reed, U. S. Department of Agriculture

Nut trees of most species commonly thrive at both latitudes and altitudes much greater than the limits of regular or even frequent crop production. This fact is seldom fully appreciated by prospective planters, particularly in the North, who, not unnaturally, assume that the presence of a group of vigorous appearing trees, or even of a single tree, particularly in a fruitful year, is sufficient evidence of local hardiness to justify commercial planting. However, practically all of our native species of nut-bearing trees are indigenous well beyond the range of regular crop production. This is made possible by occasional seasons favorable to seed production which enable such species to reproduce themselves. A crop once in a quarter century would be sufficient for this purpose.

Taking the pecan as an illustration of how a species may be affected by latitude, it has been found that, as the limits of hardiness are approached, the ill effects on the species in approximate order are:

- (1) reduction in size of nut, especially with oblong varieties
in length,
- (2) increased proportion of faulty kernels,
- (3) increased irregularity of crop,

- (4) practical crop failure, and lastly the
- (5) partial, then complete, destruction of the tree.

On the other hand, the fact that a tree is subject to occasional winter injury, or that it bears irregularly, or not at all in a particular site, is not necessarily to be taken that the same tree in a different site or under slightly changed environment would not perform satisfactorily, even in the same locality. A change in exposure or of cultural treatment, or of rootstock, or of variety, or a modified association of varieties, might and frequently does bring about entirely different results. Sometimes a southern exposure causes trees to respond to mild weather, in winter or early spring, and to be caught by subsequent, violent drops in temperature. Some of the best known and best performing Persian walnut trees in the East are on a northwestern exposure, yet the species is commonly not hardy in the temperate portions of this country.

To a certain extent the ability of orchard trees to withstand frost injury is subject to control. The danger is greatest with trees which have grown late or those which have become devitalized for some reason or with those which are in poorly drained soils. The kind of root stock which has been used, is known to have had an influence in some cases. Doubtless this will be better understood as different stocks are used by the leaders in pecan breeding. Varieties also are known to differ greatly in their degree of hardiness. However, failure upon the part of otherwise normal trees to bear paying crops with regularity is not necessarily due to low temperatures. Other factors, such as self-sterility, may be wholly responsible for at least the lightness of crops.

So far as the orchardist is concerned, a tree is not hardy unless it is capable of bearing crops the average of which are profitable. On the other hand, occasional winter injury does not prove that a species cannot be

grown successfully in the same locality. Neither the peach nor the apple industries of the North nor those of the citrus in the South and California nor, in fact, any of the other horticultural commodities of this country are wholly unaffected by frost damage. Our forest trees may be more subject to winter killing than we suspect. A certain amount of winter-injury is to be expected in any part of the country no matter what the species of plant may be.

The frequency with which winter or spring injury is definitely known to occur gives color to a rising theory that freezing temperatures may play a vastly greater part in the development of the nut industry over the entire country than is commonly supposed. Much of the evidence of damage from this cause is of such nature as to be easily overlooked or attributed to other causes. Trees and plants of many kinds have become so accustomed to injury by freezing that they are able to recover without the injury always being apparent. A few illustrations of this which have come to the writer's attention might be cited.

In December 1919, a sudden drop in temperature of from 32 deg.F to 24 deg.F occurred at McMinnville, Oregon, with fatal result to cultivated trees and shrubs of many kinds. The damage was greatest in flat bottoms, especially those where neither land nor air drainage was good. Under such conditions, numerous apple orchards were killed outright. Prunes and Persian walnuts were so badly injured to the snow-line that subsequently great numbers of trees were cut down. Both staminate and pistillate buds of filberts above the snow were practically all destroyed. Later on, the entire tops of many of the older-bearing filbert trees succumbed. An instance of particular interest, in so far as this discussion is concerned, was afforded by the behavior of a shagbark hickory tree in McMinnville, some 20 or 30 years old, which had been grown from a Missouri seed. In February, when examination was made of the condition of this tree, it was found that all

visible buds had been killed, yet the bark on the branches between the buds was in apparently perfect condition. The question as to what the tree would do, therefore, became one of great interest. The following September, when revisited, this tree was found to have such a wealth of luxuriant foliage that the observer felt that the accuracy of his February records was challenged. However, closer inspection showed that growth had entirely taken place from adventitious buds, and that the dead buds and spurs were still in evidence. There were no nuts on the tree but otherwise the casual observer would not have suspected that the tree had been affected in any way. In all likelihood, the owner of the tree would deny that it had been injured.

Another case of somewhat similar kind occurred early during the present year in a pecan orchard in South Georgia. The trees had been set in 1917, and in 1919, a portion selected by the Bureau of Plant Industry for conducting a series of fertilizer and cover-crop experiments. The summer of 1923 was extremely dry. This was followed by warm rains in the late fall and early winter. On January 6, during a period of high wind, the mercury dropped to within a few degrees of zero, official reports recording temperatures of from 6 to 8 degrees above zero at various nearby stations.

On March 31, Dr. J. J. Skinner, of the Office of Soil Fertility Investigations, in attending to the spring fertilizer applications, discovered that a high proportion of the trees had been badly winter injured, as indicated by the usual characteristic evidence. These included a considerable exudence of sour and frothy sap from the trunks of the trees, particularly those having smooth bark. This invariably occurred on the west side. Shot-hole borers, which not infrequently follow such injury, were already at work.

This situation was at once called to the attention of the owner of the orchard who lived some 50 miles away. He replied that although he made

frequent visits to the orchard, the matter had not attracted his attention, nor had it been reported to him. On April 17, he inspected the orchard and the day following, reported to the Bureau by special delivery that as a result of a rather hasty inspection, he was convinced that from 16 to 20 per cent of the trees in the experimental tract were injured, but that in the rest of this orchard the injury was insignificant, probably not exceeding 4 per cent. His not unnatural deduction was that the high fertilization of the soil in the experimental tract had caused tender growth which, under the extreme conditions of the previous months, had been unable to survive.

On April 24, a careful record of the condition of all trees in this tract and of a representative number of those in adjacent parts of the orchard, was made by Mr. J. L. Pelham of the Bureau of Plant Industry and the writer, in company with the owner of the orchard and his superintendent. It was found that in the experimental tract, 50 per cent of the trees had been visibly injured, thus exceeding the owner's maximum estimate by about 30 per cent. Of the total number of trees, 20 per cent were regarded as being slightly injured, and 30 per cent severely so. Of the fertilized trees within the experimental tract, 55 per cent showed injury to some degree as compared with 58 per cent of the trees unfertilized, also within the tract.

Inspection of the trees outside of the experimental tract showed that 52.6 per cent were affected, 40.8 per cent being slightly, and 11.8 per cent severely injured. A second inspection made June 9 showed that while a few of the most severely injured trees had succumbed, the apparent condition of the majority was greatly improved. In the experimental tract 6 per cent were dead, 13.50 per cent in doubtful condition, and 80.25 per cent were apparently in good condition. Of the trees in outside tracts, the percentage dead, doubtful and apparently sound were 2.80, 9.008 and 87.42, respectively.

The lesson of present importance from this narrative is that afforded by the illustration not only of the ease with which the matter all but escaped the attention of a careful grower but of the difficulty of even impressing upon him the full gravity of the situation. In spite of a prejudice which he conceded was in his mind, when he first inspected the trees on April 17, he underestimated the number affected by from one-third to one-half.

This grower was not alone in his failure to detect evidence of winter injury as was subsequently proven by the negative replies to a general inquiry to growers in many sections sent out in May, together with numerous reports of severe injury received during June and early July. The fact is that winter injury was more or less general in the pecan orchards of much of the South. Had it been possible to observe further, it is highly probable that a direct relation would have been found between this damage and the lightness in the set of the crop of nuts in 1924 over the general pecan district.

Other instances of damages to nut trees which have largely escaped notice might be cited, but these will perhaps be sufficient to call similar cases to the minds of other observers. Of particular interest in the northern part of the country are specific instances of the behavior of individual species and their varieties with reference to ability to withstand local climatic conditions. To cite a few: Mr. E. A. Riehl, of Godfrey, Ill., 8 miles from Alton, reports that during his 60 years of residence on a high bluff overlooking the Mississippi, the pecan trees in the river bottoms of the immediate neighborhood have fruited with exceeding irregularity. A correspondent from Evansville, who cleared 200 acres of forest land along the Ohio of all growth other than pecan, reports that the yields have been disappointing. F. W. McReynolds of Washington, D. C. has 50 or more grafted trees now 8 or 10 years old, 10 miles north of the District, which, although in otherwise thrifty condition, have not fruited.

T. P. Littlepage of Washington, D. C., has some 30 acres of pecan trees, also grafted, on his farm near Bowie, Md., which have borne some nuts during the last three years, but the product has been undersized, poorly-filled and distinctly inferior. Mr. Littlepage reports that during the past spring, these trees suffered appreciable injury in the freezing back of the fruit spurs and that the nuts which formed were from a second set of spurs. His trees bore in the neighborhood of a bushel of nuts which looked more promising than usual until the middle of October when freezing temperature occurring between the 14th and the 24th, completely destroyed the crop. At Bell Station, near Glenndale, Md., about three miles nearer Washington than Bowie, at Marietta, a colonial plantation, there is a clump of pecan trees dating back to the days of Thomas Jefferson. These are apparently hardy except in the matter of yields. Dr. M. B. Waite, of the Bureau of Plant Industry, who has long known these trees, states that they bore heavily in one year, about 1912, but that since that time, they have borne very little.

On the other hand, Mr. Albert Stabler of Washington, has 6 or 8 trees of varieties similar to those in the plantings of Messrs. Littlepage and McReynolds and of about the same age, on a farm not far from that of the latter, one variety of which, Major, in 1923 bore some very fair quality nuts. Although small, they were typical for that variety both in respect to size and high quality. The crop of 1924 was practically a failure, the set being very light. In the test orchard of Mr. J. F. Jones of Lancaster, Pa., young trees of several of the better known varieties are making a good start in the way of beginning to yield and in showing no appreciable signs of winter injury. Most of these trees bore light crops last year, (1923) but are practically barren this year.

South of Waynesboro, Pa., on a farm belonging to Mr. G. H. Lesher, there are 7 seedling pecan trees some 50 years old, which not only show no signs of winter injury outwardly visible, but have the reputation of bearing fairly

well on alternate years. The present (1924) being the favorable year, the trees had a good sprinkling of nuts in clusters of as many as 5 each, when seen on July 23. A few miles farther north, in the town of Mont Alto, at an altitude of about 1000 feet, near the location of the State Forestry School of Pennsylvania, another tree said to be 65 years old, and having a girth at breast height of 65 inches, on the residence grounds of Mr. H. B. Verdeer, is apparently as hardy as are the indigenous species of the neighborhood. It is claimed to have recently borne three pecks of nuts in a single season, and it now has a very good crop. Numerous other instances of pecan trees in the North might be cited, but these suffice to establish not only the uncertainty of hardiness of the pecan in the North, but also the probability of nut crops in occasional years or oftener, well beyond the generally accepted range of the species.

The hardiness of the Persian walnut is difficult to define. To again quote Dr. Waite, "*Juglans regia*, as we know it in the east and north, frequently succeeds over long intervals of time under conditions of climate, soil, elevation, and general environment suitable for the peach. It is perhaps a trifle more subject to injury by radical drops in temperature, but it recuperates with decidedly greater difficulty." Dr. Waite points out that there is a striking similarity between the requirements of local environment of the Persian walnut and the sweet cherry. It develops that this is a familiar comparison in southwestern British Columbia. Both require good drainage of air and soil, or the benefit of moderating influence such as is afforded by large bodies of water. Also both are endangered by warm spells during the dormant months.

These statements cover the situation quite correctly, as it is seen by the writer, although it might be added that beyond or west of the Ohio River, in the middle portion of the country, this species is seldom able to survive for more than one or two winters. Many trees have been planted in Michigan,

but the great majority have passed out entirely even where peaches normally succeed. However, it is the experience of a few growers in Sanilac County, bordering Lake Huron, that within a half mile of the lake, there is a greater profit in Persian walnuts than in peaches. One grower at Lockport, New York, has found Persian walnuts to pay better than other orchard crops which he has raised at equal expense or upon equal areas of land. An orchard at East Avon, widely known at one time and visited by the Northern Nut Growers' Association in 1915, practically succumbed entirely after having borne but one good crop in about 35 years. Mr. F. A. Bartlett, of Stamford, Conn., who knows intimately many dozen trees of this species within a radius of 50 miles of New York City, finds that few bear significant crops except at long intervals. From Stamford, Conn., near the Atlantic Seaboard, south to Norfolk, Va., Persian walnut trees are not uncommon in door-yards. They are fairly frequent in southern Pennsylvania west over practically half the length of the State and through Maryland west to Hagerstown. There are perhaps more productive trees in Lancaster County, than in any other county in either Pennsylvania or Maryland, with the possible exception of some county of the Eastern Shore of the latter state, which section already has been referred to. In Lancaster county yields are sufficient to give considerable profit from trees not occupying expensive land.

The Japanese walnut affords a curious analogy in regard to hardiness. During normal years, it succeeds over practically the same range as that of the black walnut, yet it freezes in early fall, mild winter or late spring when conditions are adverse, even when black walnut and pecan nearby are not visibly affected. Mr. Jones finds the Lancaster heartnut, a variety originating in his county, to be subject to injury by spring freezing to such an extent that he has largely discontinued its propagation. Mr. Edwin A. Surprise, of Boston, reports that this variety grows well in summer but freezes back in winter about as much as it grows in summer. Mr. Bartlett

regards it as one of the most valuable acquisitions in his nut planting at Stamford, Conn., as it is a handsome, vigorous grower, and promises to bear well. As a safer variety in the Lancaster district Mr. Jones has substituted the Faust from Bamberg, S. C., which vegetates later in spring and thus far has proved less subject to injury.

The twigs of young black walnut trees are occasionally injured by freezing in winter, but recorded instances of such damage are rare. This is a field which should be investigated, as there is evidently no data showing even the regularity with which the black walnut bears in any section, much less the extent to which fruiting is restricted by destruction of the buds or spurs as a result of severe temperatures in winter or spring. This also applies to hardiness of the butternut, the hickories and of introduced species of chestnut.

In conclusion, it is pointed out that planters should not assume that the presence of a healthy tree is proof of sufficient hardiness to warrant extensive plantings, neither should they over-look the fact that an occasional satisfactory crop may be but slim evidence of commercial possibilities. It requires years of trial before a species or variety can fully establish its hardiness. Yet, on the other hand, to wait to find a kind of nut a hundred per cent hardy under all conditions, would be not to plant at all. No varieties of any species are immune to winter injury over any great portion of the United States. The planting of nut trees in the northern part of the country is certain to go forward, but for the present, east of the Rockies, large orchards of nut trees of any species or variety must be regarded as fields promising for experimentation rather than of sound commercial investment.

A common error in the minds of the American people is the assumption that to be a success, a thing must be performed upon a large scale. To

develop a nut industry, it is imagined that there must be great orchards of hundreds of acres. It is not realized that a great proportion of the walnuts, almonds, filberts, and chestnuts annually imported from Europe, are from roadside, hillside and door-yard trees which could as well have been grown in this country on what is now idle land in thickly populated agricultural districts. No one need expect to attain great wealth from the products of door-yard or waste land trees but the by-product which could readily be salvaged from nut trees, would likely be very acceptable when interest and taxes or other bills come due.

WALNUT GRAFTING INVESTIGATIONS

T. J. Talbert, Professor of Horticulture, University of Missouri, College of Agriculture

These investigations are to determine the best varieties of the improved black walnut for Missouri. Valuable information is also being procured in reference to the topworking or cleft grafting of the native seedling black walnut to the improved sorts.

Since practically every Missouri farm contains some waste land upon which the native walnut and other nut trees may be growing, it is believed that it is possible to topwork these seedling sorts to improved kinds which will not only supply a larger quantity of thinner shelled, more highly flavored nuts for home use, but a surplus for the market. There is a growing demand for the seedling black walnut.

At the present time Missouri leads all other states in the production of this nut. The results which are being obtained in this experiment are proving to be of unusual interest and profit to Missouri growers.

The investigation has been extended to include, besides black walnuts, pecans, hickories, hazel nuts, chinkapins and chestnuts. With each of these nuts our object is to determine better varieties for Missouri conditions, more

profitable and economical methods of production and more satisfactory methods of culture, as well as to stimulate an interest in the marketing and larger use of these products.

The improved varieties of seedling black walnut have been found to be exceedingly easy to propagate by cleft grafting the native or common seedlings. The cleft graft has been used successfully upon seedling trees ranging in diameter from 1-1/2 inches to as much as 8 or 10 inches. In general, however, it has been found best to cleft graft branches or limbs of no greater diameter than from 4 to 6 inches. Such wounds, if properly handled, usually heal over completely within 3 or 4 years. When larger branches are used, decay is much more apt to develop in the wound before healing over is accomplished.

The cleft grafting work is accomplished in the usual way. The limb or branch is removed by sawing it off. The end of the branch is then split with a regular grafting implement used for this purpose; or the work may be accomplished with an axe. If the branch is large a wedge is driven in the center to hold the split cavity apart and to relieve the pressure upon the scions which are to be inserted. Wood of the last season's growth is procured from the variety which it is desired to propagate and the lower end of the scion, which is made about 4 inches long, is whittled to a wedge shape, after which it is inserted in the slit made upon the stock. Where the stock is more than 2 inches in diameter, it is usually advisable to place 2 scions; and where the stock is as large as 4 to 6 inches or more in diameter 4 scions should generally be used. After the placing of the scions all the cut surfaces should be carefully covered with grafting wax. Paper sacks are often used in our experimental work to cover the grafts and cut surfaces for a week or 10 days. It has been found that the inclosing of the grafted branches in paper sacks for this period lessens greatly the evaporation, and more of the inserted scions are apt to grow.

The scions may grow very rapidly, in which case it is usually necessary to brace them by tying a stick or branch to the stock and allowing it to extend for 2 or 3 feet above the point at which the grafting work was done. The inserted scions are then tied to this support. It is very important that the grower examine grafts after wind storms in order to repair damage which may have been done.

Investigations at this station have shown that grafts usually bear fruit in 4 years after the grafting operation. We receive some fruit, occasionally, in 3 years after the work is performed. It is also interesting to note that when seedling walnuts of the same size are selected, some topworked and others untreated, the grafted trees after 5 years' growth generally grow tops equally as large as the tops of the ungrafted trees.

The principal improved varieties of black walnut which are being used at this Station are as follows: Stabler, Ohio, Thomas and Ten Eyck.

(Note by the editor.—The cleft graft described by Prof. Talbert has been superseded in the East by other methods, chiefly the bark and the modified cleft grafts).

CARE AND PREPARATION OF NUTS FOR SEED PURPOSES

By Prof. E. R. Lake, U. S. Department of Agriculture

A nut is a seed, and a seed, normally, is an embryo plant asleep. To keep a nut-seed asleep and safely resting against the favorable time when it may awake, arise and go forth, as a vigorous seedling bent upon a career of earth conquest, requires no great or unusual attention and care save that which is necessary to maintain such conditions as will insure the complete maturing, ripening and curing of the seed, its protection against the ravages of rodents or other nut-eating animals, undue moisture and an unfavorably high temperature. In other words harvest the nuts as soon after they are mature as is possible, insure their complete curing, store them where they will be kept constantly so cool that germination cannot take place, and some nuts, as the black walnut and butternut, may germinate at a temperature just above zero (centigrade(?) Ed.) and keep them moist enough to prevent undue hardening of the tissues or enclosing structures (shell), at the same time prevent them from becoming saturated with moisture and thus rotting. Summarized, these conditions are: (a) a temperature just too low for vegetative activity. (b) A moisture content of the nut just below turgidity. (c) An immunity against ants, rats, mice and squirrels.

Curing. A man-devised method for hastening the ripening of a matured seed or fruit, is usually carried on in a more or less enclosed space where the moisture and temperature conditions are kept carefully regulated, or in a place where the seeds are kept away from direct contact with sunlight and the earth. Ordinarily, the nuts are placed in trays 2" to 3" deep, 2' to 2-1/2' wide and 5' to 6' long. The bottom tray is then placed upon a pair of sawhorses or other device, in a shady place and 2' to 2-1/2' above the ground then the other trays are placed on and above the first one until all the nuts are in the tier of trays, or until it is 2' to 3' tall. Sometimes a current of heated, circulating air is used to doubly hasten the curing process, but this practice is to be discouraged as too often the undue heating of the nut germ while in this stage of ripening injures it, and thus the nuts are rendered unfit for reproduction. The nuts in the trays should be frequently stirred or turned over during the first week or ten days while curing.

In the case of chestnuts, the crop should be harvested as soon as possible after the first nuts fall so that the damage from weevils may be kept at a minimum. Immediately after the nuts are surface-dried they should be treated to an application of carbon disulphide, one ounce to a tightly closed capacity content of an apple barrel; time of treatment about 24 hours. While this treatment probably will not kill all the weevils it will insure a much larger percentage of germination than there would be otherwise.

After fumigating the nuts should be spread out on wire-cloth bottom trays and placed under a shed or trees, where a free circulation of air will in a few days sufficiently cure the nuts, so that they may be stratified and set away in a pit in the ground on the north side of a building, wall, hedge-row or evergreen trees, thus insuring them ample moisture and protection against sudden changes of temperatures and the ravages of rodents and other pests.

Other nuts of the temperate zone may, in a general way, be treated without any special care other than that required to keep them from getting moist and warm, or destroyed by rodents or other nut-eating animals, or by fungous troubles.

On the whole probably the best method of treatment for the amateur or small grower of seedling nut trees, is to stratify the nuts as soon as harvested, assuming that the nuts have been fairly well cured by a few days' exposure to drying air currents.

Stratification consists in layering the nuts in clean, sharp sand, light loam or sawdust and placing them in a cold, moist place, as a well drained and shaded north hillside, where their contact with the soil and protection from the direct rays of the sun will insure complete dormancy and at the same time prevent the development of fungous troubles. To this end the common practice is to dig a somewhat shallow trench and place in it, one layer deep, the "flats" in which the nuts are stratified. The flat usually employed is a shallow, wooden box in which the bottom is provided with ample, narrow drainage cracks and the top covered with wire cloth that will keep out mice or larger rodents. Not infrequently the bottom is a wire cloth one instead of wood. Dimensions of the flats vary, somewhat, but a convenient size is 30" long, 15"-16" wide, 3"-4" deep, sides ends and bottom being made of lumber strips (creosoted for preservation purposes) 3\4" thick and 3"-4" wide.

In these flats the nuts are placed layer upon layer, with sand, loam or sawdust between, something as follows: one inch of sand or other medium on the bottom, then a single layer of nuts, another inch layer of sand, etc., until the flat is full, when it is covered with the wire cloth, placed in the trench, covered with a few inches to a foot of leaves, moist hay, cornstalks or even soil, and left for the winter. At the time the medium for layering the

nuts is being prepared, it will be well, if ants are present in the section where the nuts are to be stored, or later placed in nursery bed, to mix a liberal percentage of unleached wood ashes with the sand, sawdust or loam, say one part in five, more or less.

Other flats are placed alongside or end to end in the trench until the stock is all in, when the whole may be covered uniformly. The layer of leaves or hay next to the wire cover of the flats assists in the work of uncovering when the inspections are made for the purpose of ascertaining the state of dormancy or germination.

One step more and the seed stage passes into the province of the seedling. As soon as the stratified nuts begin to germinate they should be removed from the flats and planted in the nursery or propagating bed. The site for this purpose should be one that is well drained, open to air and sunshine and possessing a clean, fine, mellow and rather light loamy soil. The size of this plat will vary to meet the needs of the quantity of nuts in hand and should be prepared, preferably the fall before, by stirring the soil deeply and thoroughly working into it a goodly supply of well rotted stable compost.

The rows for hand culture may be 18"-30" apart; for loose hoeing, 3' to 3-1/2' and should lie along north and south lines. The distance and depth of the nuts in the row will vary with their size. In general, one may say that a nut should be planted the length of the lateral diameter below the surface of the soil, when it has settled, or about double that depth when the soil is freshly worked over it. The distance apart in the row will vary somewhat with the rapidity of growth of the species; six to eight inches being a fair average for walnuts and chestnuts, and 4 to 6 for hickories and pecans.

Fifteenth Annual Convention of the Northern Nut Growers' Association, New York City,

September 3, 4, 5, 1924

Species Variety Exhibitor Address Origin

1. Black walnut J. A. Neilson Vineland, Ont. St. Thomas, Ont. 2. Black walnut " " " " " Niagara-on-Lake. 3. Black walnut Walsh " " " " " Simcoe, Ont. 4. Black walnut " " " " " Electric, Ont. 5. Black walnut " " " " " Villoria, Ont. 6. Black walnut Ohio J. F. Jones Lancaster, Pa. 7. Black walnut Stabler " " " " " 8. Black walnut Thomas " " " " " 9. Persian walnut J. A. Neilson Vineland, Ont. Carpathian Mts. 10. Persian walnut " " " " " Grimsley, Ont. 11. Persian walnut " " " " " St. Catherines, Ont. 12. Persian walnut Alpine J. F. Jones Lancaster, Pa. 13. Persian walnut Mayette seedling " " " " " 14. Persian walnut Sinclair " " " " " 15. Persian walnut Wiltz Mayette " " " " " 16. Heartnut J. A. Neilson Vineland, Ont. Near Jordon, Ont. 17. Heartnut " " " " " Near Hamilton, Ont. 18. Heartnut " " " " " Near Scotland, Ont. 19. Heartnut Faust J. F. Jones Lancaster, Pa. 20. Heartnut Lancaster " " " " " 21. Heartnut Ritchey " " " " " 22. Sieboldiana walnut J. A. Neilson Vineland, Ont. Hamilton, Ont. 23. Sieboldiana walnut " " " " " OAC Campus, Guelph. 24. Shagbark J. A. Neilson Vineland, Ont. Electric, Ont. 25. Shagbark " " " " " Norfolk Co., Ont. 26. Shagbark hybrid Beaver J. F. Jones Lancaster, Pa. 27. Shagbark hybrid Siers " " " " " 28. Pecan J. A. Neilson Vineland, Ont. 15 miles N. of Toronto 29. Almond " " " " " Gellatly, B. C. 30. Filbert Tray of mixed " " " " " Gellatly, B. C. 31. Filbert White aveline J. F. Jones Lancaster, Pa. 32. Filbert Barcelona " " " " " 33. Filbert Cosford " " " " " 34. Filbert Daviana " " " " " 35. Filbert Du Chilly " " " " " 36. Filbert Giant de Halle " " " " " 37. Filbert Italian Red " " " " " 38. Filbert Merribrooke " " " " " 39. Filbert Noci Lunghe " " " " " 40. Filbert Rush " " " " " 42. Filbert hybrid Rush x Barcelona " " " " " 43. Filbert hybrid Rush x Barcelona " " " " " 44. Filbert hybrid Rush x Barcelona " " " " " 45. Filbert hybrid Rush Cosford " " " " " 46. Filbert hybrid Rush Cosford " " " " " 47. Filbert hybrid Rush Giant de

Halle " " " " 48. Filbert hybrid Rush Giant de Halle " " " " 49. Filbert hybrid Rush Giant de Halle " " " " 50. Filbert hybrid Rush Italian Red " " " " 51. Photograph—Walnut-cracking machine Black Walnut Company, 509-11-13, Spruce St., St. Louis, Mo. 52. Budding Knife

[Transcriber's note: No. 41 is missing in the original]

Among those present at the Fifteenth Annual Convention of the Northern Nut Growers' Association, were the following:

Dr. N. L. Britton, Director of the N. Y. Botanical Gardens.
Dr. Fred E. Brooks, Entomologist, U. S. Dept. of Agriculture.
Dr. and Mrs. Frank L. Baum, Boyertown, Pa.
Mr. Willard G. Bixby, Baldwin, N. Y.
Dr. A. F. Blakeslee, Cold Spring Harbor, L. I.
Mr. and Mrs. F. A. Bartlett, Stamford, Conn.
Miss H. T. Bennett, Boston, Mass.
Prof. J. Franklin Collins, Providence, R. I.
Dr. John E. Cannaday, Charleston, W. Va.
Mr. G. M. Coddling, Mt. Vernon, N. Y.
Prof. A. S. Colby, University of Illinois, Urbana, Ill.
Dr. W. C. Deming, Hartford, Conn.
Mr. Zenas H. Ellis, Fair Haven, Vt.
Mrs. W. D. Ellwanger, Rochester, N. Y.
Mr. Ammon P. Fritz, 55 E. Franklin St., Ephrata, Pa.
Mr. A. F. Graf, Bardonia, N. Y.
Mrs. B. W. Gahn, U. S. Dept. of Agriculture.
Mr. and Mrs. Karl W. Greene, Washington, D. C.
Dr. M. A. Howe, Assistant to Director, N. Y. Botanical Gardens.
Mr. Henry Hicks, Baldwin, L. I. (Hicks' Nurseries).
Mr. John W. Hershey, E. Downingtown, Pa.

Mr. Lee Whitaker Jaques, 74 Waverly St., Jersey City, N. J.
Mr. J. F. Jones, Lancaster, Pa.
Mr. M. G. Kains, Suffern, N. Y.
Mr. Thomas W. Little, Cos Cob, Conn.
Dr. Robt. T. Morris, Cos Cob, Route 28, Box 95, Stamford, Conn.
Dr. L. H. MacDaniels, N. Y. State College of Agriculture, Ithaca, N. Y.
Prof. Jas. A. Neilson, Horticultural Exp. Station, Vineland, Ont., Can.
Mr. Ralph T. Olcott, Ed. American Nut Journal, Rochester, N. Y.
Mrs. R. T. Olcott, Rochester, N. Y.
Mr. P. H. O'Connor, Bowie, Md.
Mr. C. A. Reed, U. S. Department of Agriculture
Mr. John Rick, Reading, Pa.
Dr. J. Russell Smith, Columbia University, New York, N. Y.
Dr. Oscar Stapf, F. R. S., late Curator of the Herbarium, Royal Botanic
Gardens, Kew, London, England.
Mr. Harry R. Weber, Cincinnati, Ohio.
Mrs. Laura E. Woodward, West Chester, Pa.
Dr. and Mrs. G. A. Zimmerman, Piketown, Pa.

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This space is paid for by Jas. L. Brooke, Pleasantville, Ohio, who is only too anxious at any time to assist in encouraging and promoting Nut Culture in the North.

While he has only recently taken up this work, and is therefore a practical stranger on the roster of The Northern Nut Growers' Association, he will

only be too anxious and willing at any time to contribute to the cause in any way possible.

He is making a thorough search in his neighborhood where chestnuts, hickory nuts and black walnuts grow in abundance, for nuts of approved merit for propagation.

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LANCASTER, PA.

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FIFTEENTH ANNUAL MEETING ***

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